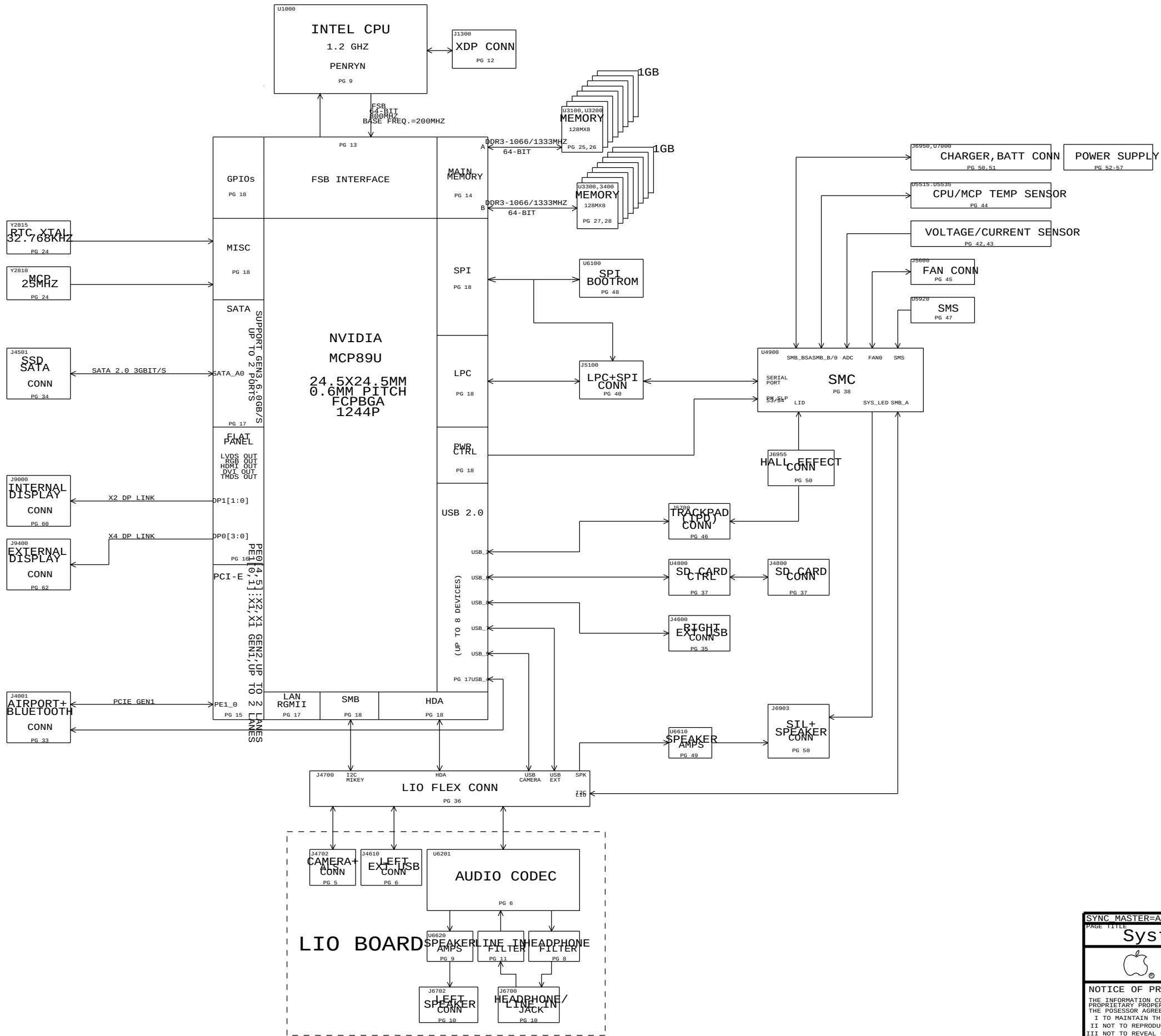


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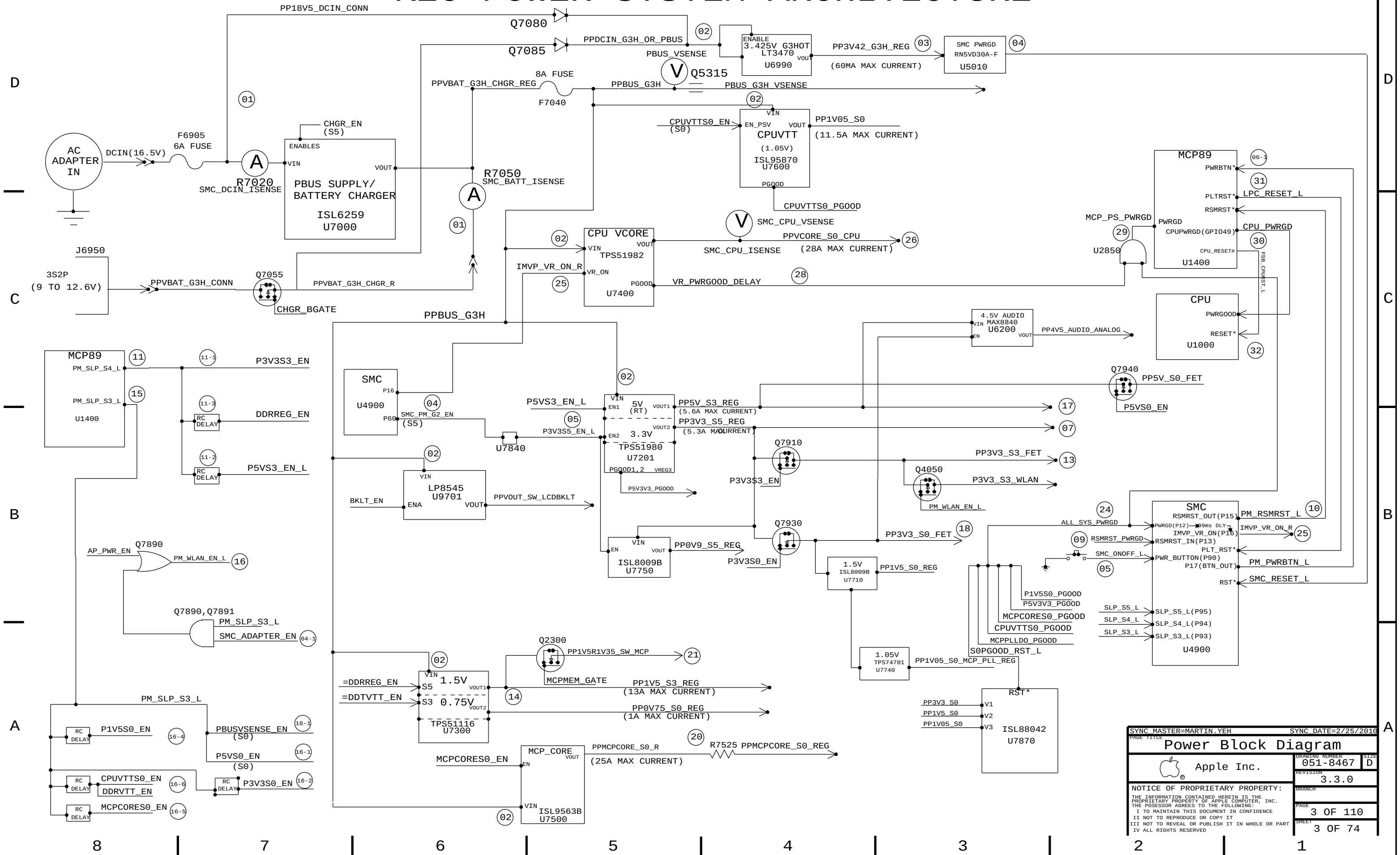
B

A



SYNC MASTER=ALAN.DAI		SYNC DATE=02/26/2010	
PAGE TITLE			
System Block Diagram			
 Apple Inc.		DRAWING NUMBER	051-8467
		STATE	D
		REVISION	3.3.0
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K16 POWER SYSTEM ARCHITECTURE



SYNC MASTER=MARTIN.YEH		SYNC DATE=2/25/2016	
PAGE TITLE		Power Block Diagram	
Apple Inc.		DRAWING NUMBER	051-8467
		REVISION	3.3.0
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K16 BOM Variants on following page

DRAM CFG CHART

VENDOR	CFG 0	CFG 1
HYNIX	0	0
SAMSUNG	0	1
MICRON	1	0
ELPIDA	1	1

SIZE	CFG 2
2GB	0
4GB	1

DIE REV	CFG 3
A	0
B	1

D

C

B

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BOM Groups

BOM GROUP	BOM OPTIONS
K16_COMMON	COMMON, ALTERNATE, PROJ:K16,K16_MISC,MCP89U:A03,K16_DEBUG:ENG,K16_PROGPARTS,SPI:41MHZ,LVDDR3:YES,WLAN_PCTL:HW,IPD_5V:S5_INT,IPD_3V3:S5
K16_MISC	DP_ESD,DP_PWR:SMC,VFRQ:SLPS3,HVDDLDO:FIXED,MCPHVD0:P2V5,MCPPLL_R:REG,S0PG00D_BJT,ISL6259_SCREENED:YES,DPI2C:SMC
K16_PROGPARTS	BOOTROM:UNLOCKED,SMC:PROG
K16_DEVEL:ENG	BKLT:ENG,BMON:ENG,XDP_CONN,LPCPLUS,VREFMRGN:YES,EFI_DEBUG,S0PG00D_ISL,MCPPLL_LDO,S3_S0_LED
K16_DEVEL:PVT	LPCPLUS
K16_DEBUG:ENG	DEVEL_BOM,SMC_DEBUG:YES,XDP
K16_DEBUG:PVT	DEVEL_BOM,BKLT:PROD,BMON:PROD,SMC_DEBUG:YES,XDP,VREFMRGN:NO
K16_DEBUG:PROD	BKLT:PROD,BMON:PROD,SMC_DEBUG:YES,XDP,VREFMRGN:NO
DDR3:HYNIX_2GB	DRAM_CFG0:L,DRAM_CFG1:L,DRAM_CFG2:L,DRAM_CFG3:L,DRAM_TYPE:HYNIX_2GB
DDR3:SAMSUNG_2GB	DRAM_CFG0:L,DRAM_CFG2:L,DRAM_CFG3:L,DRAM_TYPE:SAMSUNG_2GB
DDR3:MICRON_2GB	DRAM_CFG0:H,DRAM_CFG1:L,DRAM_CFG2:L,DRAM_CFG3:L,DRAM_TYPE:MICRON_2GB
DDR3:ELPIDA_2GB	DRAM_CFG0:H,DRAM_CFG2:L,DRAM_CFG3:L,DRAM_TYPE:ELPIDA_2GB
DDR3:ELPIDA_4GB	DRAM_CFG0:H,DRAM_CFG2:H,DRAM_CFG3:L,DRAM_TYPE:ELPIDA_4GB
DDR3:HYNIX_4GB	DRAM_CFG0:L,DRAM_CFG1:L,DRAM_CFG2:H,DRAM_CFG3:L,DRAM_TYPE:HYNIX_4GB
DDR3:SAMSUNG_4GB	DRAM_CFG0:L,DRAM_CFG2:H,DRAM_CFG3:L,DRAM_TYPE:SAMSUNG_4GB
DDR3:MICRON_4GB	DRAM_CFG0:H,DRAM_CFG1:L,DRAM_CFG2:H,DRAM_CFG3:L,DRAM_TYPE:MICRON_4GB
CAPS:SS	SS_CAP_2_2UF,SS_CAP_10UF,SS_CAP_1UF,SS_CAP_22UF
CAPS:MU	MU_CAP_2_2UF,MU_CAP_10UF,MU_CAP_1UF,MU_CAP_22UF
CAPS:TY	TY_CAP_2_2UF,TY_CAP_10UF,TY_CAP_1UF,TY_CAP_22UF

Programmable Parts

338S0563	1	IC,SMC,HS8/2117,9X9MM,TLP,HF	U4900	CRITICAL	SMC:BLANK
335S0610	1	IC,FLASH,SPI,32MBIT,3.3V,86MHZ,8-SOP	U6100	CRITICAL	BOOTROM:BLANK

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
-------------	---------------------------	------------	---------	-----------

138S0681	138S0638		ALL	TAIYO YUDEN AS ALTERNATE
152S0874	152S0516		ALL	MAGLAYERS AS ALTERNATE
152S0847	152S0586		ALL	MAGLAYERS AS ALTERNATE
353S2987	353S2988	HVDDLDO:FIXED	ALL	TPS7172S ALTERNATE FOR U2590
104S0023	104S0018		ALL	CYNTEC/DALE AS ALTERNATES
107S0139	107S0075		ALL	CYNTEC AS ALTERNATE
138S0671	138S0673		ALL	TAIYO AS ALTERNATE
155S0578	155S0367		ALL	TAIYO AS ALTERNATE
376S0926	376S0610		ALL	FAIRCHILD AS ALTERNATE
155S0457	155S0329		ALL	MAGLAYERS AS ALTERNATE
377S0107	377S0066		ALL	ON SEMI AS ALTERNATE

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S3820	1	IC,MCP89U-A01,24.5MMX24.5MM,1244FCBGA	U1400	CRITICAL	MCP89U:A01
337S3868	1	IC,MCP89U-A02,24.5MMX24.5MM,1244FCBGA	U1400	CRITICAL	MCP89U:A02
337S3938	1	IC,MCP89U-A03,24.5MMX24.5MM,1244FCBGA	U1400	CRITICAL	MCP89U:A03

333S0552	4	HYNIX,LVDDR3,1GBIT,7.5X11.0	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:HYNIX_2GB
333S0552	4	HYNIX,LVDDR3,1GBIT,7.5X11.0	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:HYNIX_2GB
333S0552	4	HYNIX,LVDDR3,1GBIT,7.5X11.0	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:HYNIX_2GB
333S0552	4	HYNIX,LVDDR3,1GBIT,7.5X11.0	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:HYNIX_2GB
333S0553	4	SAMSUNG,LVDDR3,1GBIT,7.5X11.0	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:SAMSUNG_2GB
333S0553	4	SAMSUNG,LVDDR3,1GBIT,7.5X11.0	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:SAMSUNG_2GB
333S0553	4	SAMSUNG,LVDDR3,1GBIT,7.5X11.0	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:SAMSUNG_2GB
333S0553	4	SAMSUNG,LVDDR3,1GBIT,7.5X11.0	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:SAMSUNG_2GB
333S0554	4	MICRON,LVDDR3,1GBIT,8X11.5	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:MICRON_2GB
333S0554	4	MICRON,LVDDR3,1GBIT,8X11.5	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:MICRON_2GB
333S0554	4	MICRON,LVDDR3,1GBIT,8X11.5	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:MICRON_2GB
333S0554	4	MICRON,LVDDR3,1GBIT,8X11.5	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:MICRON_2GB
333S0565	4	ELPIDA,LVDDR3,1GBIT,7.5X10.6	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:ELPIDA_2GB
333S0565	4	ELPIDA,LVDDR3,1GBIT,7.5X10.6	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:ELPIDA_2GB
333S0565	4	ELPIDA,LVDDR3,1GBIT,7.5X10.6	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:ELPIDA_2GB
333S0565	4	ELPIDA,LVDDR3,1GBIT,7.5X10.6	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:ELPIDA_2GB
333S0566	4	ELPIDA,LVDDR3,1GBIT,7.5X10.6	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:ELPIDA_4GB
333S0566	4	ELPIDA,LVDDR3,2GBIT,7.5X10.6	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:ELPIDA_4GB
333S0566	4	ELPIDA,LVDDR3,2GBIT,7.5X10.6	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:ELPIDA_4GB
333S0566	4	ELPIDA,LVDDR3,2GBIT,7.5X10.6	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:ELPIDA_4GB
333S0555	4	HYNIX,LVDDR3,2GBIT,9X11.1	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:HYNIX_4GB
333S0555	4	HYNIX,LVDDR3,2GBIT,9X11.1	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:HYNIX_4GB
333S0555	4	HYNIX,LVDDR3,2GBIT,9X11.1	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:HYNIX_4GB
333S0555	4	HYNIX,LVDDR3,2GBIT,9X11.1	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:HYNIX_4GB
333S0556	4	SAMSUNG,LVDDR3,2GBIT,7.5X11.0	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:SAMSUNG_4GB
333S0556	4	SAMSUNG,LVDDR3,2GBIT,7.5X11.0	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:SAMSUNG_4GB
333S0556	4	SAMSUNG,LVDDR3,2GBIT,7.5X11.0	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:SAMSUNG_4GB
333S0556	4	SAMSUNG,LVDDR3,2GBIT,7.5X11.0	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:SAMSUNG_4GB
333S0557	4	MICRON,LVDDR3,2GBIT,9X11.5	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:MICRON_4GB
333S0557	4	MICRON,LVDDR3,2GBIT,9X11.5	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:MICRON_4GB
333S0557	4	MICRON,LVDDR3,2GBIT,9X11.5	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:MICRON_4GB
333S0557	4	MICRON,LVDDR3,2GBIT,9X11.5	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:MICRON_4GB
353S2392	1	IC,ISL6259,BATCHARGER,4X4MM,QFN28	U7000	CRITICAL	ISL6259_SCREENED:NO
353S2929	1	IC,ISL6259,BATCHARGER,3%,4C4MM,QFN28	U7000	CRITICAL	ISL6259_SCREENED:YES

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SYNC MASTER=K6_MLB

SYNC DATE=12/11/2009

PAGE TITLE

BOM Configuration



Apple Inc.

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051-8467

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BOM Variants			Bar Code Labels / EEE #'s					
BOM NUMBER	BOM NAME	BOM OPTIONS	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
639-1070	PCBA, MLB, 1.86GHZ HY 2GB, MU CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCWQ, CAPS:MU, DDR3:HYNIX_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCWM]	CRITICAL	EEEE:DCWM
639-0837	PCBA, MLB, 1.86GHZ, HY 2GB, SS CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCXW, CAPS:SS, DDR3:HYNIX_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCWN]	CRITICAL	EEEE:DCWN
639-1096	PCBA, MLB, 1.86GHZ HY 2GB, TY CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCXN, CAPS:TY, DDR3:HYNIX_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCWP]	CRITICAL	EEEE:DCWP
639-1101	PCBA, MLB, 1.86GHZ HY 4GB, MU CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCXV, CAPS:MU, DDR3:HYNIX_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCWQ]	CRITICAL	EEEE:DCWQ
639-1098	PCBA, MLB, 1.86GHZ HY 4GB, SS CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCXQ, CAPS:SS, DDR3:HYNIX_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCWR]	CRITICAL	EEEE:DCWR
639-1068	PCBA, MLB, 1.86GHZ HY 4GB, TY CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCWN, CAPS:TY, DDR3:HYNIX_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCWT]	CRITICAL	EEEE:DCWT
639-1083	PCBA, MLB, 1.86GHZ MI 2GB, MU CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCX6, CAPS:MU, DDR3:MICRON_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCWV]	CRITICAL	EEEE:DCWV
639-1078	PCBA, MLB, 1.86GHZ MI 2GB, SS CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCX1, CAPS:SS, DDR3:MICRON_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCWW]	CRITICAL	EEEE:DCWW
639-1090	PCBA, MLB, 1.86GHZ MI 2GB, TY CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCXG, CAPS:TY, DDR3:MICRON_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCWX]	CRITICAL	EEEE:DCWX
639-1088	PCBA, MLB, 1.86GHZ MI 4GB, MU CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCXD, CAPS:MU, DDR3:MICRON_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCWY]	CRITICAL	EEEE:DCWY
639-1067	PCBA, MLB, 1.86GHZ MI 4GB, SS CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCWM, CAPS:SS, DDR3:MICRON_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCX0]	CRITICAL	EEEE:DCX0
639-1077	PCBA, MLB, 1.86GHZ MI 4GB, TY CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCX0, CAPS:TY, DDR3:MICRON_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCX1]	CRITICAL	EEEE:DCX1
639-1080	PCBA, MLB, 1.86GHZ SA 2GB, MU CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCX3, CAPS:MU, DDR3:SAMSUNG_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCX2]	CRITICAL	EEEE:DCX2
639-1095	PCBA, MLB, 1.86GHZ SA 2GB, SS CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCXM, CAPS:SS, DDR3:SAMSUNG_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCX3]	CRITICAL	EEEE:DCX3
639-1071	PCBA, MLB, 1.86GHZ SA 2GB, TY CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCWR, CAPS:TY, DDR3:SAMSUNG_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCX4]	CRITICAL	EEEE:DCX4
639-1097	PCBA, MLB, 1.86GHZ SA 4GB, MU CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCXP, CAPS:MU, DDR3:SAMSUNG_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCX5]	CRITICAL	EEEE:DCX5
639-1084	PCBA, MLB, 1.86GHZ SA 4GB, SS CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCX7, CAPS:SS, DDR3:SAMSUNG_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCX6]	CRITICAL	EEEE:DCX6
639-1091	PCBA, MLB, 1.86GHZ SA 4GB, TY CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DCXH, CAPS:TY, DDR3:SAMSUNG_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCX7]	CRITICAL	EEEE:DCX7
639-1092	PCBA, MLB, 2.13GHZ HY 2GB, MU CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCXJ, CAPS:MU, DDR3:HYNIX_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCX8]	CRITICAL	EEEE:DCX8
639-1082	PCBA, MLB, 2.13GHZ, HY 2GB, SS CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCX5, CAPS:SS, DDR3:HYNIX_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCX9]	CRITICAL	EEEE:DCX9
639-1085	PCBA, MLB, 2.13GHZ HY 2GB, TY CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCX8, CAPS:TY, DDR3:HYNIX_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DXC0]	CRITICAL	EEEE:DCXC
639-1089	PCBA, MLB, 2.13GHZ HY 4GB, MU CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCXF, CAPS:MU, DDR3:HYNIX_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCXD]	CRITICAL	EEEE:DCXD
639-1075	PCBA, MLB, 2.13GHZ HY 4GB, SS CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCWX, CAPS:SS, DDR3:HYNIX_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCXF]	CRITICAL	EEEE:DCXF
639-1079	PCBA, MLB, 2.13GHZ HY 4GB, TY CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCX2, CAPS:TY, DDR3:HYNIX_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCXG]	CRITICAL	EEEE:DCXG
639-1099	PCBA, MLB, 2.13GHZ MI 2GB, MU CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCXR, CAPS:MU, DDR3:MICRON_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCXH]	CRITICAL	EEEE:DCXH
639-1087	PCBA, MLB, 2.13GHZ MI 2GB, SS CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCXC, CAPS:SS, DDR3:MICRON_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCXJ]	CRITICAL	EEEE:DCXJ
639-1069	PCBA, MLB, 2.13GHZ MI 2GB, TY CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCWP, CAPS:TY, DDR3:MICRON_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCKK]	CRITICAL	EEEE:DCXK
639-1100	PCBA, MLB, 2.13GHZ MI 4GB, MU CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCXT, CAPS:MU, DDR3:MICRON_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCTL]	CRITICAL	EEEE:DCXL
639-1093	PCBA, MLB, 2.13GHZ MI 4GB, SS CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCXK, CAPS:SS, DDR3:MICRON_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DXXM]	CRITICAL	EEEE:DCXM
639-1076	PCBA, MLB, 2.13GHZ MI 4GB, TY CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCWY, CAPS:TY, DDR3:MICRON_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DXXN]	CRITICAL	EEEE:DCXN
639-1074	PCBA, MLB, 2.13GHZ SA 2GB, MU CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCWW, CAPS:MU, DDR3:SAMSUNG_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DXXP]	CRITICAL	EEEE:DCXP
639-1072	PCBA, MLB, 2.13GHZ SA 2GB, SS CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCWT, CAPS:SS, DDR3:SAMSUNG_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DXXQ]	CRITICAL	EEEE:DCXQ
639-1086	PCBA, MLB, 2.13GHZ SA 2GB, TY CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCX9, CAPS:TY, DDR3:SAMSUNG_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DXXR]	CRITICAL	EEEE:DCXR
639-1073	PCBA, MLB, 2.13GHZ SA 4GB, MU CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCWV, CAPS:MU, DDR3:SAMSUNG_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DXXT]	CRITICAL	EEEE:DCXT
639-1081	PCBA, MLB, 2.13GHZ SA 4GB, SS CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCX4, CAPS:SS, DDR3:SAMSUNG_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DXXV]	CRITICAL	EEEE:DCXV
639-1094	PCBA, MLB, 2.13GHZ SA 4GB, TY CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DCXL, CAPS:TY, DDR3:SAMSUNG_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DXXW]	CRITICAL	EEEE:DCXW
639-1450	PCBA, MLB, 1.86GHZ EL 2GB, MU CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DG4W, CAPS:MU, DDR3:ELPIDA_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG4W]	CRITICAL	EEEE:DG4W
639-1451	PCBA, MLB, 1.86GHZ EL 2GB, SS CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DG4Y, CAPS:SS, DDR3:ELPIDA_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG4Y]	CRITICAL	EEEE:DG4Y
639-1455	PCBA, MLB, 1.86GHZ EL 2GB, TY CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DG53, CAPS:TY, DDR3:ELPIDA_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG53]	CRITICAL	EEEE:DG53
639-1453	PCBA, MLB, 2.13GHZ EL 2GB, MU CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DG51, CAPS:MU, DDR3:ELPIDA_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG51]	CRITICAL	EEEE:DG51
639-1454	PCBA, MLB, 2.13GHZ EL 2GB, SS CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DG52, CAPS:SS, DDR3:ELPIDA_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG52]	CRITICAL	EEEE:DG52
639-1452	PCBA, MLB, 2.13GHZ EL 2GB, TY CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DG50, CAPS:TY, DDR3:ELPIDA_2GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG50]	CRITICAL	EEEE:DG50
639-1458	PCBA, MLB, 1.86GHZ EL 4GB, MU CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DG5P, CAPS:MU, DDR3:ELPIDA_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG5P]	CRITICAL	EEEE:DG5P
639-1463	PCBA, MLB, 1.86GHZ EL 4GB, SS CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DG5W, CAPS:SS, DDR3:ELPIDA_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG5W]	CRITICAL	EEEE:DG5W
639-1460	PCBA, MLB, 1.86GHZ EL 4GB, TY CAP, K16	K16_CMNPTS, CPU:1.86GHZ, EEEE:DG5T, CAPS:TY, DDR3:ELPIDA_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG5T]	CRITICAL	EEEE:DG5T
639-1462	PCBA, MLB, 2.13GHZ EL 4GB, MU CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DG5Q, CAPS:MU, DDR3:ELPIDA_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG5Q]	CRITICAL	EEEE:DG5Q
639-1459	PCBA, MLB, 2.13GHZ EL 4GB, SS CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DG5V, CAPS:SS, DDR3:ELPIDA_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG5V]	CRITICAL	EEEE:DG5V
639-1461	PCBA, MLB, 2.13GHZ EL 4GB, TY CAP, K16	K16_CMNPTS, CPU:2.13GHZ, EEEE:DG5R, CAPS:TY, DDR3:ELPIDA_4GB	825-7557	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DG5R]	CRITICAL	EEEE:DG5R
607-6915	CMN PTS, PCBA, MLB, K16	K16_COMMON						
085-1327	K16 MLB DEVELOPMENT BOM	K16_DEVEL : ENG						

SAMSUNG

MURATA

TAIYO YUDEN

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0632	1	CAP, 2.2UF, 6.3V, 20%, 0402	C4807	CRITICAL	SS_CAP_2_2UF	138S0633	1	CAP, 2.2UF, 6.3V, 20%, 0402	C4807	CRITICAL	MU_CAP_2_2UF	138S0634	1	CAP, 2.2UF, 6.3V, 20%, 0402	C4807	CRITICAL	TY_CAP_2_2UF

K16-Specific BOM Tables

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S3751	1	PDC, SLGAB, PRQ, 1.86, 17W, 1066, ED, 6M, BGA	U1000	CRITICAL	CPU:1.86GHZ
337S3758	1	PDC, SLGEQ, PRQ, 2.13, 17W, 1066, ED, 6M, BGA	U1000	CRITICAL	CPU:2.13GHZ
341T0276	1	IC ASSY, SMC EXTERNAL, K16	U4900	CRITICAL	SMC:PROG
341T0275	1	IC ASSY, EFI UNLOCKED, K16	U6100	CRITICAL	BOOTROM:UNLOCKED
341S2785	1	IC EFI ROM, PVT, LOCKED, K16	U6100	CRITICAL	BOOTROM:LOCKED

Sub-BOMs

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
085-1327	1	K16 MLB DEVELOPMENT BOM	DEVEL	CRITICAL	DEVEL_BOM
607-6915	1	CMN PTS, PCBA, MLB, K16	CMNPTS	CRITICAL	K16_CMNPTS

SYNC MASTER=N/A

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PAGE TITLE

K16 BOM Variants



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Revision History							
Proto 0 (ECO #0000876215, v1.0.0, P4 change #210266, 03/16/2010)							
v1.1.0 (P4 change #211399, 03/24/2010)							
MCP: 7742015 - Added RC to DDC pass FETs to avoid glitch (pp. 7, 93). 7788138 - Added feedback divider and BOM tables for more HVDD LD0s (pp. 4, 25). SMC: 7761747 - Added resistors to connect TCON to SMC or MCP SMBus (pp. 4, 52, 90). 7787883 - Added support for DP HPD wake / S4 state (pp. 4, 7, 8, 19, 49, 50, 78, 94). SMS: 7765466 - Added S3 pull-up to SMS_INT_L to prevent leakage path (pp. 50, 59). 7769139 - Unstuffed SMS circuit (pg. 4). General: 7787897 - Property/page fixes to reduce CheckPlus warnings/errors (pp. 7, 8, 12, 17, 74, 93, 108).							
v1.2.0 (P4 change #211839, 03/26/2010)							
USB: 7796626 - Changed port switch from TPS2052B to TPS2069 (pg. 46). SMC: 7787883 - Added PLACE_NEAR property on R5022 to avoid stub (pg. 50). SMBus: 7761747 - Added TCON I2C nets to FUNC_TEST list for J9000 (pg. 7). Power: 7796648 - Changed DP and LCD power from PP3V3_S3 to PP3V3_S5 (pp. 8, 90). 7796658 - Changed backlight driver to E00 version (pg. 97). BOM: 7796661 - Set up primary & alternate for power supply FET (pp. 4, 72). 7796654 - Consolidated SSM6N15FE to SSM6N37FE (pg. 48). 7796658 - Changed RCs on some SMC analog inputs (pg. 54). 7796683 - Stuffed RC on backlight driver PWM input (pg. 97). General: 7796631 - Sorted BOM variants for easier verification (pg. 5). 7796631 - Cosmetic clean-up (pg. 76).							
v1.3.0 (P4 change #212050, 03/26/2010)							
SMBus: 7761747 - Added isolation FET and unstuffed series R's on TCON I2C for now (pp. 4, 90, 108). Power Supply: 7796661 - Removed alternate FET, made some FETs primary to other APN (pp. 4, 72, 73, 76). 7798425 - R/C value changes for 3.42V G3Hot power supply (pg. 69). 7798399 - R/C value changes for 5V/3.3V power supply (pg. 72). 7800179 - R value changes for CPU VCore power supply (pg. 74). 7798445 - R value changes for 0.9V S5 power supply (pg. 77). 7796658 - Changed backlight driver back to non-E00 version (pp. 4, 97). BOM: 7796658 - Added alternates for two caps per GSM and removed unused alternates (pg. 4). 7798399 - Consolidated 100pF caps (pp. 74, 75).							
v1.4.0 (P4 change #212757, 03/31/2010)							
MCP SPI: 7809733 - Changed strapping to select 62.5MHz SPI bus frequency (pg. 4). SMBus: 7796631 - Added XDP connection to SMBus aliases page (pp. 13, 52). 7808530 - Changed SMC 'MGMT' SMBus pull-ups from 4.7K to 2K (pg. 52). 7761747 - Documented SMBus addresses for panel (pg. 52). SD Card: 7800415 - Changed SD Card discharge R to more standard value (pg. 48). Power Supplies: 7803283 - Changed 5V S3 regulator output from 5.02V to 5.12V nominal (pg. 72). 7809760 - Stuffed C9799 and clarified tables/BOMOPTI0Ns around these parts (pg. 97).							
Proto 1 (ECO #0000884508, v2.0.0, P4 change #212783, 03/31/2010)							
v2.1.0 (P4 change #??????, ??/??/2010)							
BOM: 7796658 - Changed OMITs to OMIT_TABLES (pp. 10-11, 14-20, 26, 31-36, 49, 61).							
NOTE: All page numbers are .csa, not PDF. See page 1 for .csa -> PDF mapping.							

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SYNC MASTER=N/A

SYNC DATE=N/A

Revision History

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Revision History

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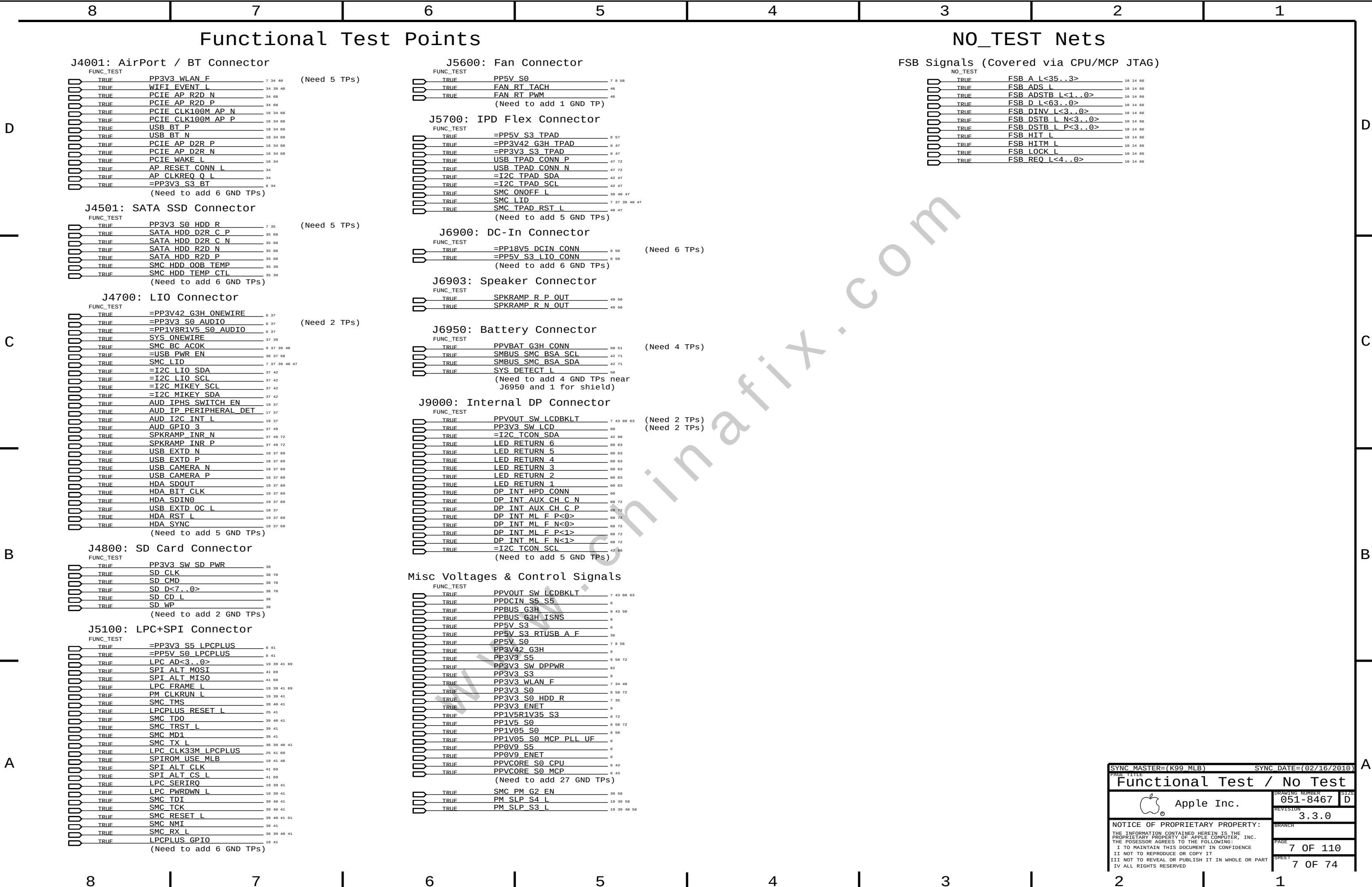
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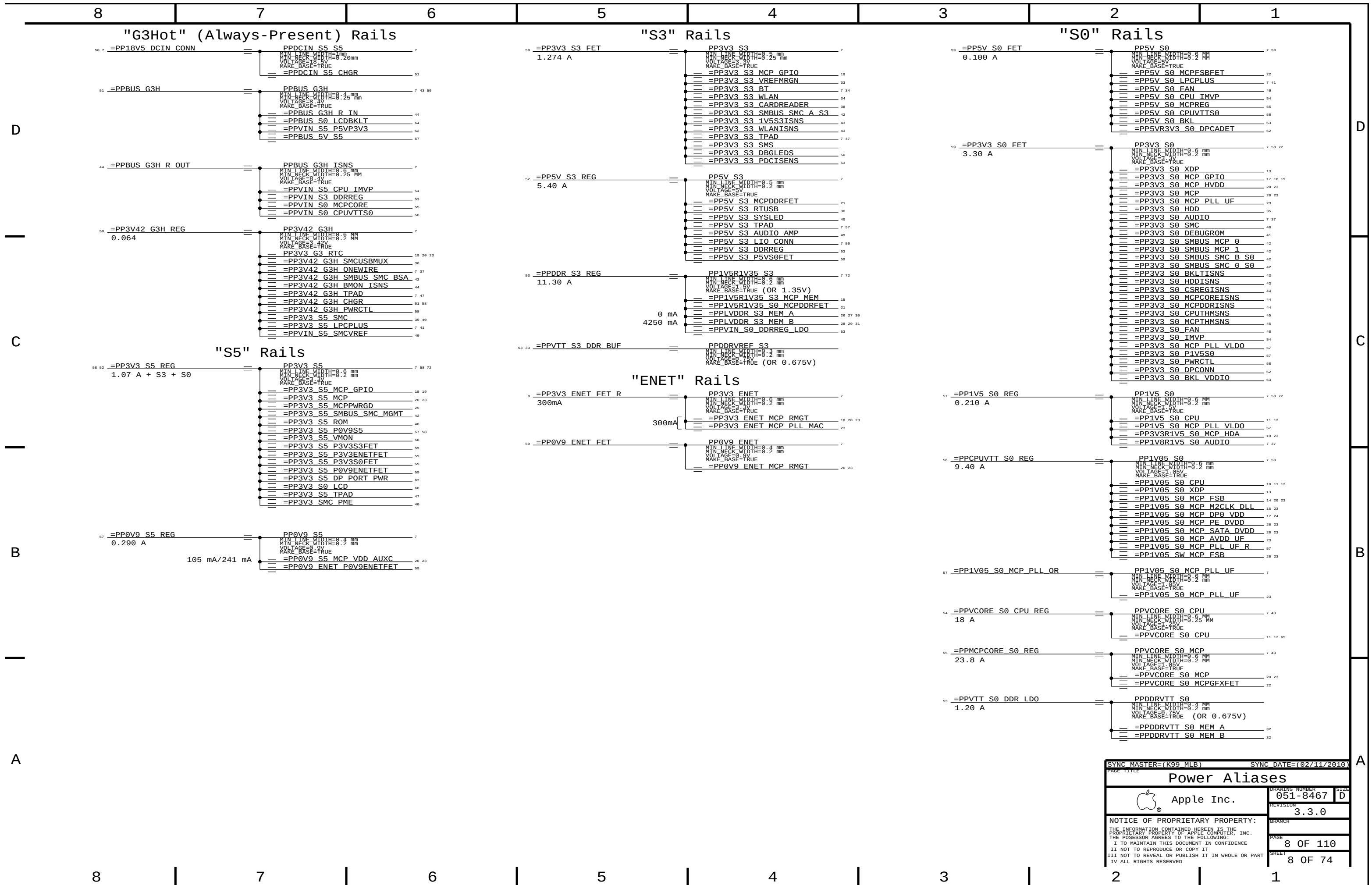
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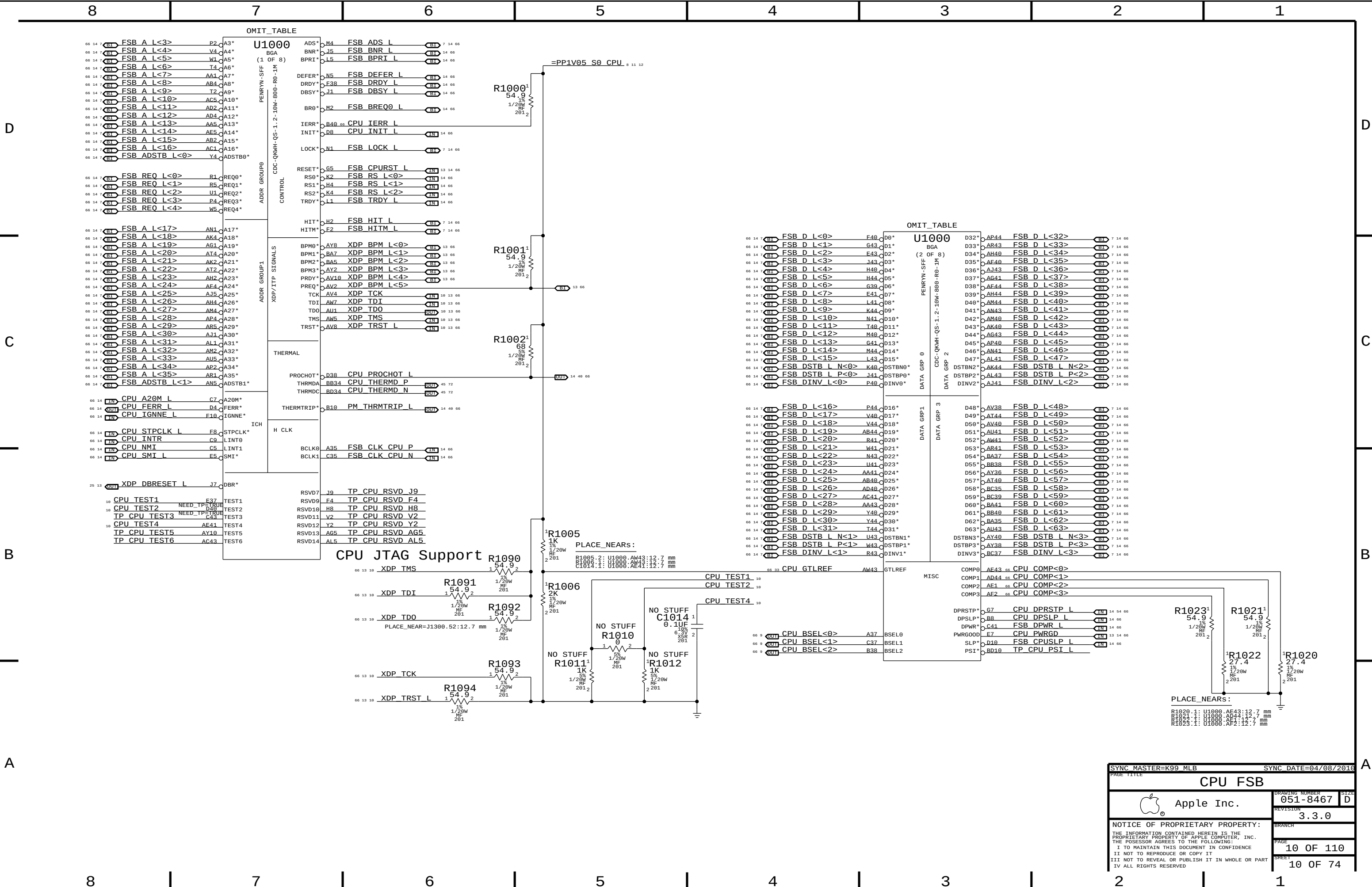
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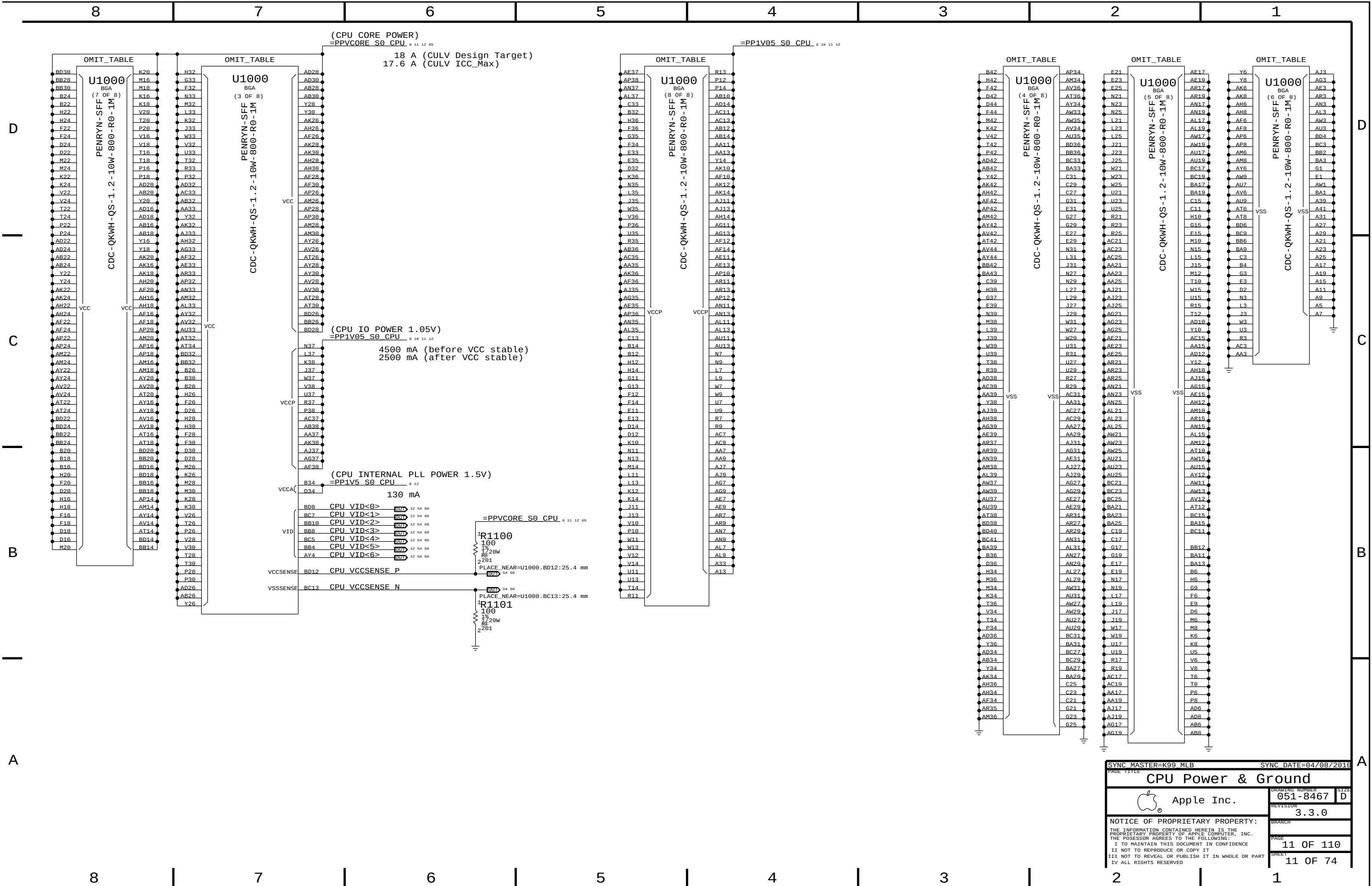
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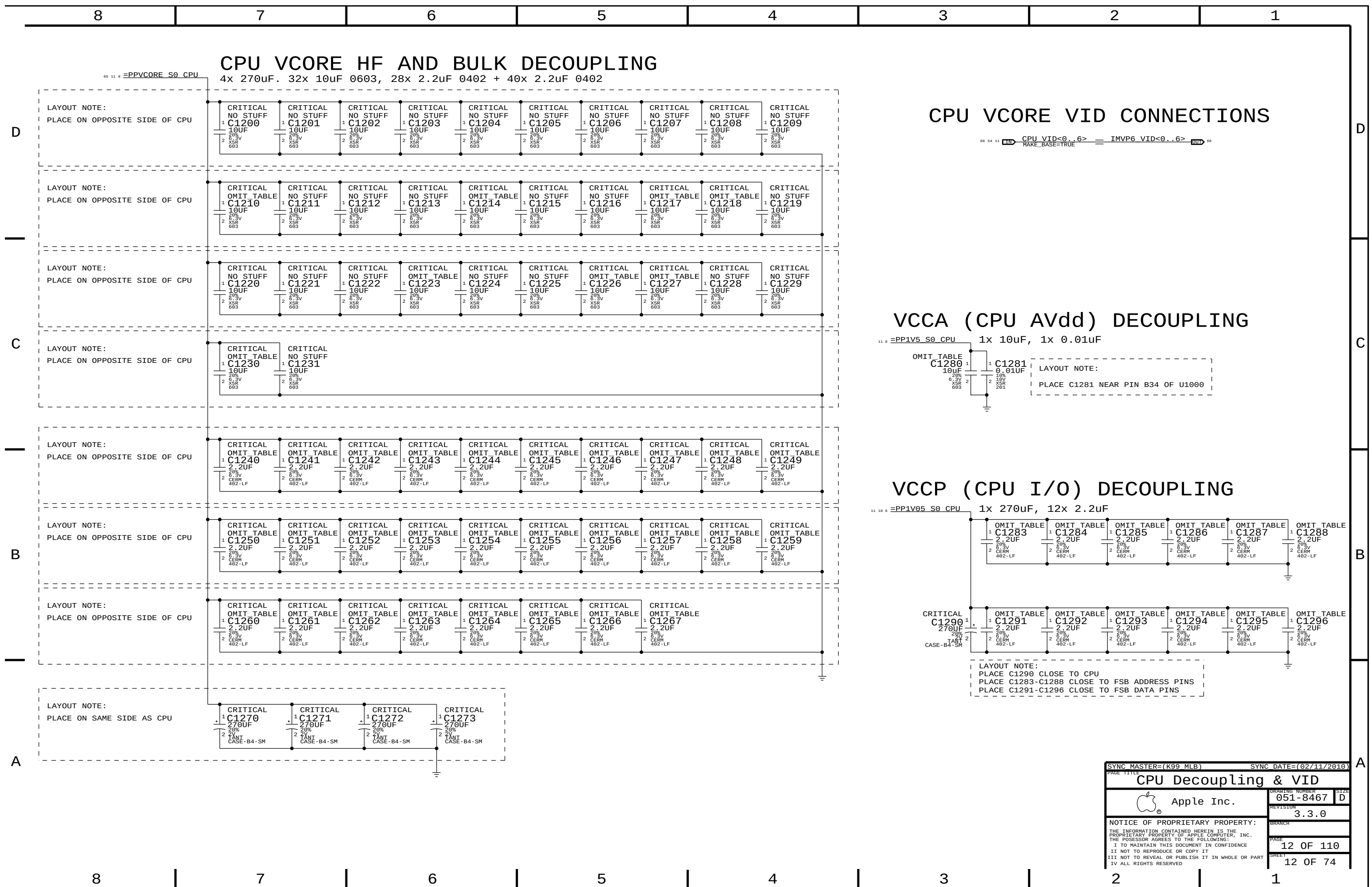


SYNC MASTER=(K99 MLB)		SYNC DATE=(02/16/2010)	
PAGE TITLE			
Functional Test / No Test		SIZE	
		DRAWING NUMBER	D
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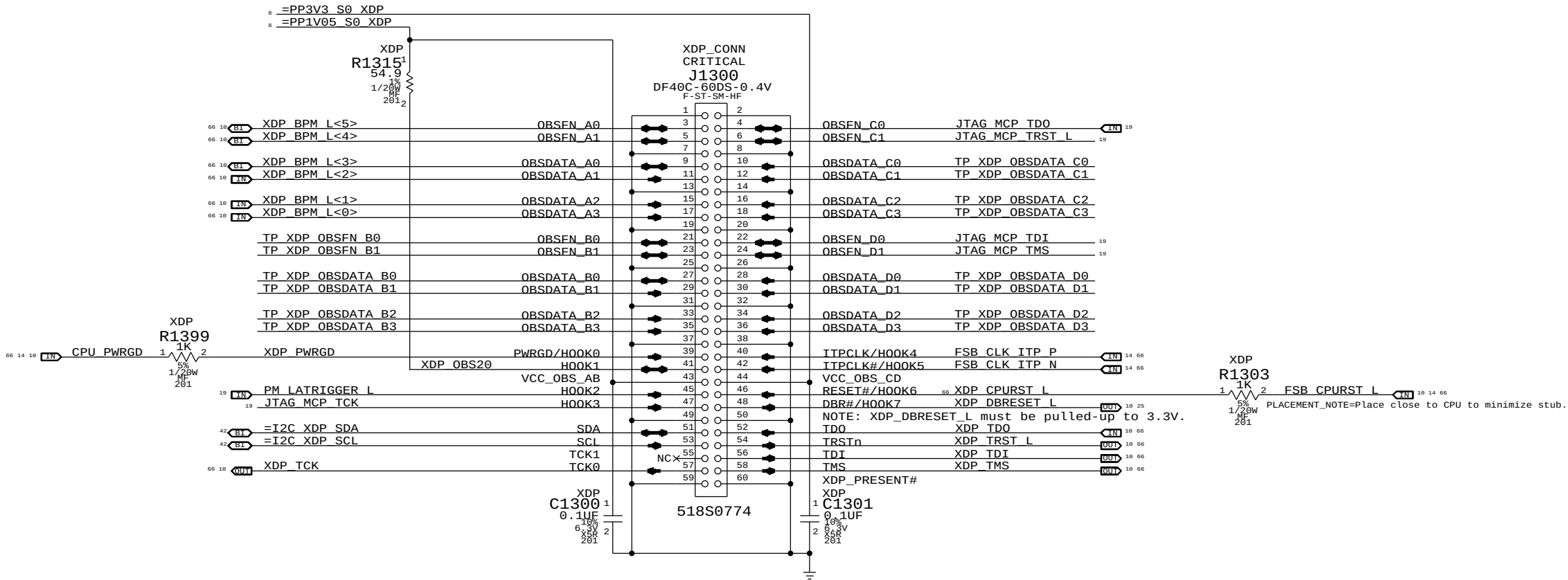




Micro2-XDP Connector

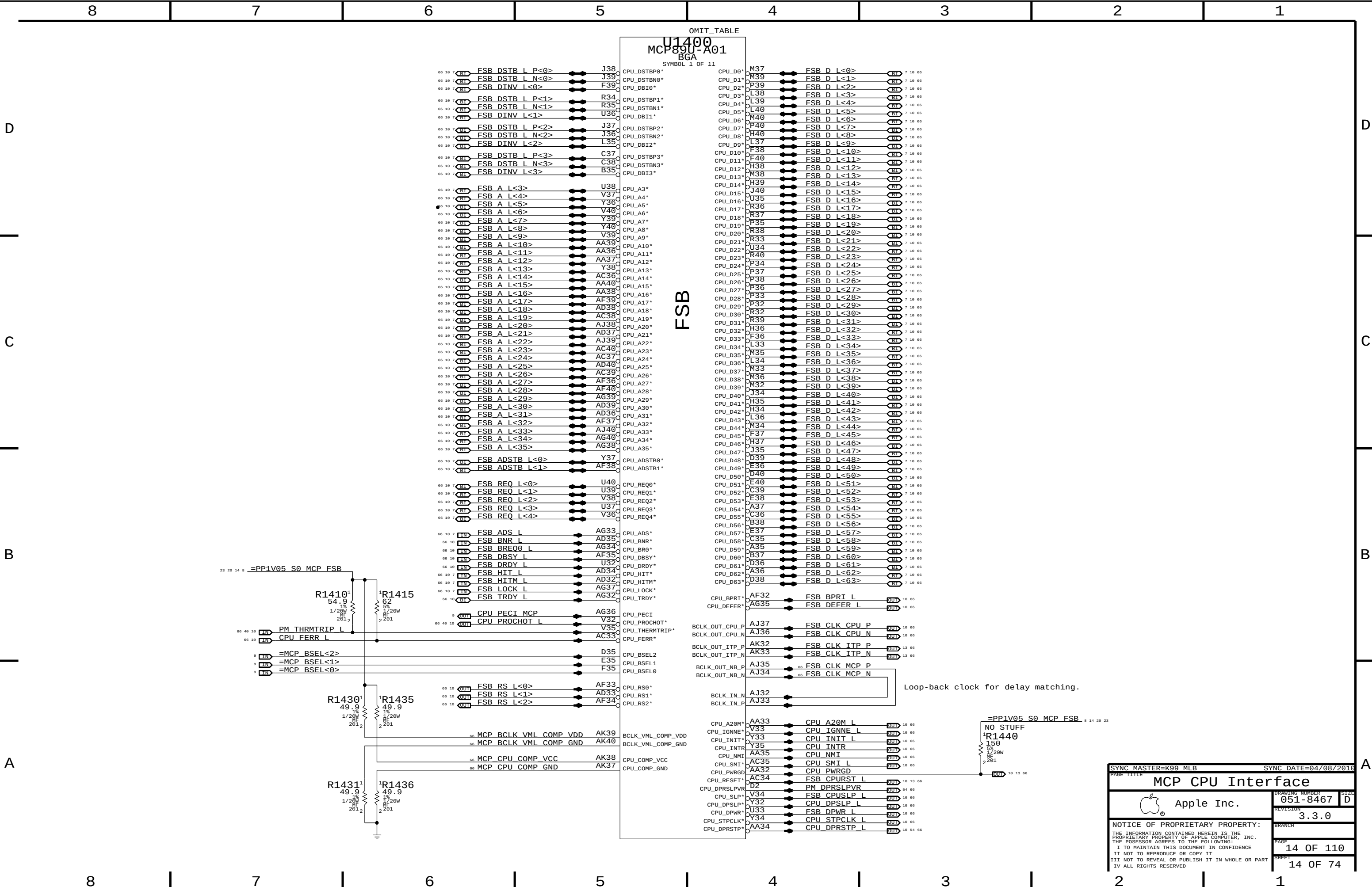
NOTE: This is not the standard XDP pinout.

Use with 920-0782 Adapter Flex to support chipset debug.

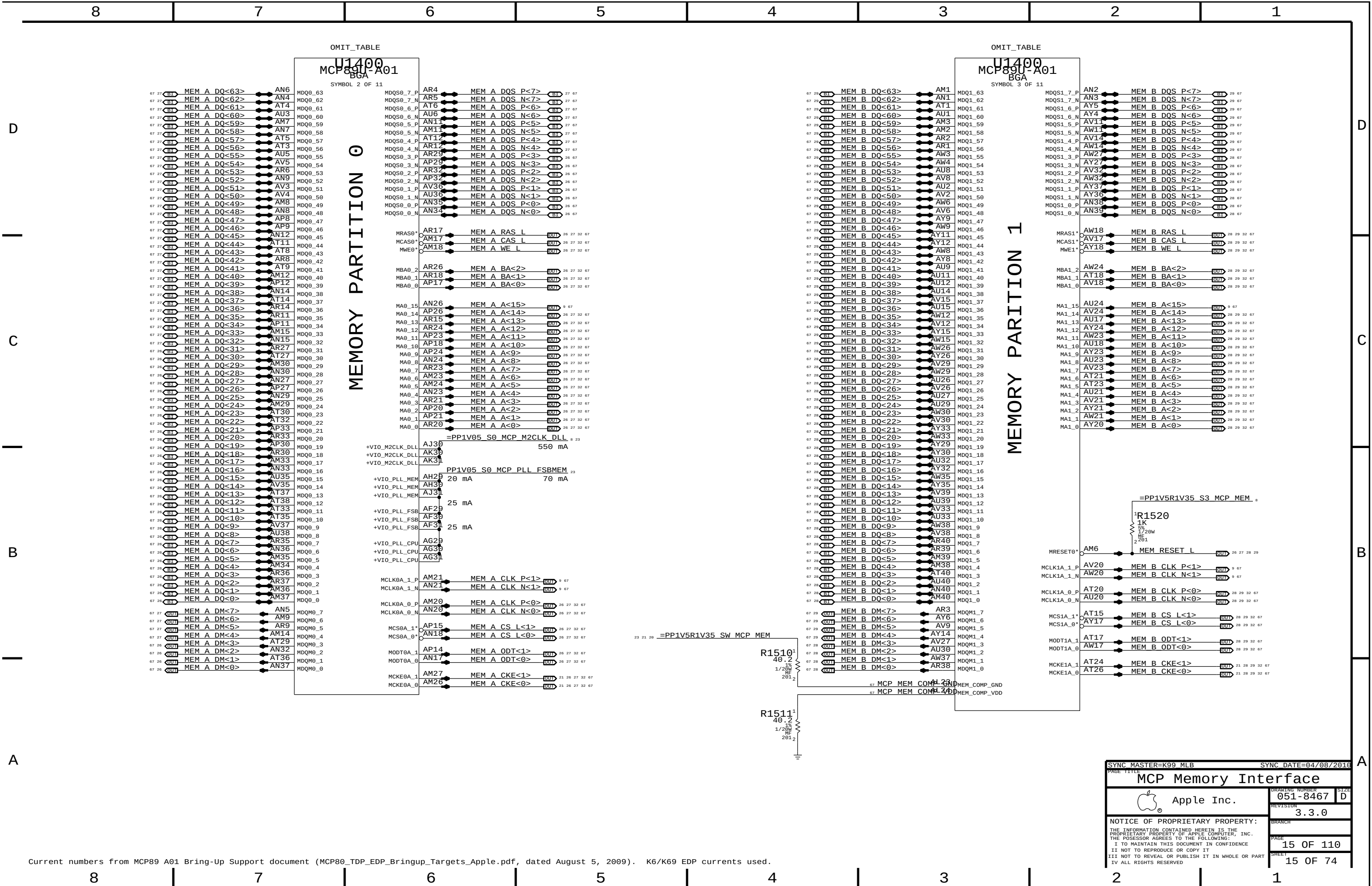


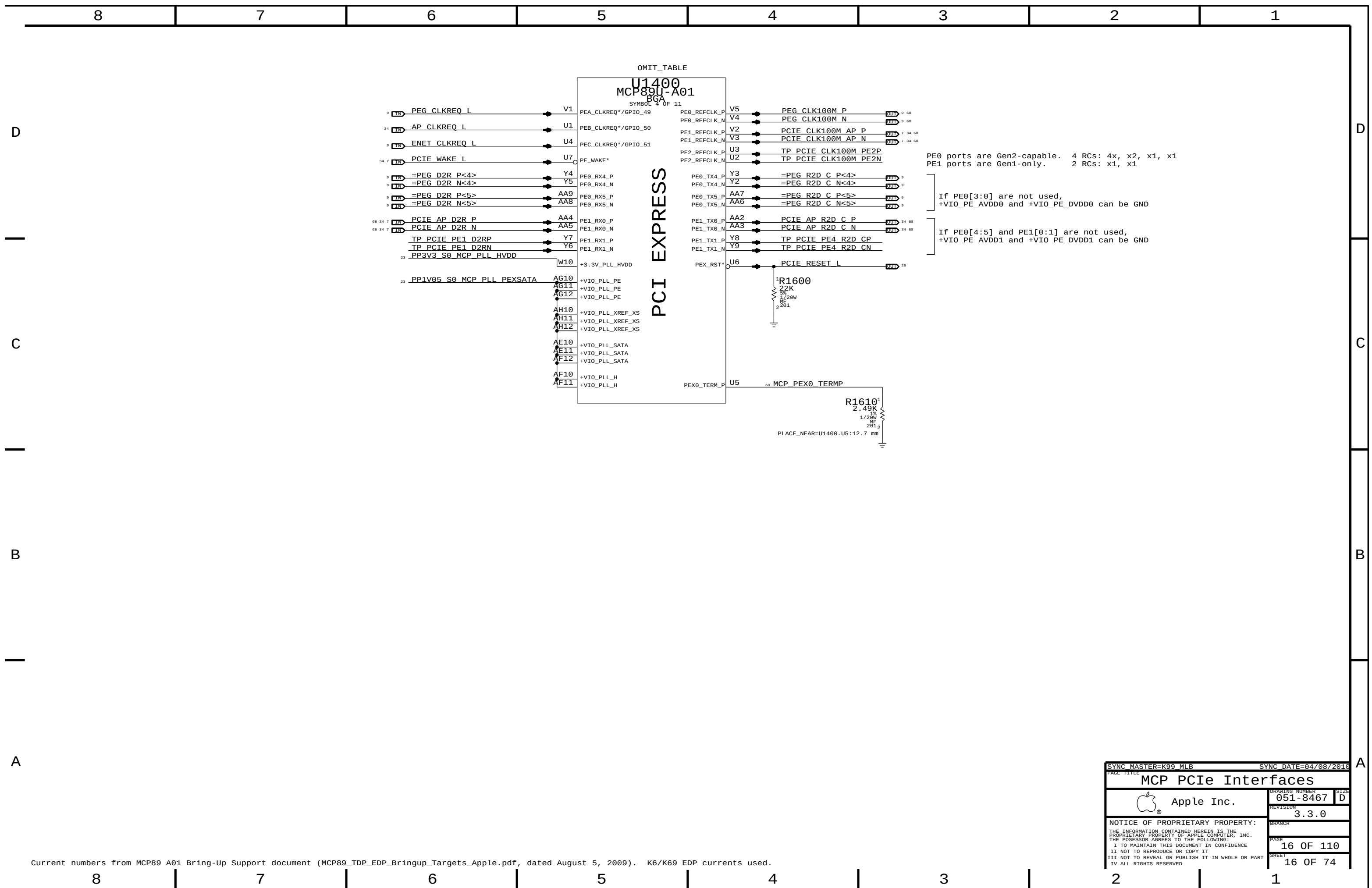
Direction of XDP adapter flex

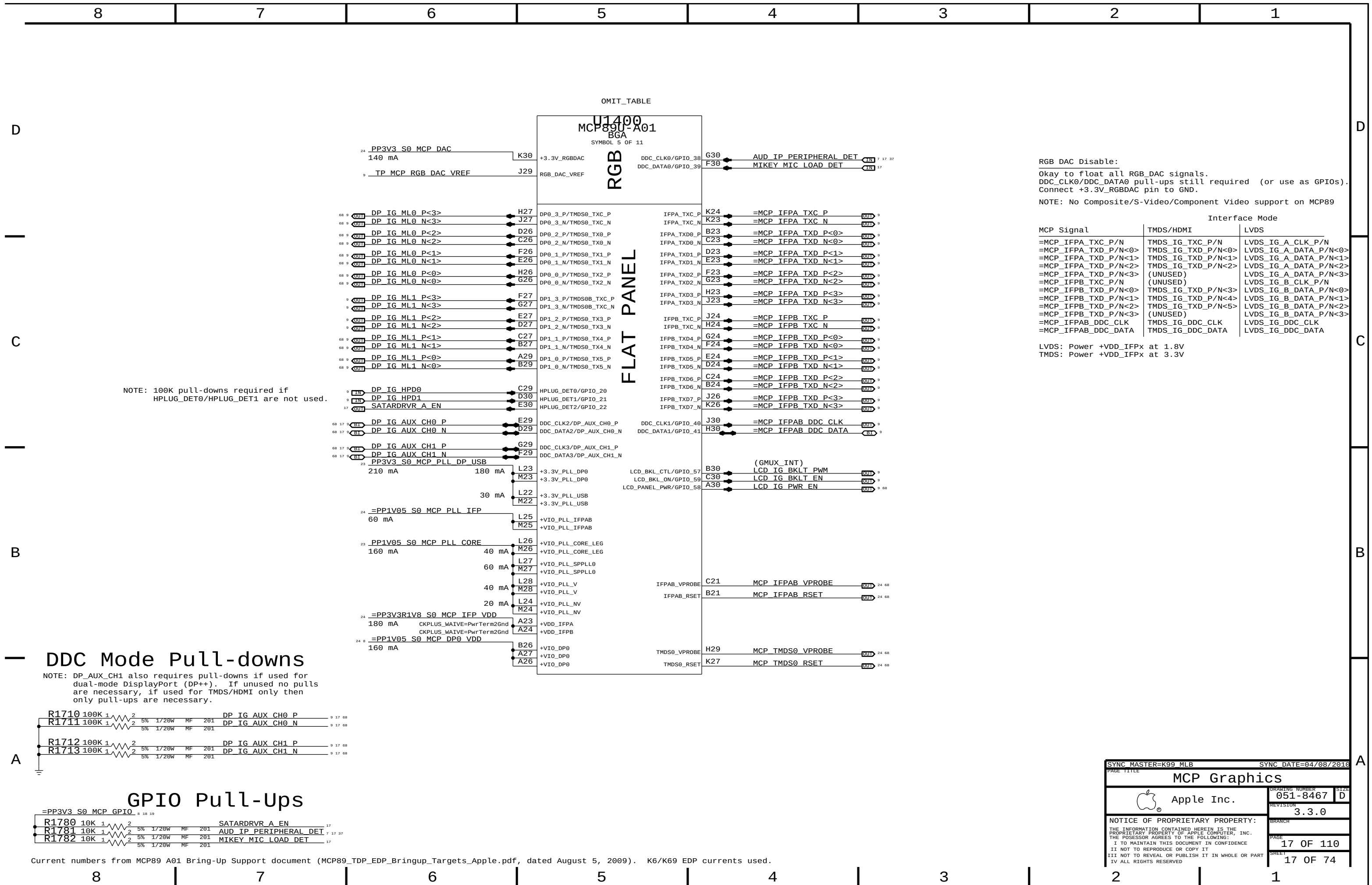
Please place J1300 within 1" of board edge with odd-numbered pins facing edge. Avoid any tall components between J1300 and edge.



SYNC MASTER=K99 MLB		SYNC DATE=04/08/2016	
PAGE TITLE		MCP CPU Interface	
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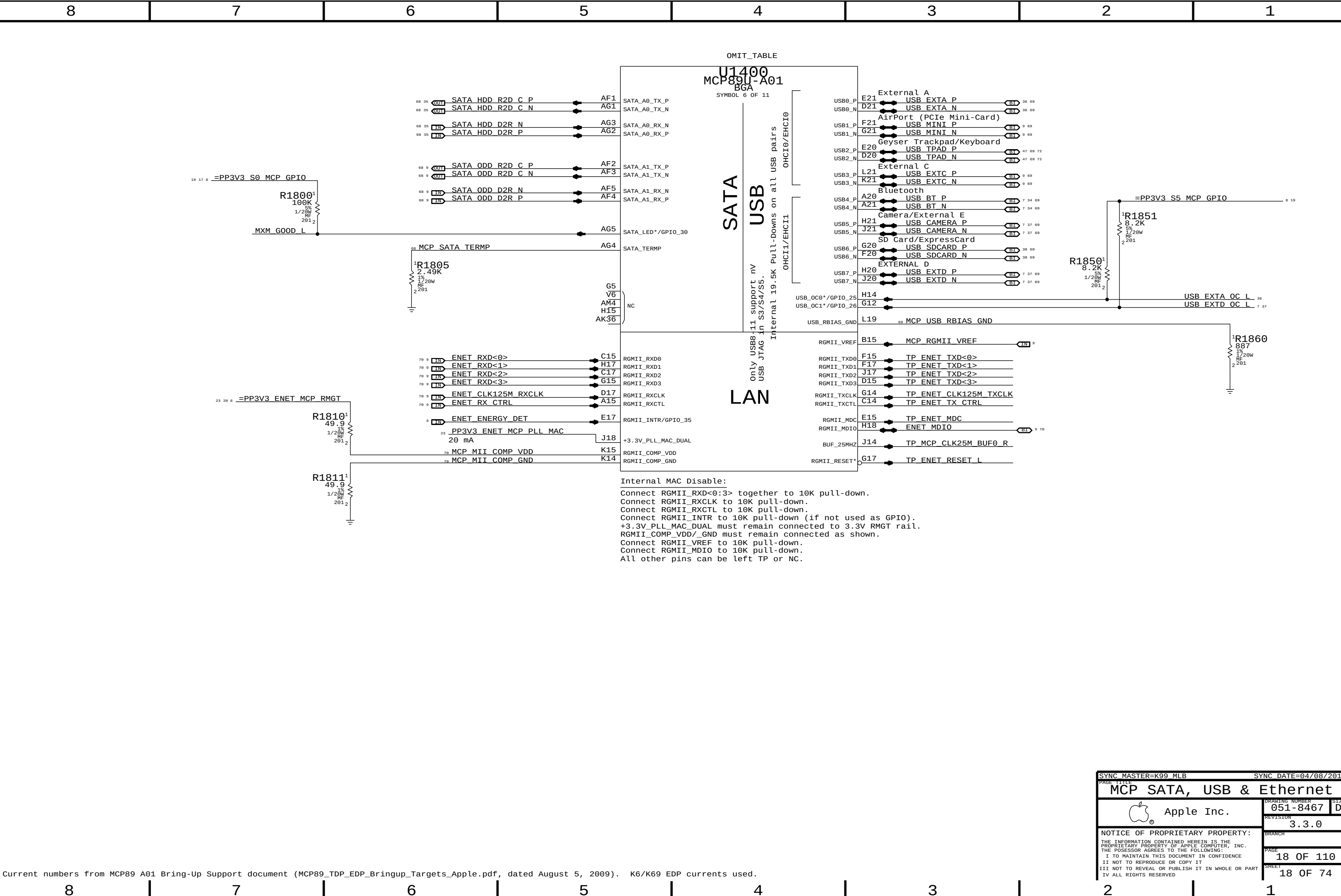
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Current numbers from MCP89 A01 Bring-Up Support document (MCP89_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

SYNC MASTER=K99 MLB		SYNC DATE=04/08/2016	
PAGE TITLE			
MCP SATA, USB & Ethernet		DRAWING NUMBER	051-8467
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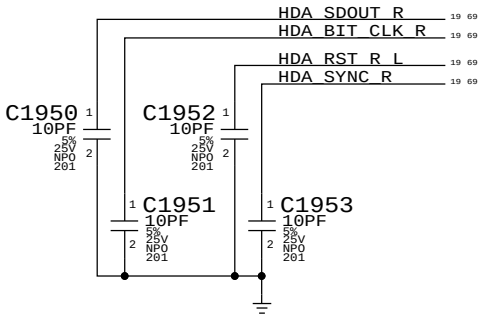
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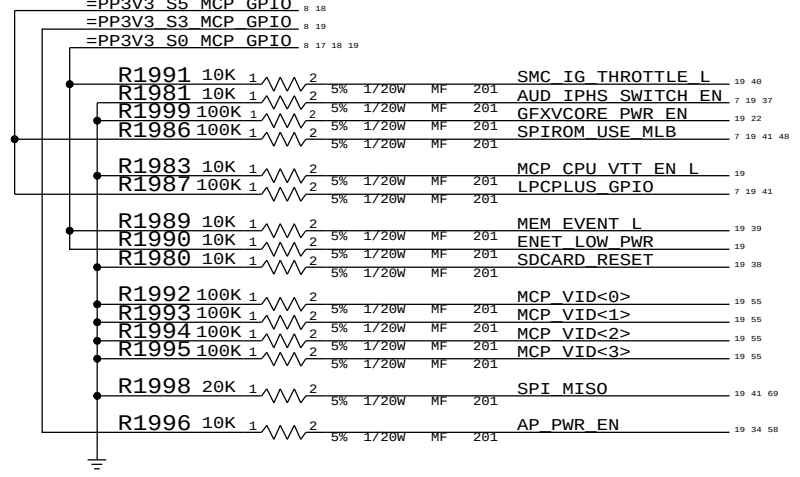
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HDA Output Caps

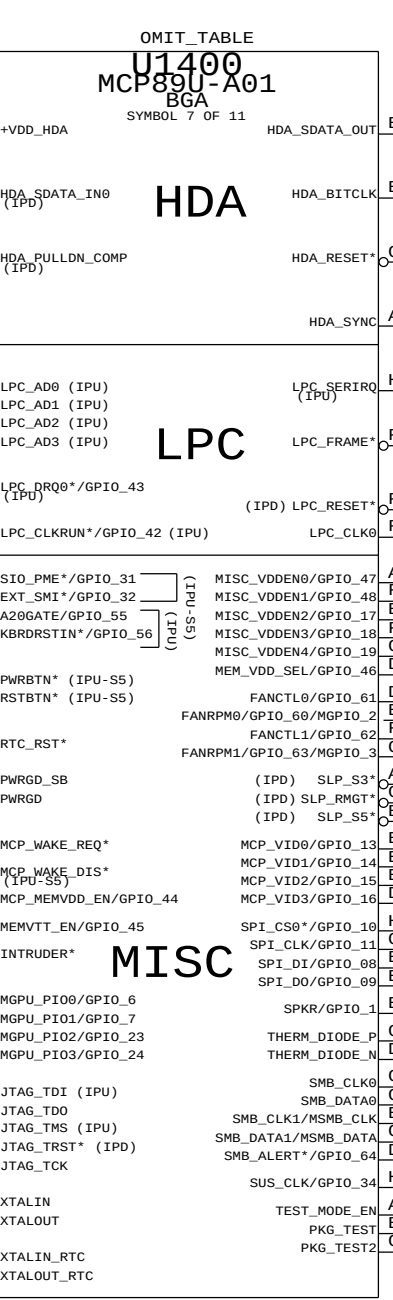
For EMI Reduction on HDA interface



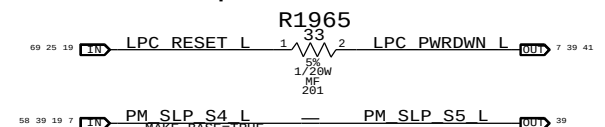
GPIO Pull-Ups/Downs



Current numbers from MCP89 A01 Bring-Up Support document (MCP89_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.



Platform-Specific Connections



NOTE: MCP SLP_S5# signal has the behavior of Intel's SLP_S4# signal

BUF_SIO_CLK Frequency	
Frequency	HDA_SYNC
24 MHz	1
14.31818 MHz	0

BIOS Boot Select	
I/F	LPC_FRAME#
LPC	0
SPI	1

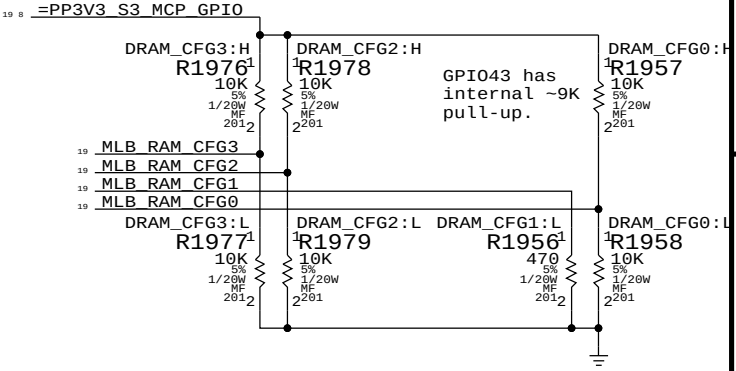
NOTE: MCP89 does not support FWH, only LPC ROMs. So Apple designs will not use LPC for BootROM override.

SPI Frequency Select		
Frequency	SPI_DO	SPI_CLK
25.0 MHz	0	0
31.2 MHz	0	1
42.7 MHz	1	0
62.5 MHz	1	1

NOTE: 42 & 62 MHz use FAST_READ command. Straps not provided on this page.

NOTE: MCP SLP_S5# pin has the behavior of Intel's SLP_S4# signal.

MCP_SPKR:
0 = USER mode (Normal boot mode)
1 = SAFE mode (For ROMSIP recovery)
Connects to SMC for automatic recovery.



SYNC MASTER=K99 MLB

SYNC DATE=04/08/2016

MCP HDA, LPC & MISC

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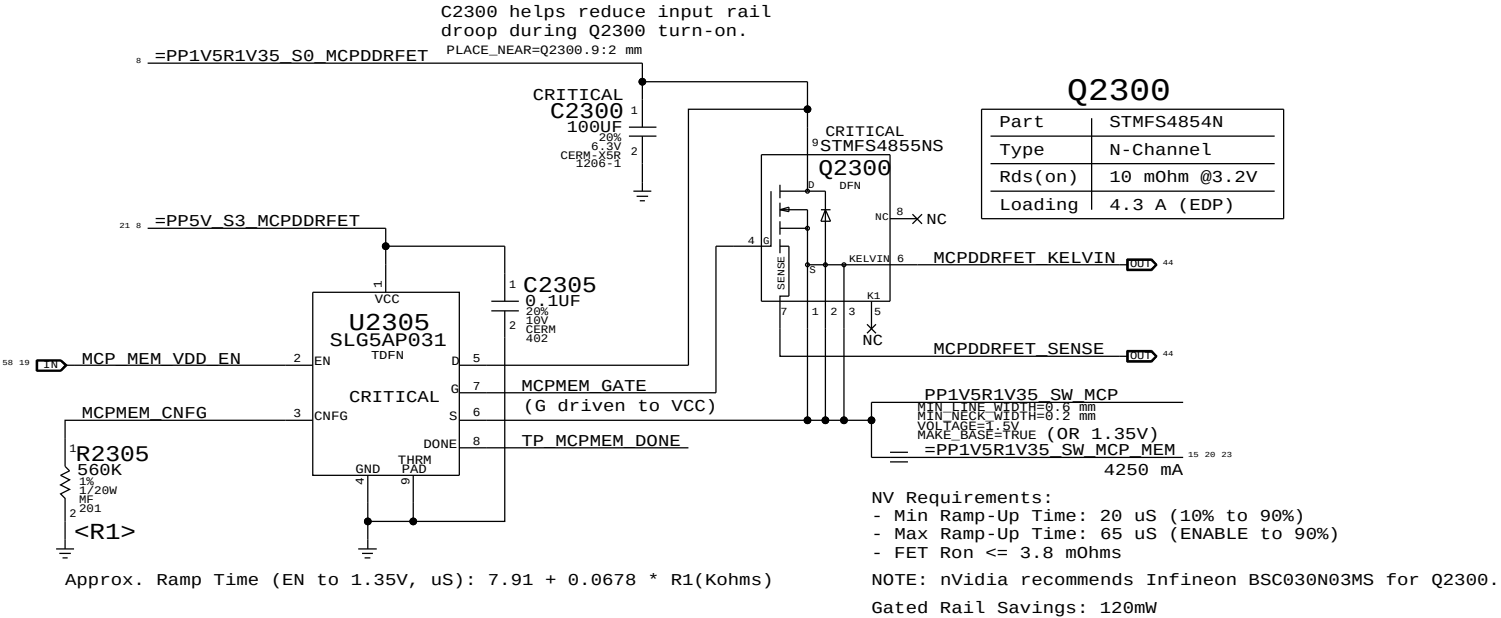
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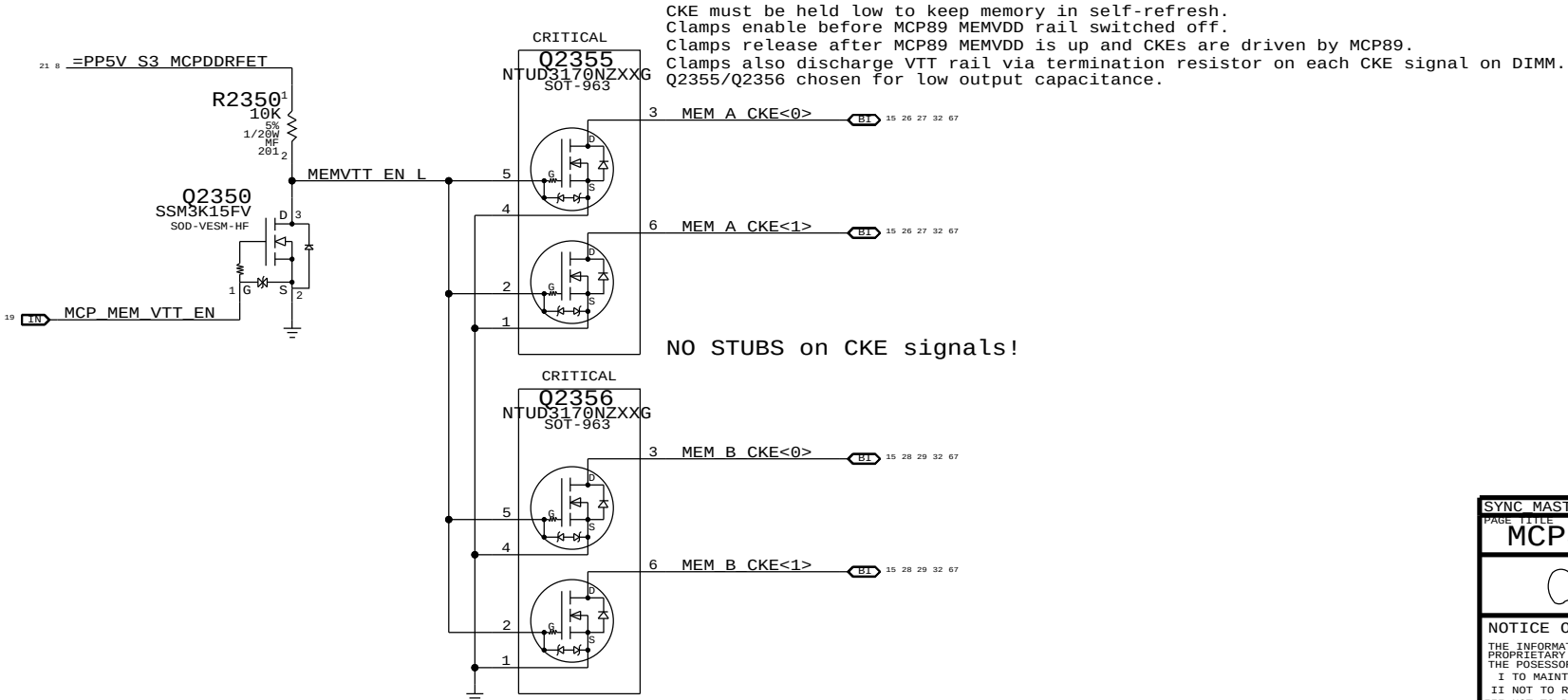
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DIMM CKE Clamps



SYNC MASTER=K99 MLB

SYNC DATE=04/08/2016

MCP89 Memory Rail Gating

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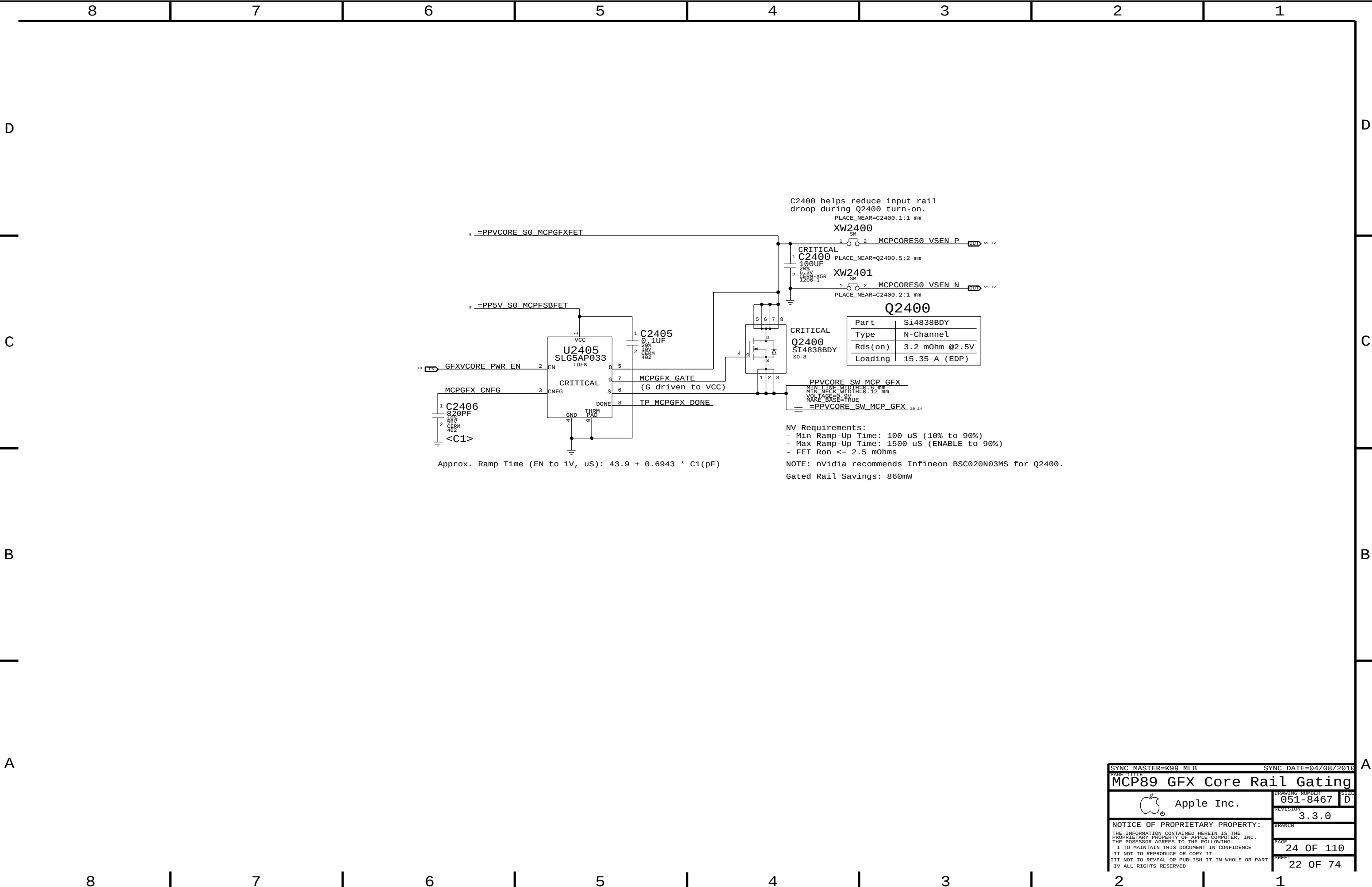
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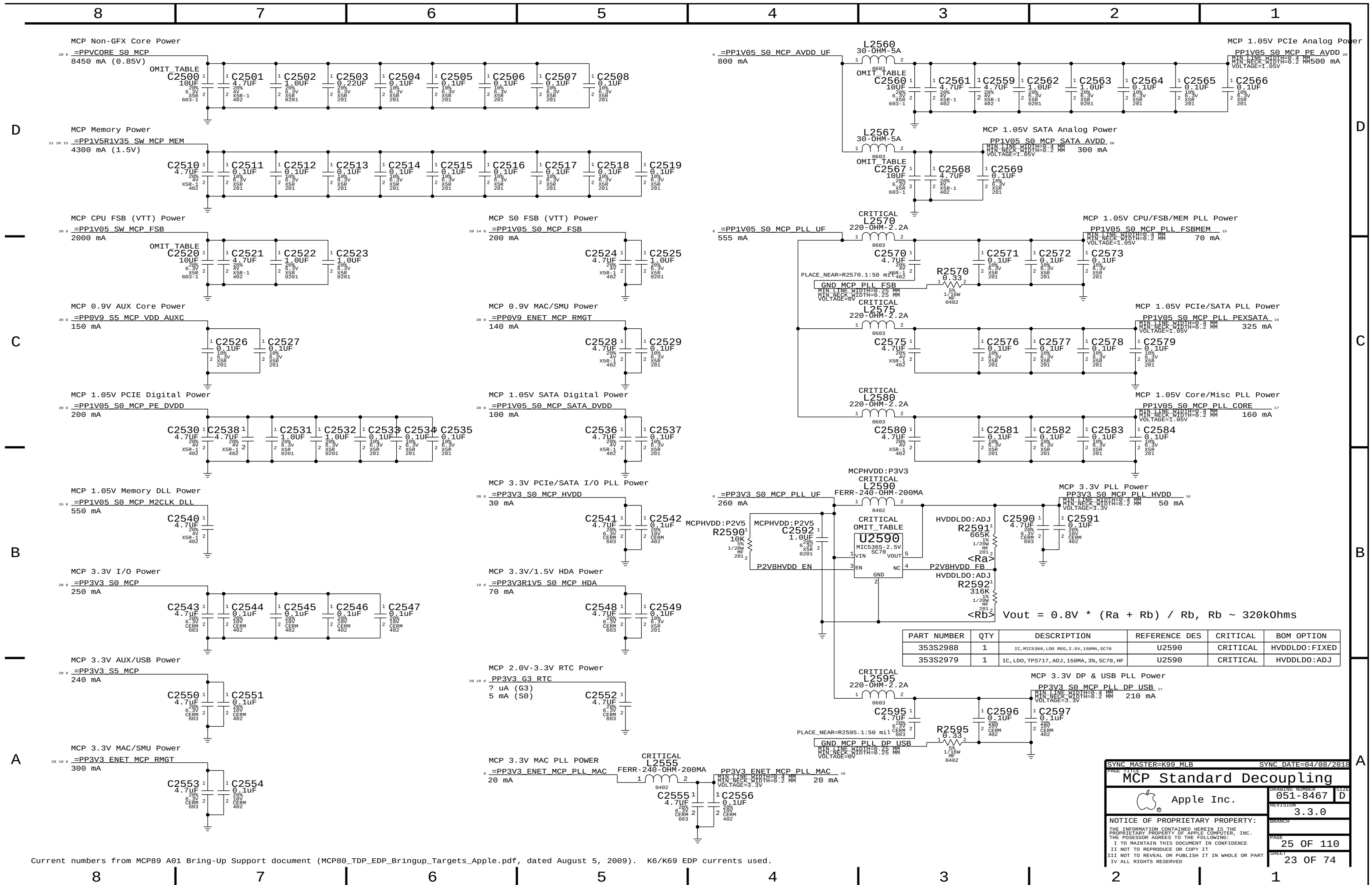
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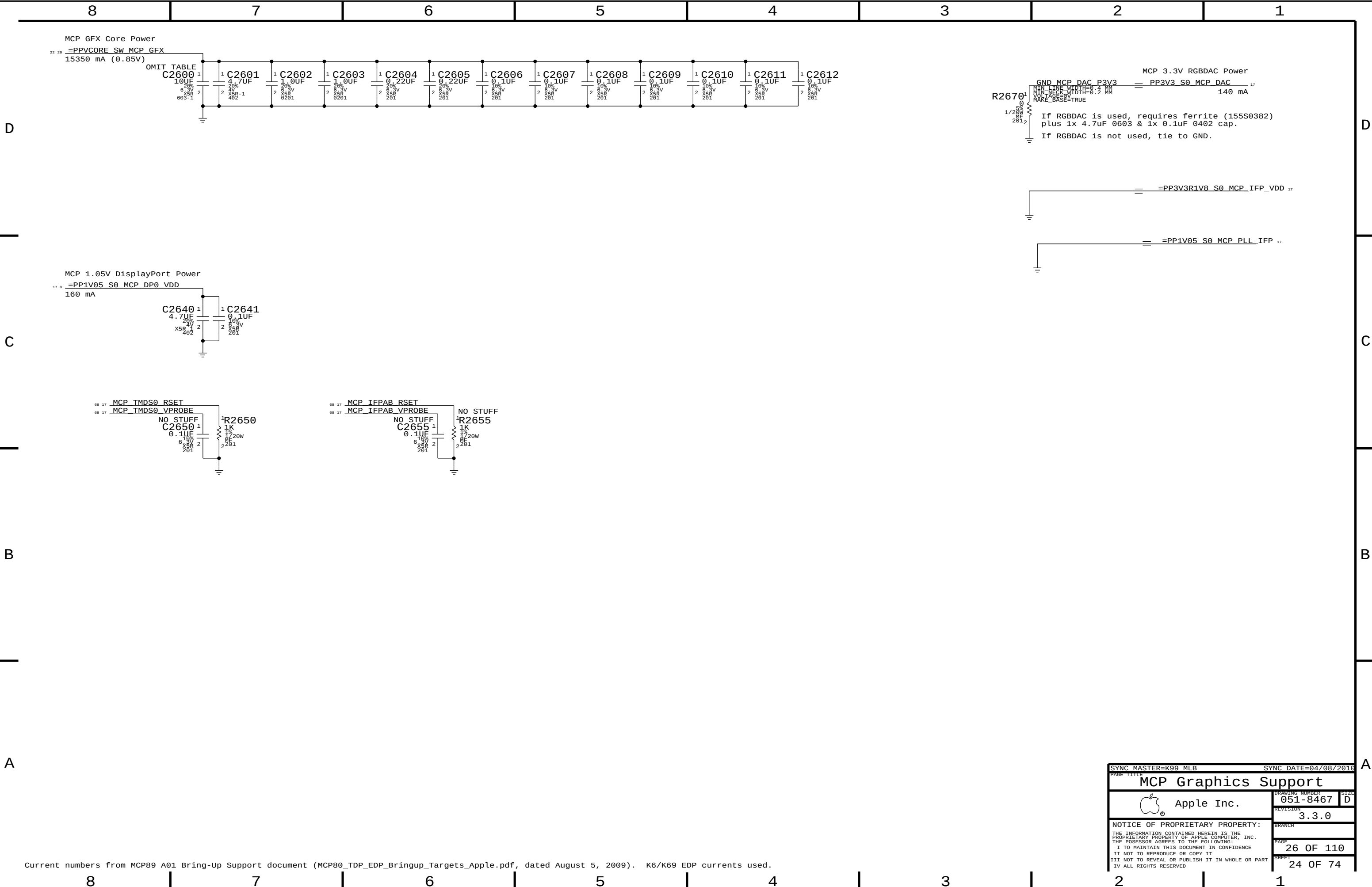
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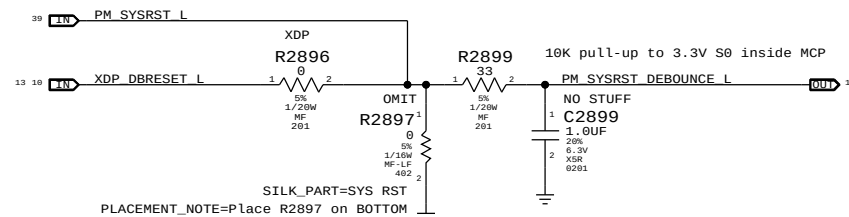
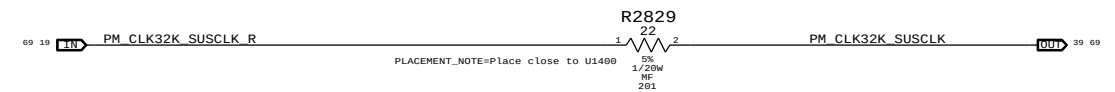
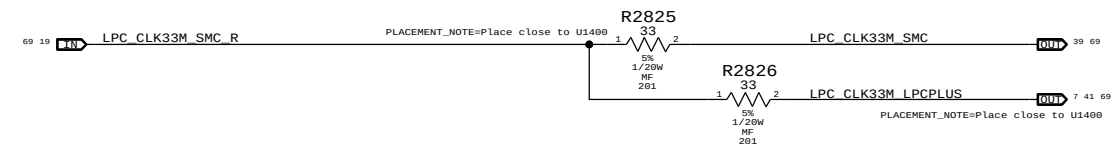
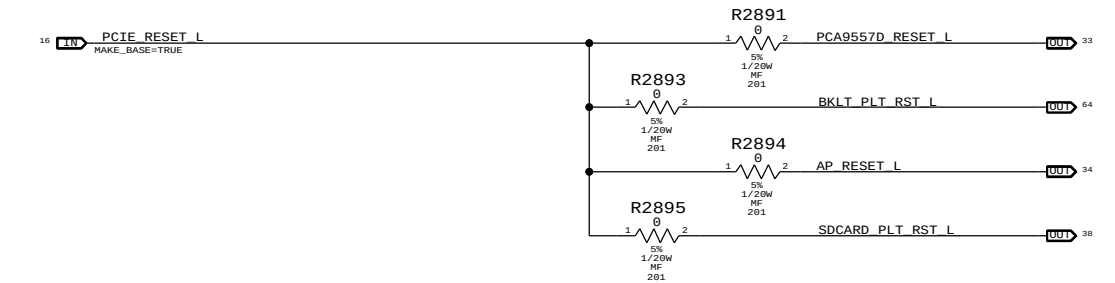
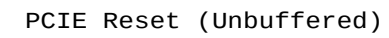
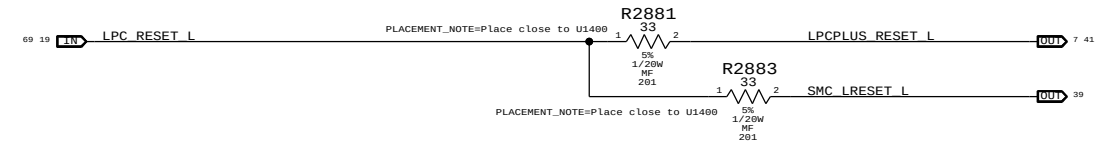
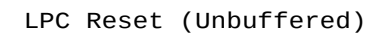
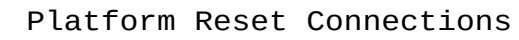
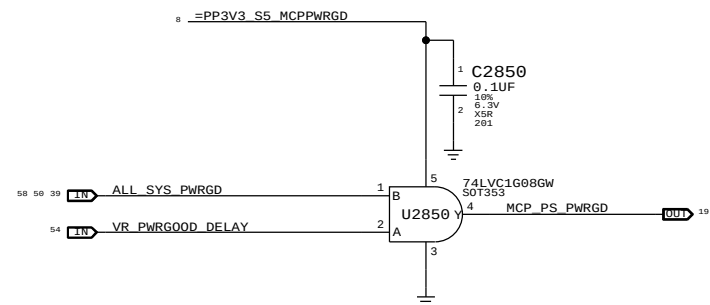
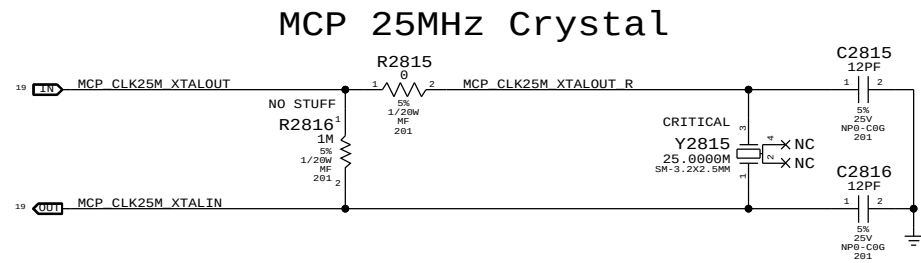
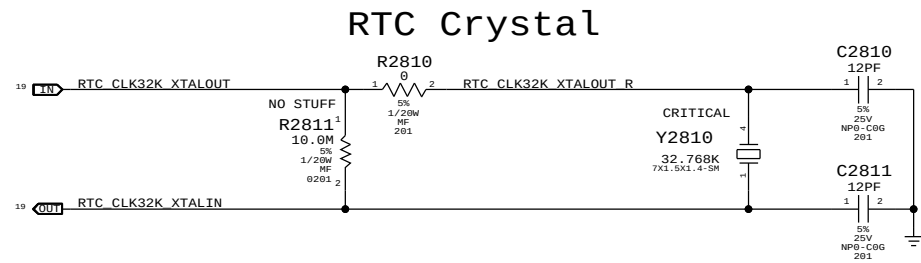






Current numbers from MCP89 A01 Bring-Up Support document (MCP80_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

SYNC MASTER=K99 MLB		SYNC DATE=04/08/2010	
PAGE TITLE			
MCP Graphics Support			
 Apple Inc.		DRAWING NUMBER	051-8467
		REVISION	3.3.0
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		PAGE	26 OF 110
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SYNC MASTER=(K99 MLB)		SYNC DATE=(02/11/2010)	
PAGE TITLE			
SB Misc			
 Apple Inc.		DRAWING NUMBER 051-8467	SIZE D
		REVISION 3.3.0	
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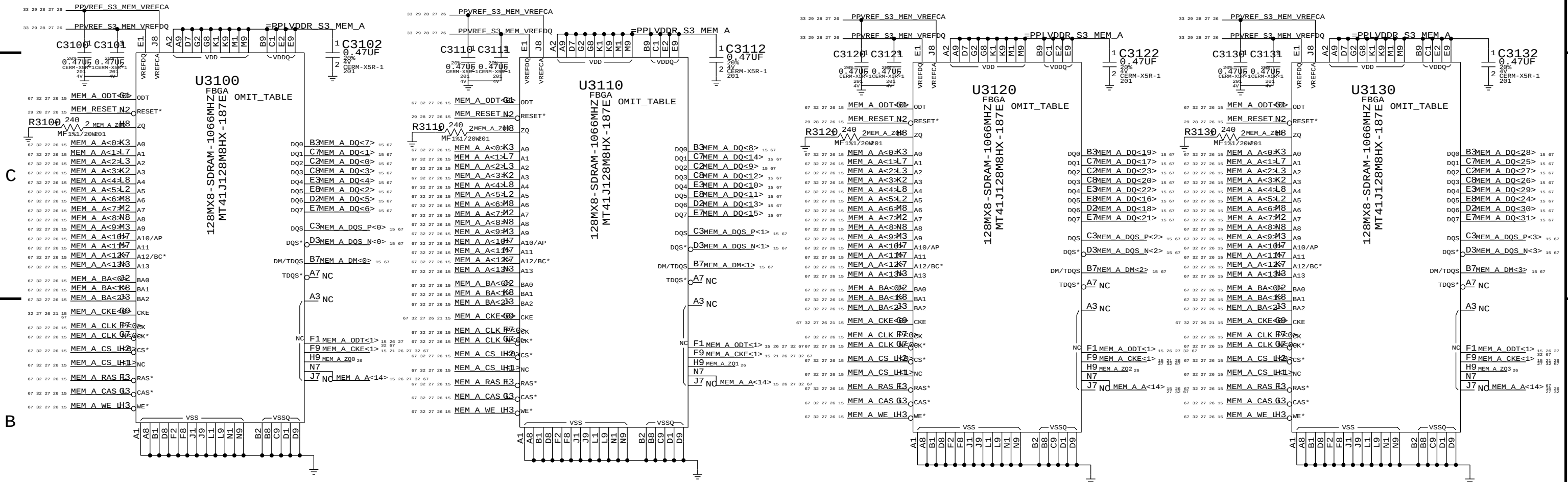
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SYNC MASTER=K99 MLB		SYNC DATE=04/08/2016	
PAGE TITLE			
DDR3 DRAM Channel A		(0-31)	
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	REVISION	3.3.0	D
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BRANCH			
PAGE	31 OF 110		
SHEET	26 OF 74		

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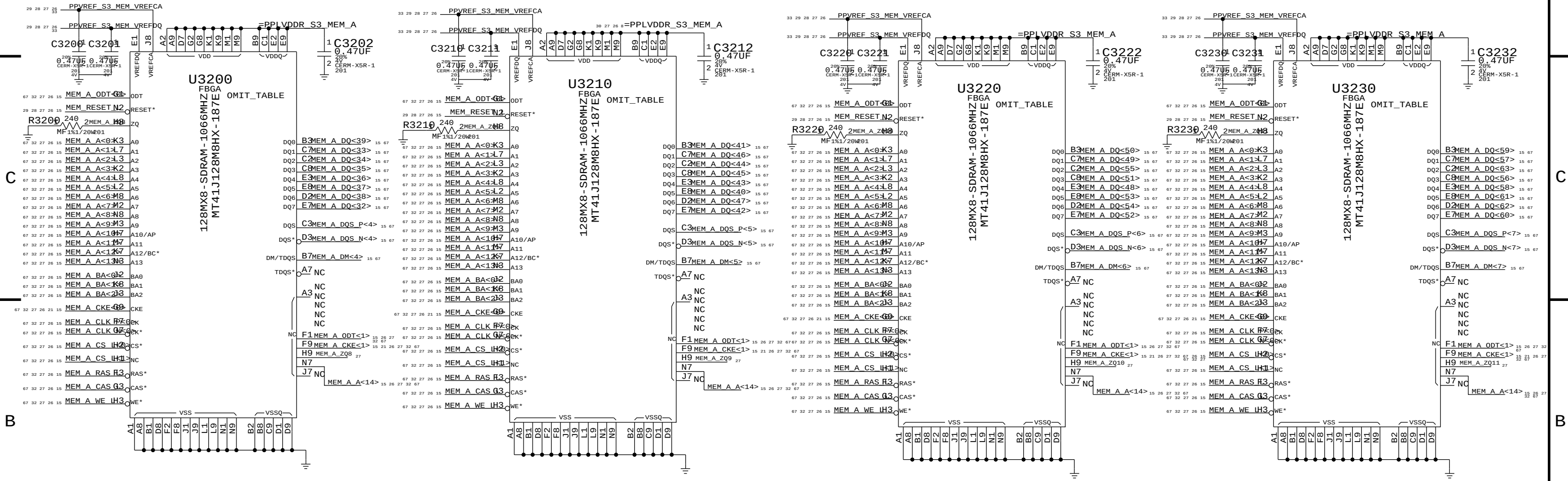
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A14/A15 FOR 2G/4G MONO ONLY
CS1 IS FOR 2G DDP RANK CONTROL

SYNC MASTER=K99 MLB		SYNC DATE=04/08/2016	
PAGE TITLE			
DDR3 DRAM Channel A		(32-63)	
 Apple Inc.	DRAWING NUMBER	051-8467	SIZE
	REVISION	D	
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		BRANCH	
		PAGE	
		32 OF 110	
		SHEET	
		27 OF 74	

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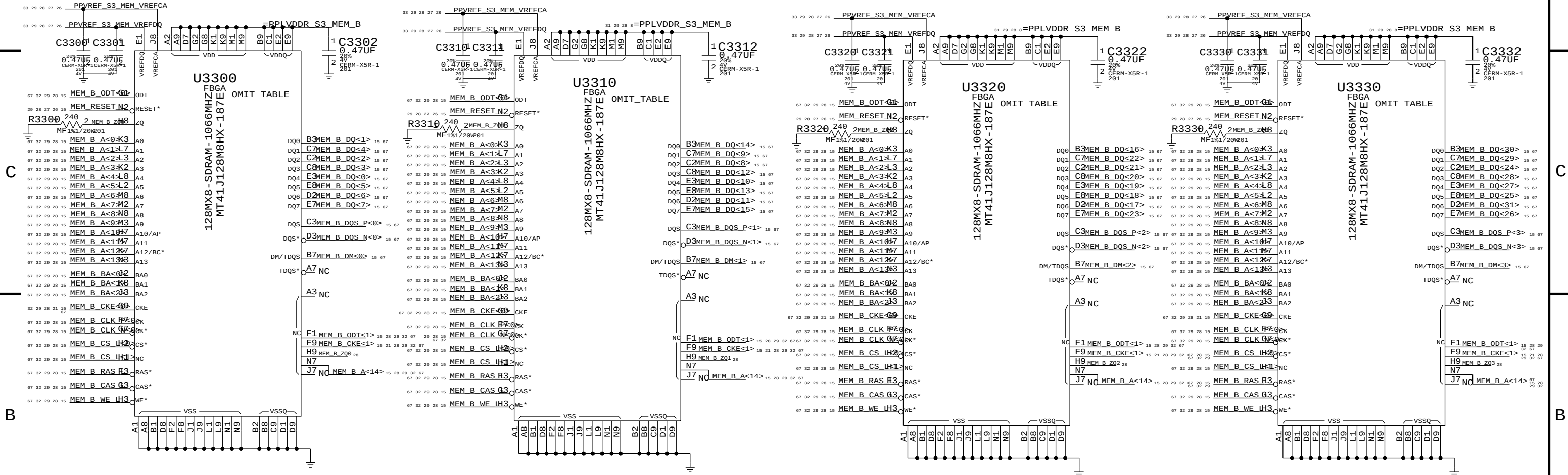
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A14/A15 FOR 2G/4G MONO ONLY
CS1 IS FOR 2G DDP RANK CONTROL

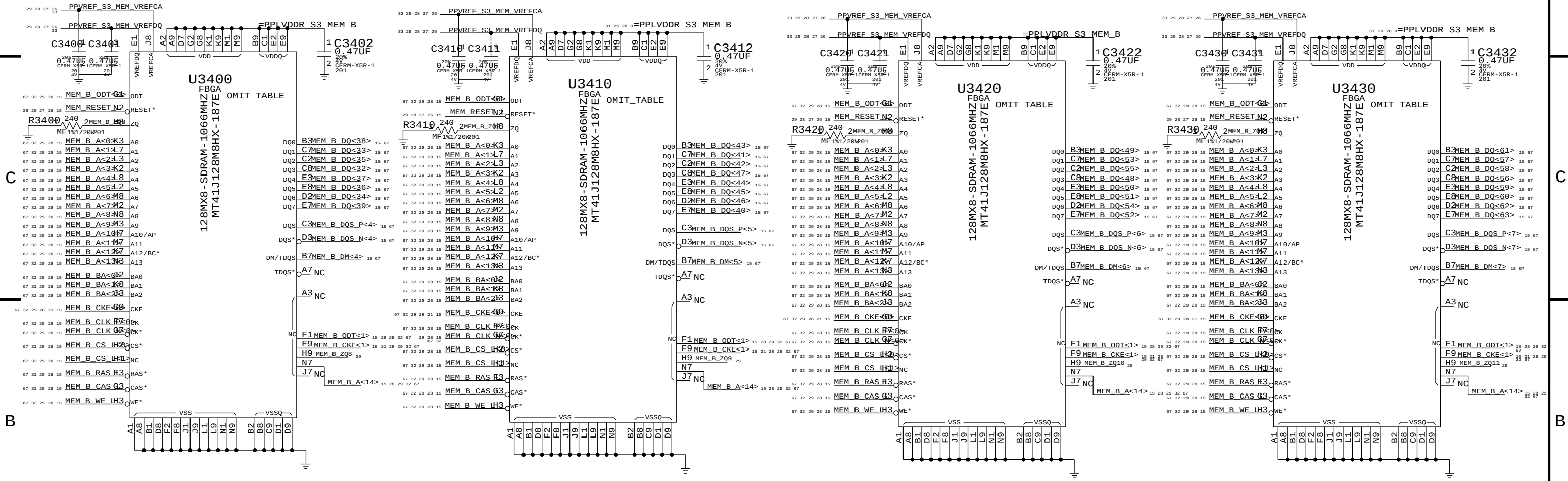
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PAGE TITLE			
DDR3 DRAM Channel B		(0-31)	
	Apple Inc.		DRAWING NUMBER
			051-8467
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		3.3.0	D
		BRANCH	
		PAGE	33 OF 110
		SHEET	28 OF 74

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A14/A15 FOR 2G/4G MONO ONLY
CS1 IS FOR 2G DDP RANK CONTROL

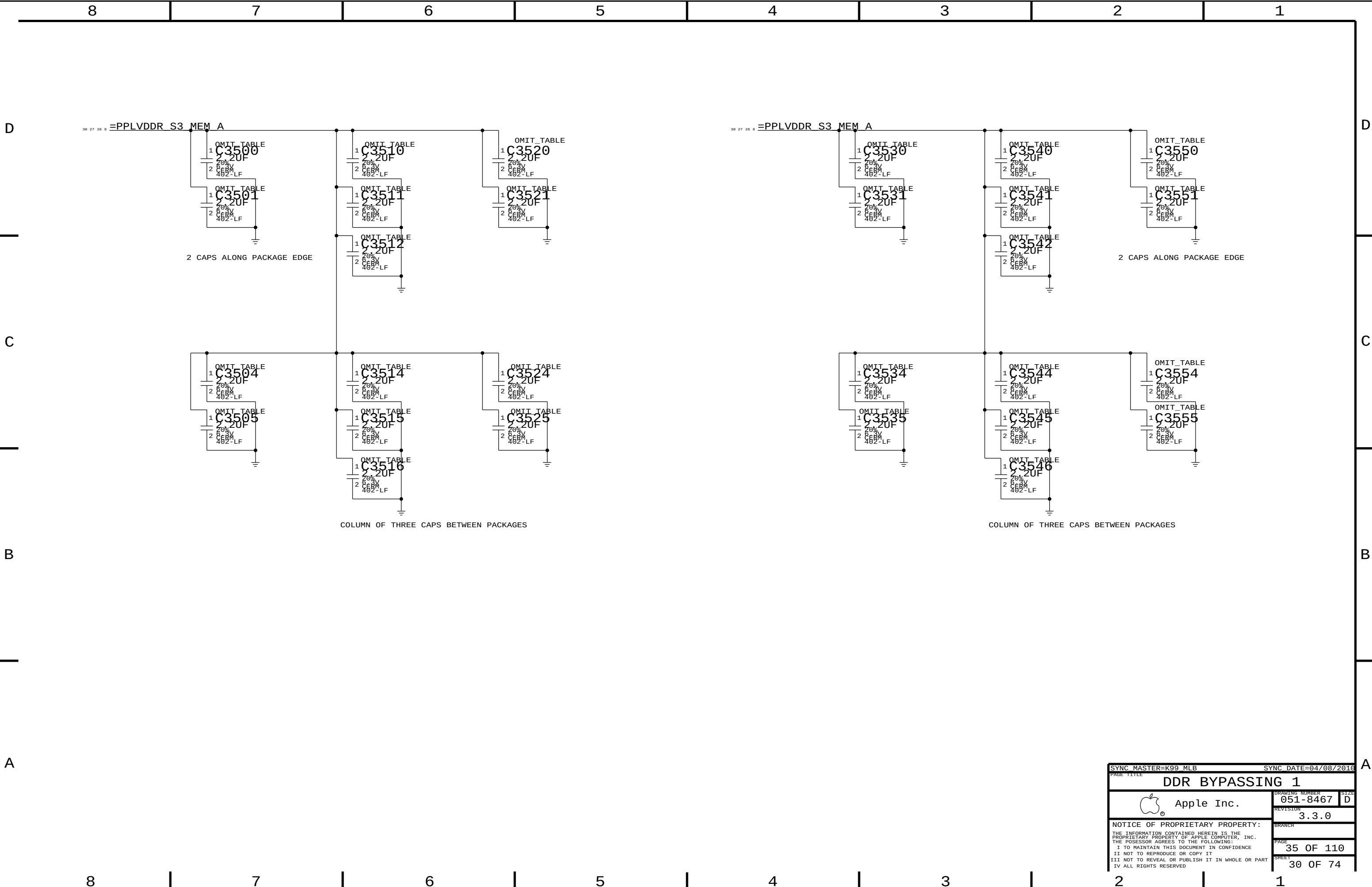
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PAGE TITLE			
DDR3 DRAM Channel B		B (32-63)	
 Apple Inc.	DRAWING NUMBER	051-8467	SIZE
	REVISION	3.3.0	D
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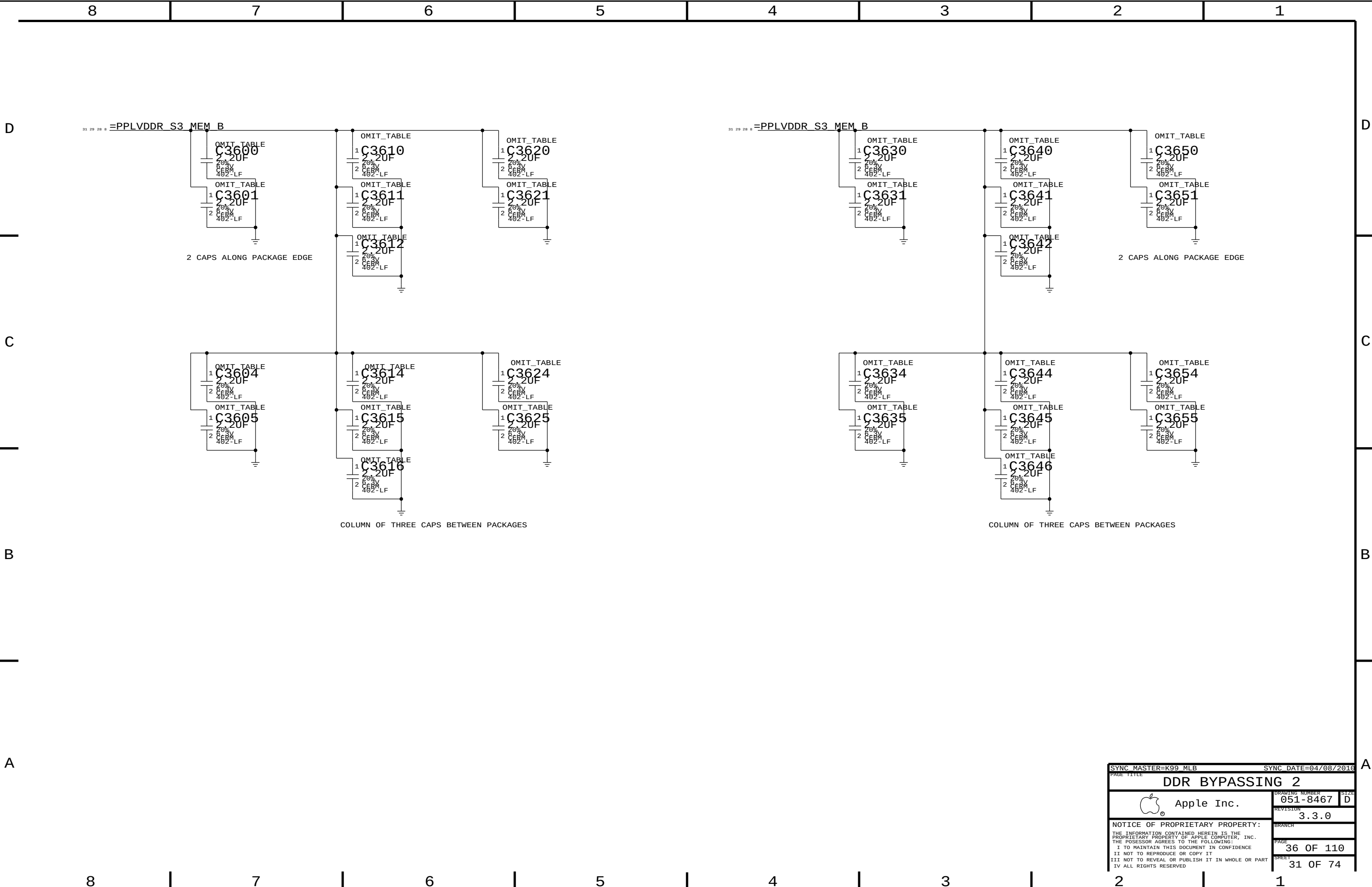
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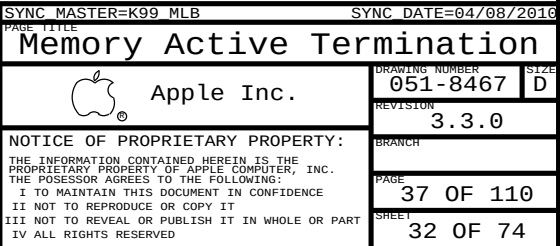
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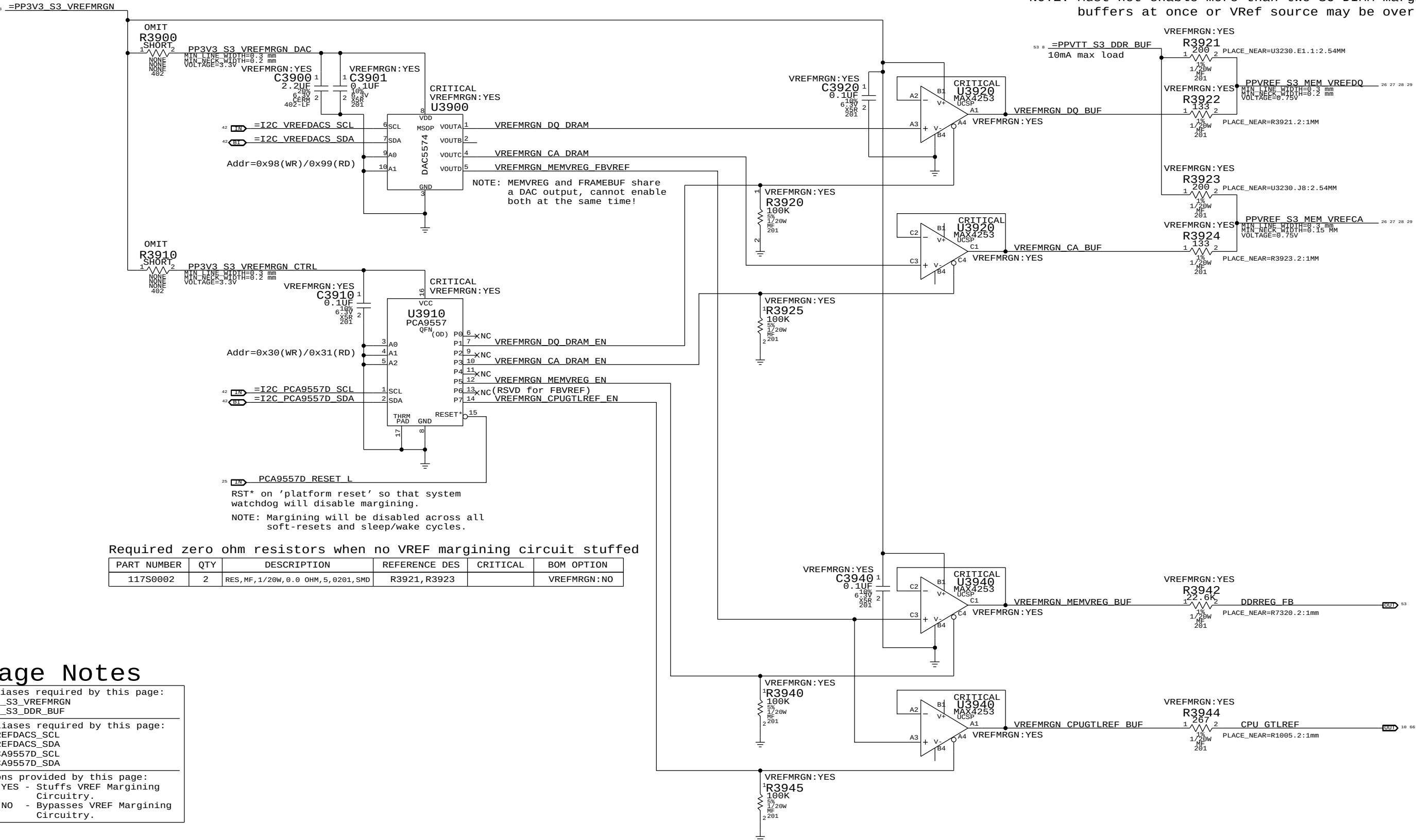
D

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NOTE: Must not enable more than two S0-DIMM margining buffers at once or VRef source may be overloaded.



Page Notes

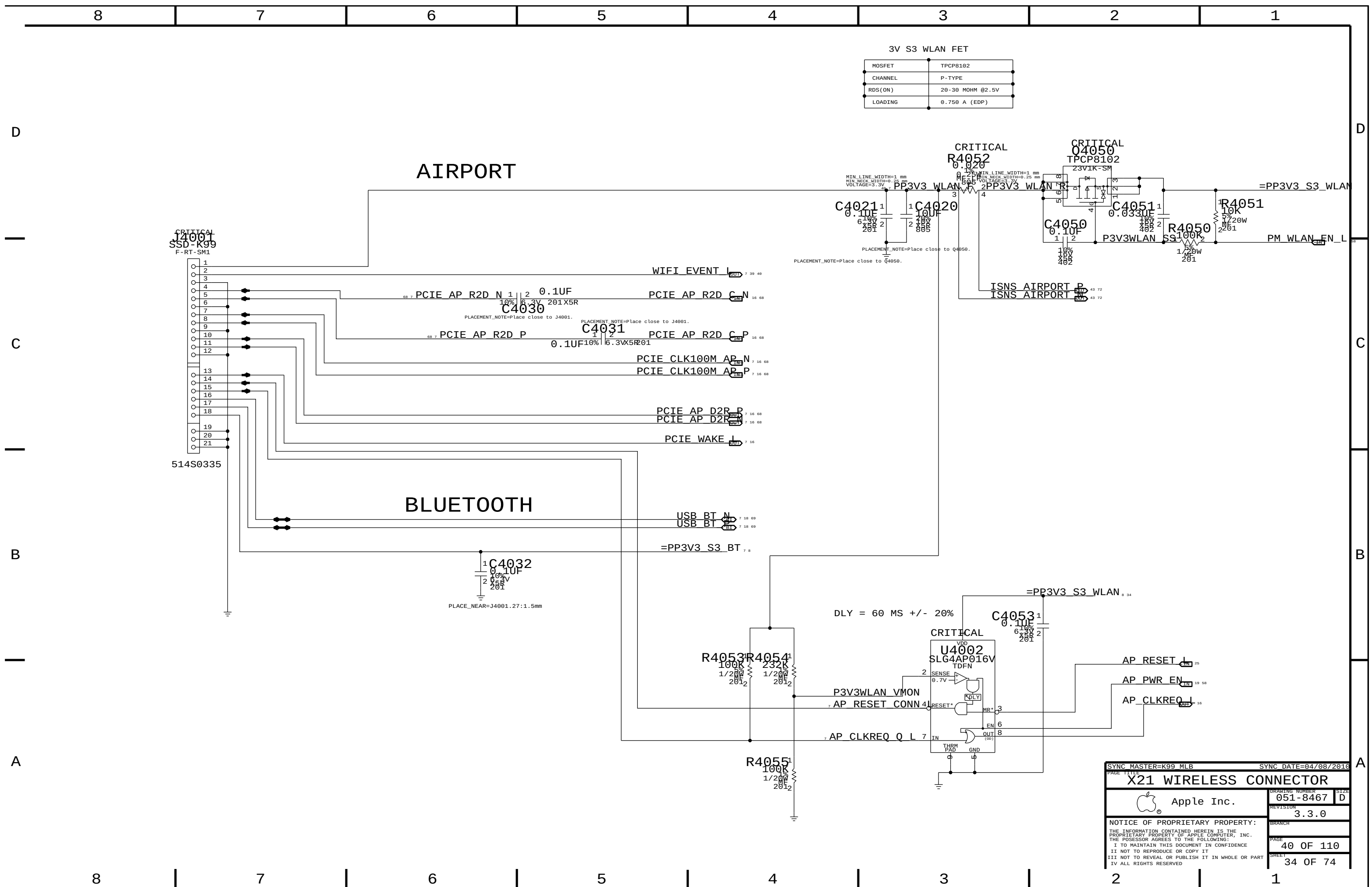
Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPVTT_S3_DDR_BUF

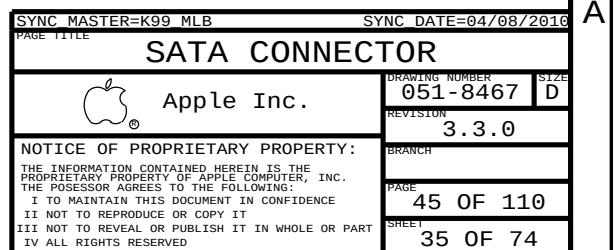
Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
VREFMRGN:YES - Stuffs VREF Margining Circuitry.
VREFMRGN:NO - Bypasses VREF Margining Circuitry.

	MEM VREF DQ	MEM VREF CA	MEM VREG	CPU GTLREF (FSB)
DAC Channel:	A	C	D	D
PCA9557D Pin:	1	3	5	7
Nominal value		0.75V (DAC: 0x3A)	1.5V (DAC: 0x3A)	0.7V (DAC: 0x8B)
Margined target:		0.300V - 1.200V (+/- 450mV)	1.998V - 1.002V (+/- 498mV)	0.200V - 1.050V (+/- 500mV)
DAC range:		0.000V - 1.501V (0x00 - 0x74)	0.000V - 1.501V (0x00 - 0x74)	0.000V - 1.191V (0x00 - 0x5C)
VRef current:		+3.4mA - -3.4mA (- = sourced)	+33uA - -33uA (- = sourced)	+750uA - -528uA (- = sourced)
DAC step size:		7.69mV / step @ output	8.59mV / step @ output	9.24mV / step @ output

SYNC MASTER=K99 MLB		SYNC DATE=04/08/2016	
PAGE TITLE			
FSB/DDR3 Vref		Margining	
 Apple Inc.		DRAWING NUMBER	051-8467
		REVISION	3.3.0
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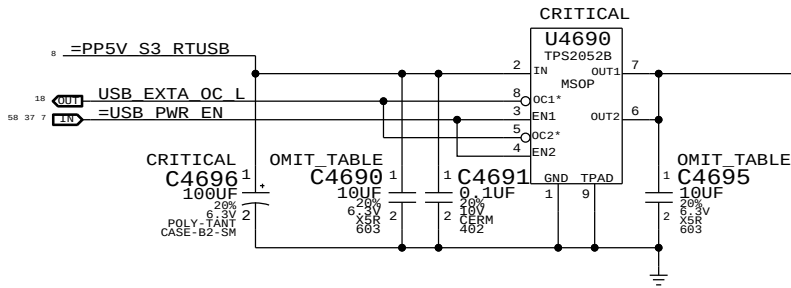
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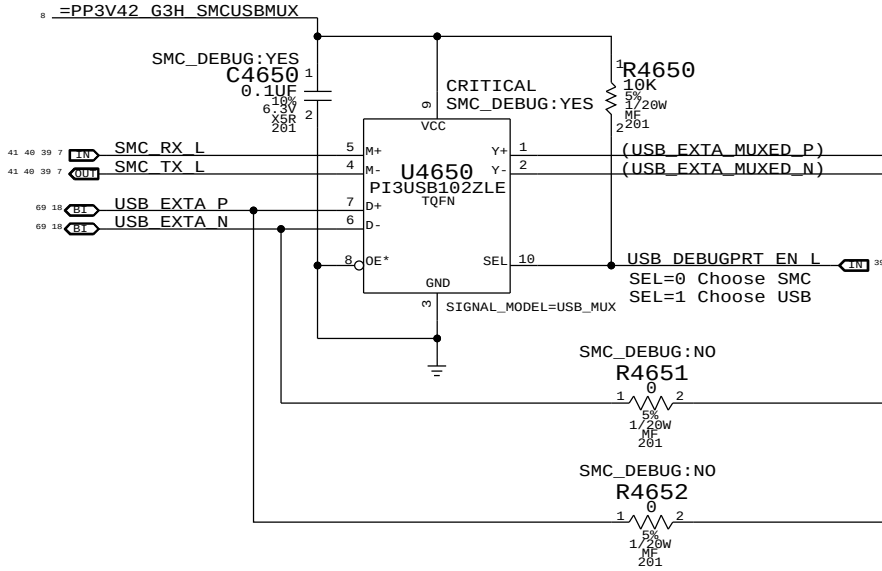
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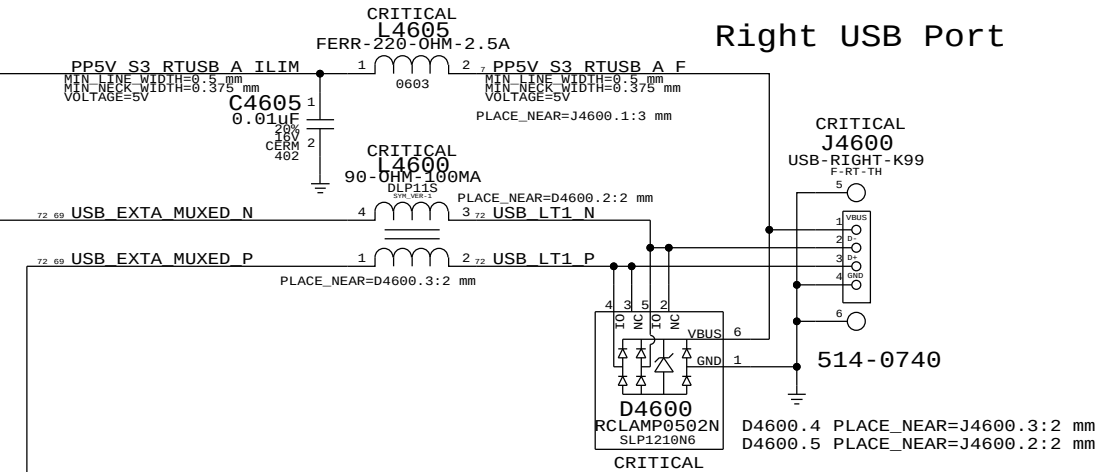
Port Power Switch



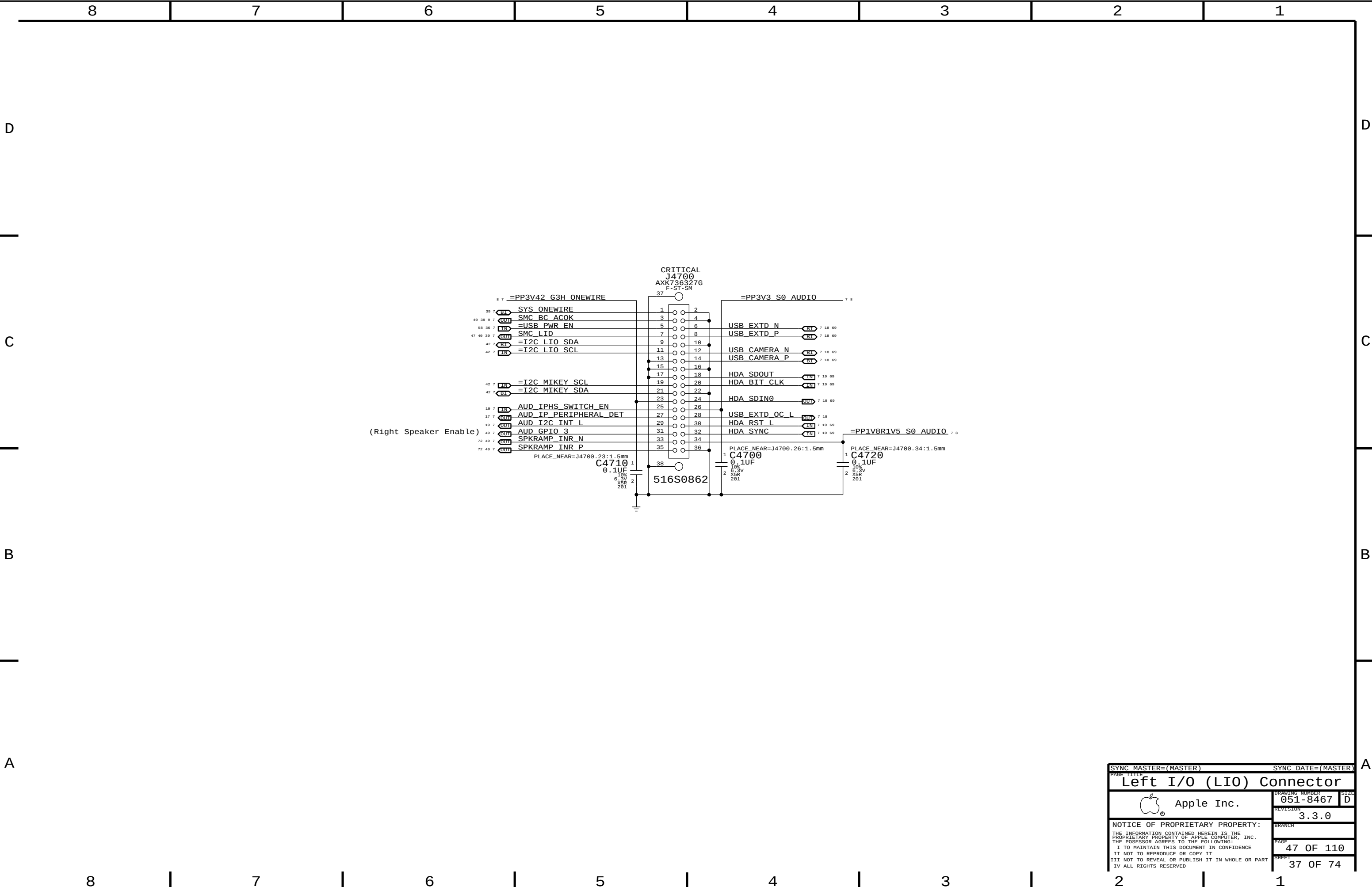
USB/SMC Debug Mux



Right USB Port

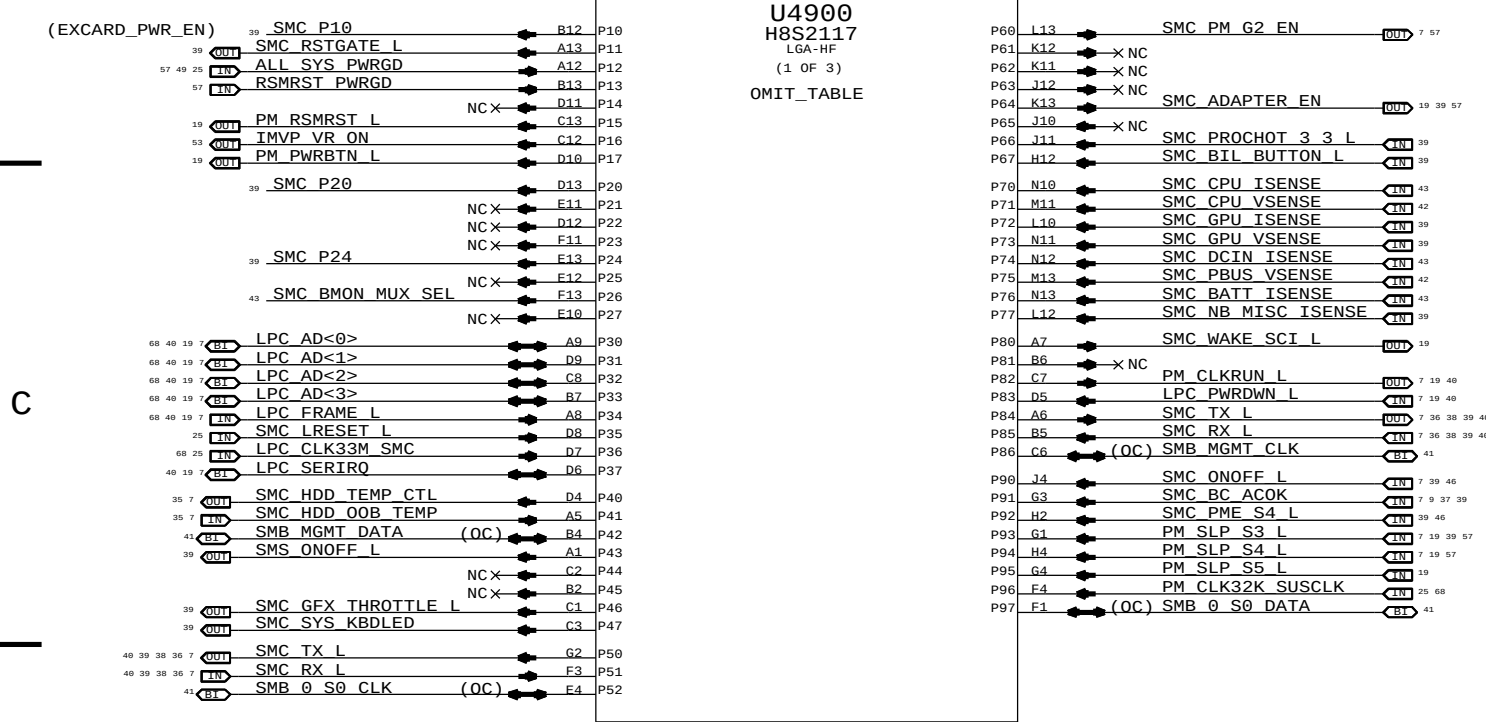


SYNC MASTER=K99 MLB		SYNC DATE=03/01/2016	
PAGE TITLE			
External USB Connectors			
 Apple Inc.		DRAWING NUMBER	051-8467
		REVISION	3.3.0
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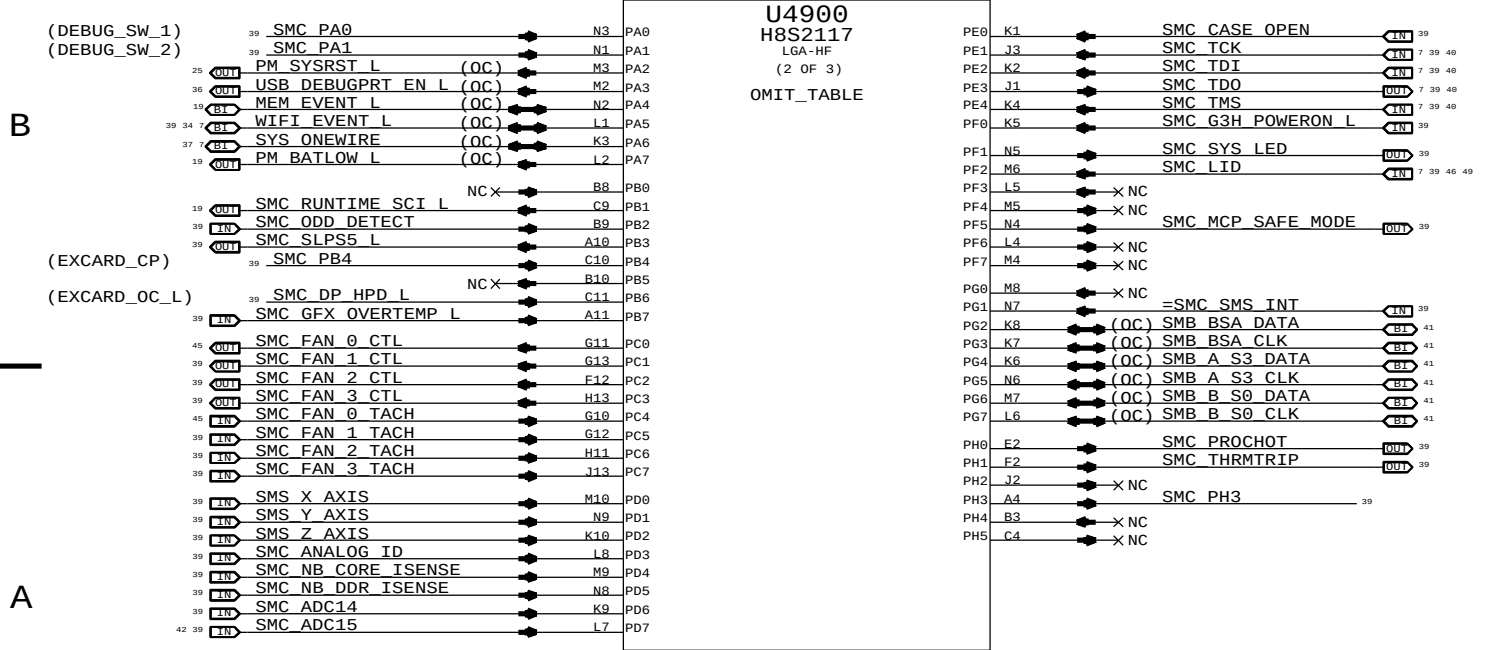


NOTE: Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

D



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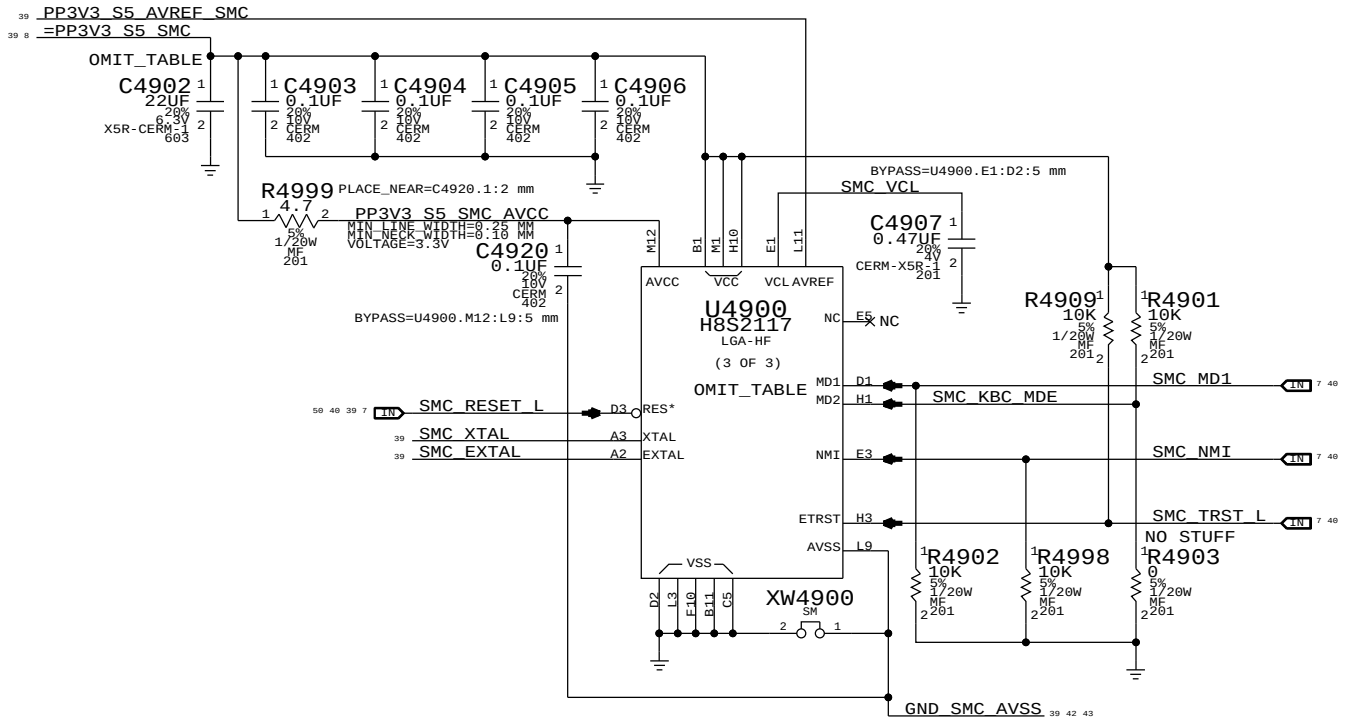


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(EXCARD_CP)

(EXCARD_OC_L)

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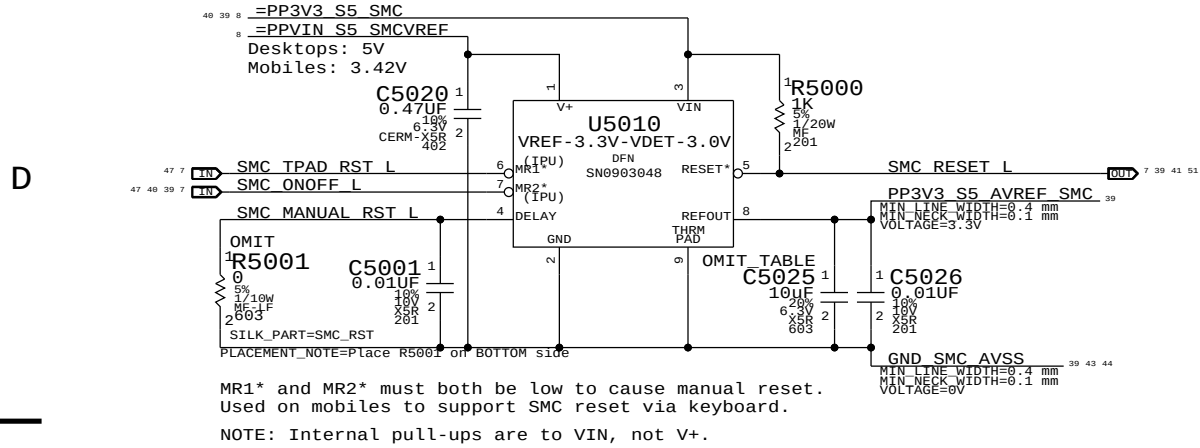


NOTE: SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

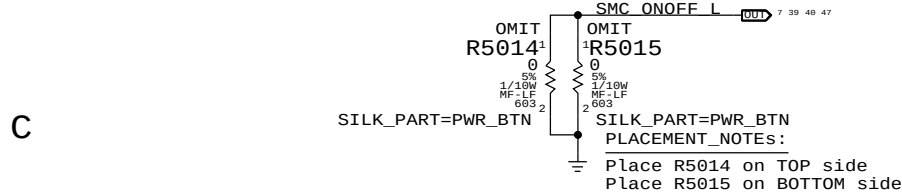
H8S2117-R:
(SMC_PECI)
(SMC_PECI_VREF)
(SMC_PECI_VSTP)

SYNC MASTER=K16_MLB		SYNC DATE=06/01/2016	
PAGE TITLE			
SMC			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-8467		D
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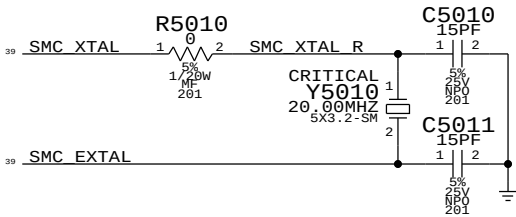
SMC Reset "Button", Supervisor & AVREF Supply



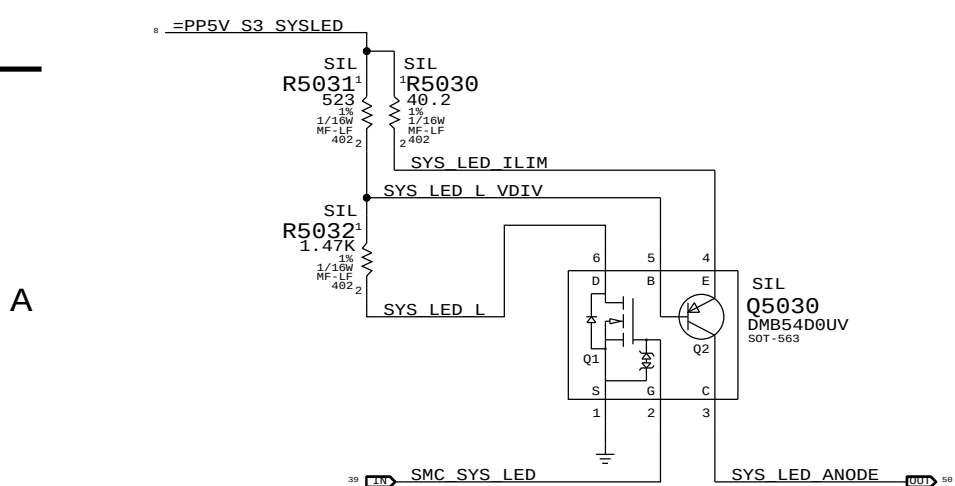
Debug Power "Buttons"



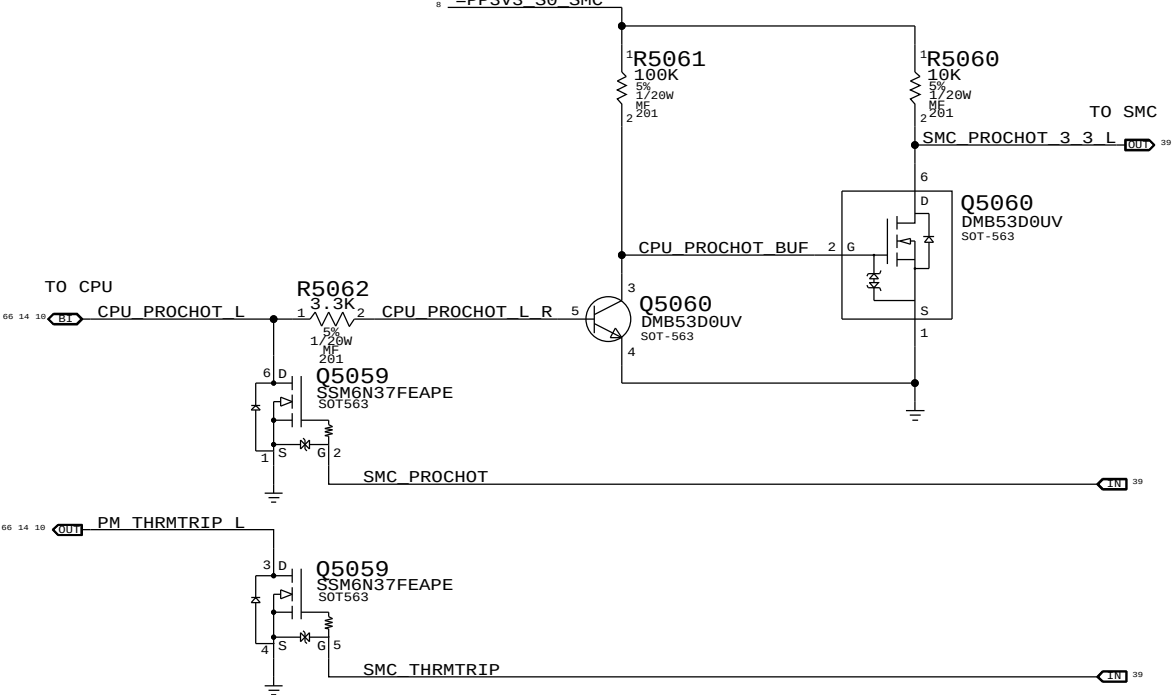
SMC Crystal Circuit



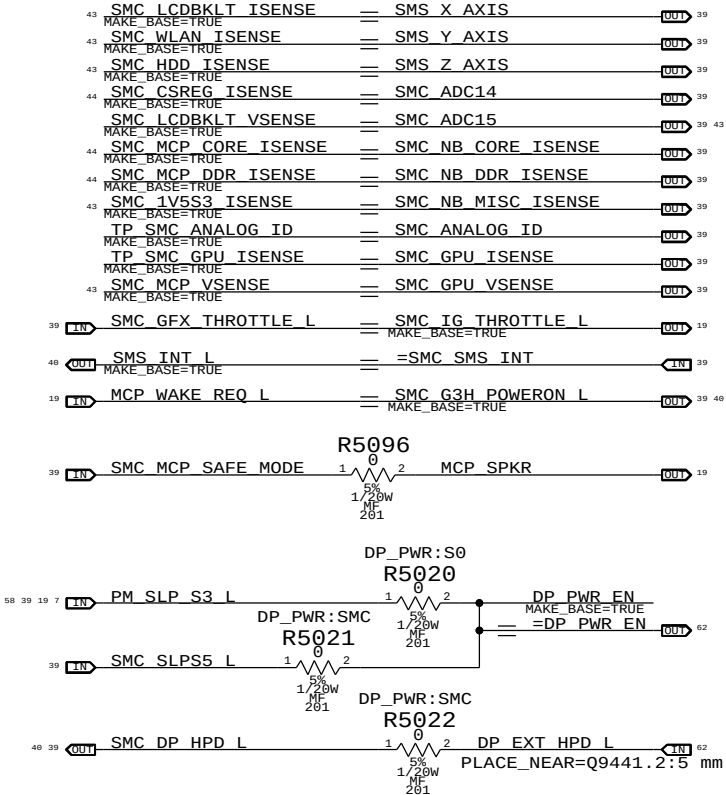
System (Sleep) LED Circuit



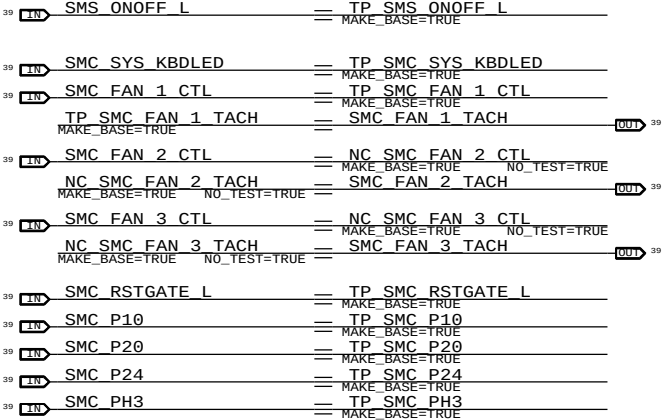
SMC FSB to 3.3V Level Shifting



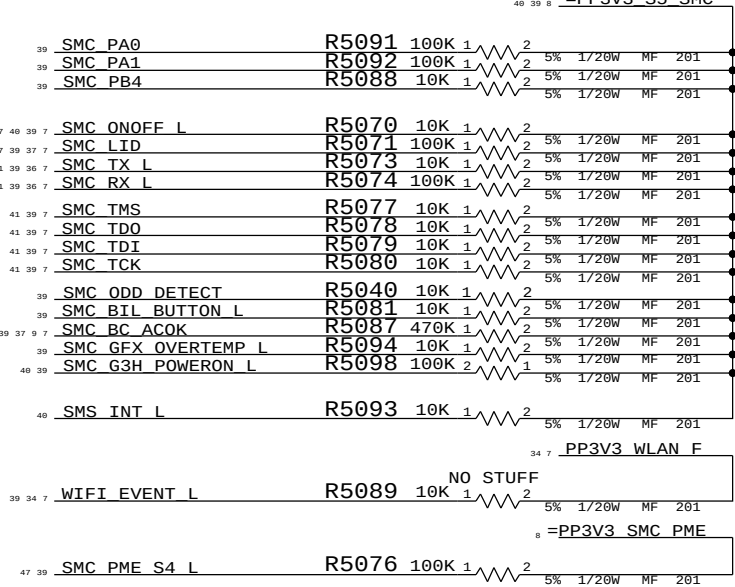
SMC Aliases



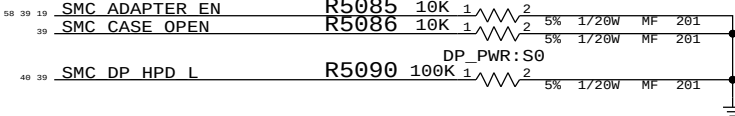
Unused Pins



SMC Pull-ups



SMC Pull-downs



SYNC_MASTER=(K99_MLB) SYNC_DATE=(03/01/2010)

SMC Support

Apple Inc.

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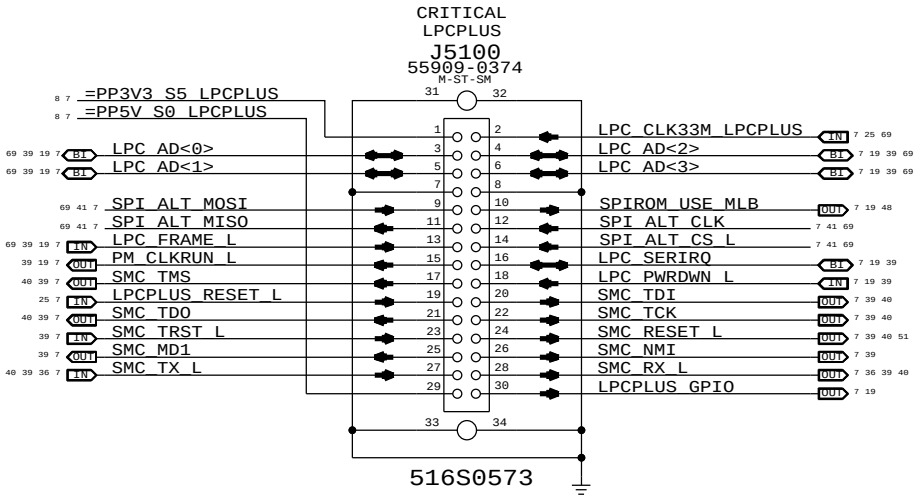
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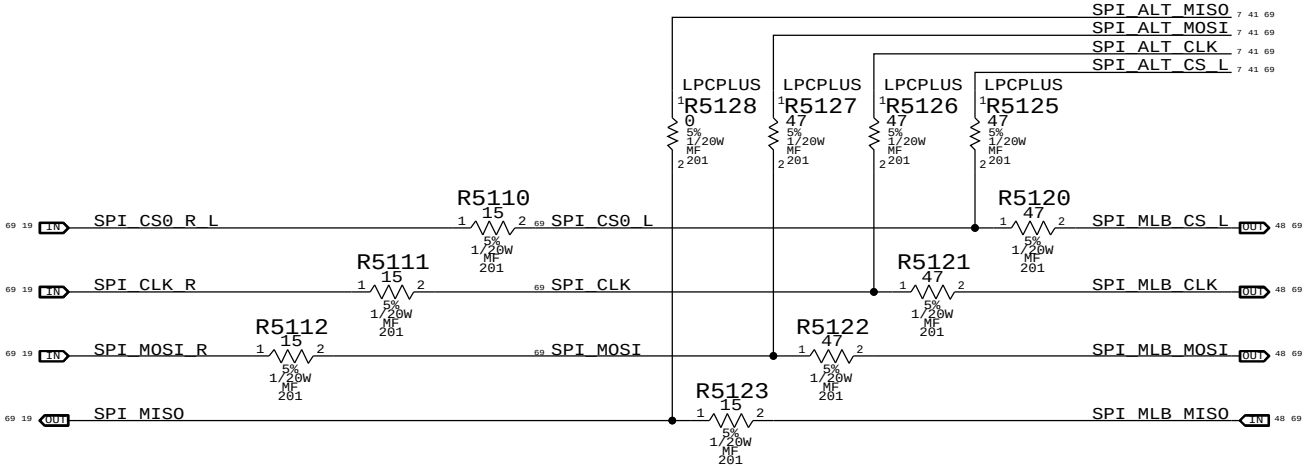
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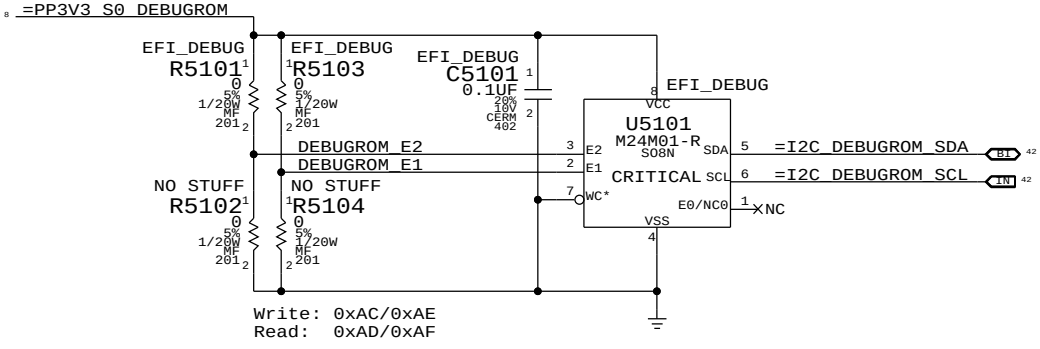
LPC+SPI Connector



SPI Bus Series Termination



EFI Debug ROM



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SYNC MASTER=K99 MLB

SYNC DATE=04/08/2016

PAGE TITLE

LPC+SPI Debug Connector

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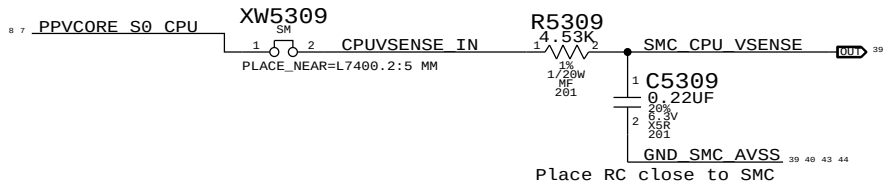
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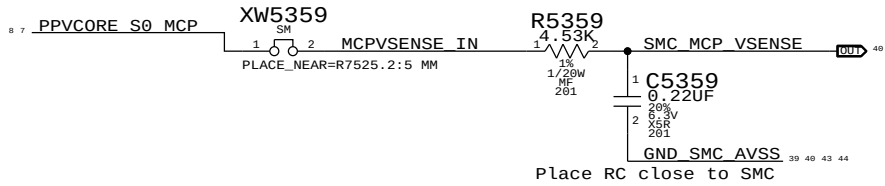
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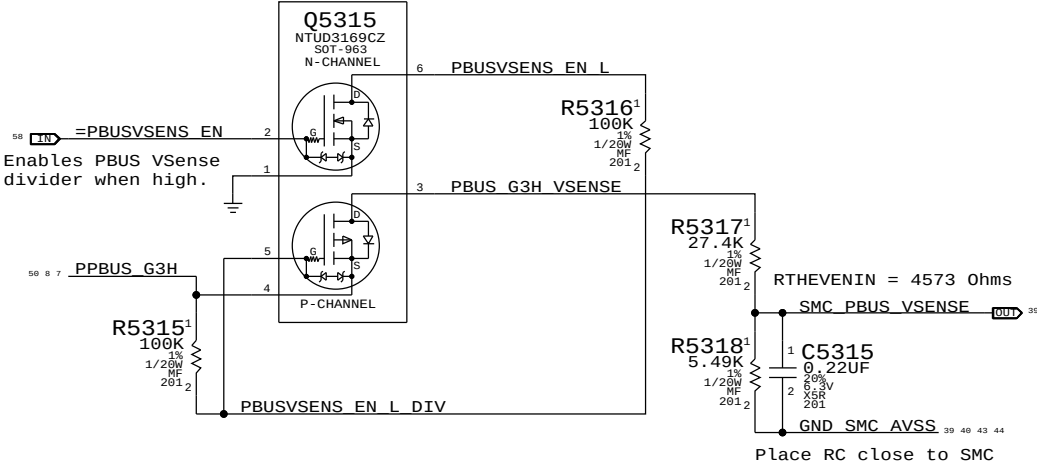
CPU Voltage Sense / Filter



MCP Voltage Sense / Filter

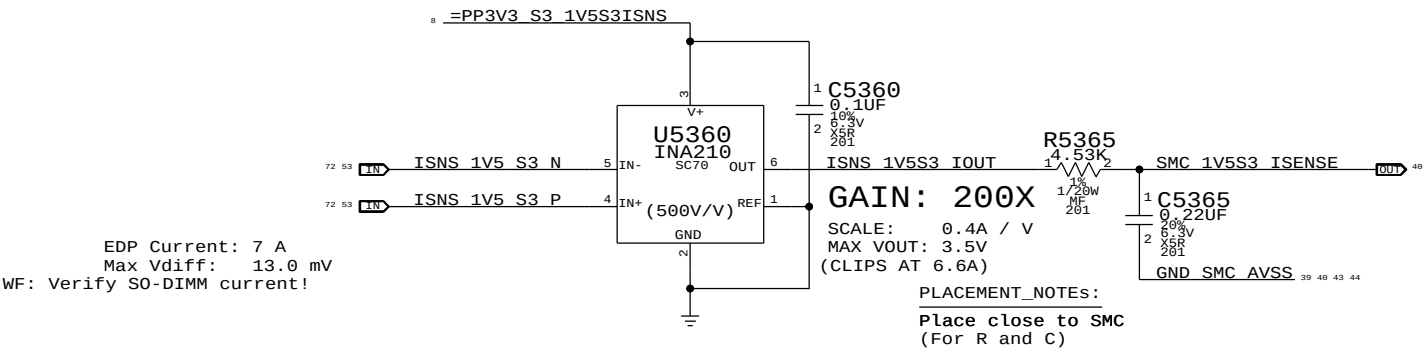


PBUS Voltage Sense Enable & Filter

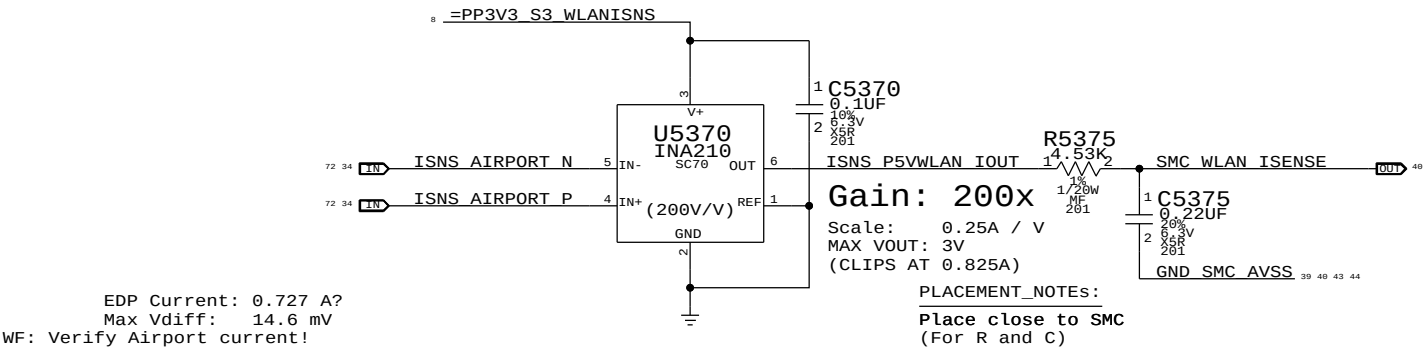


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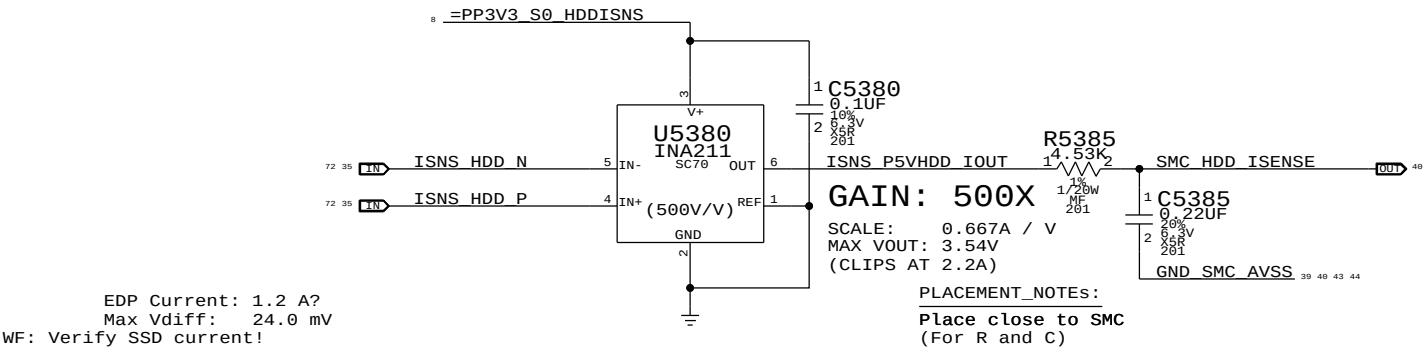
DDR3 1V5R1V35 Current Sense / Filter



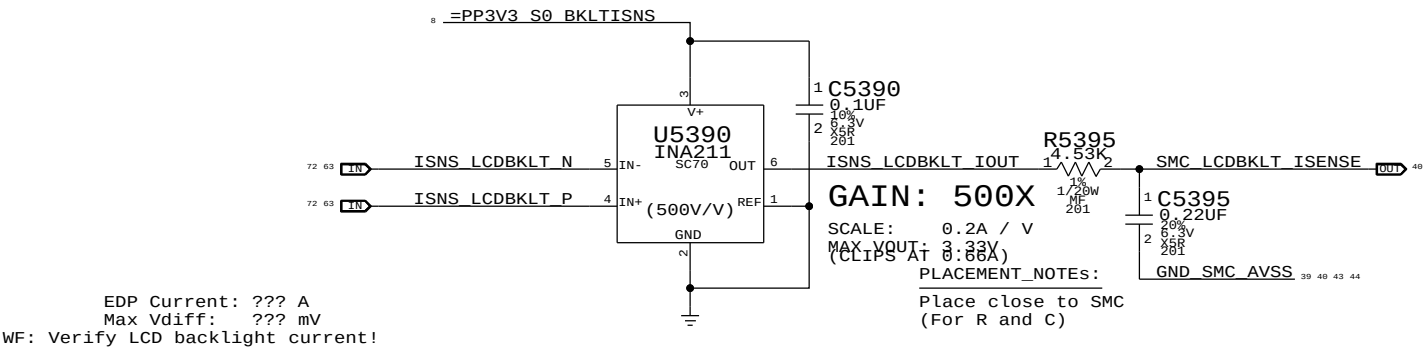
AirPort Current Sense / Filter



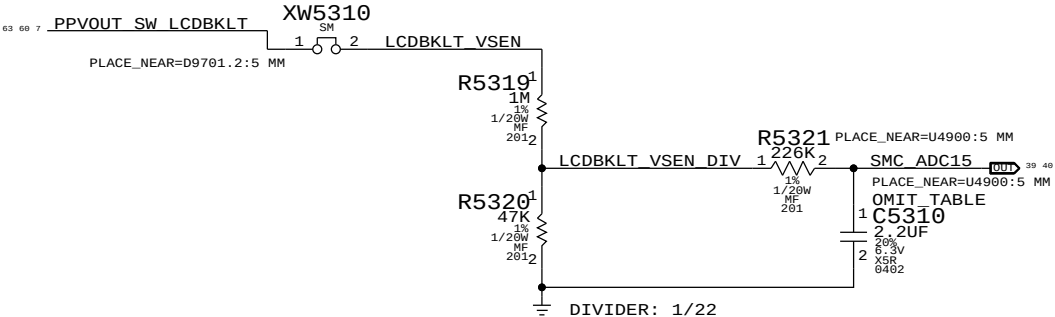
HDD Current Sense / Filter



LCD Backlight Driver Input Current Sense / Filter



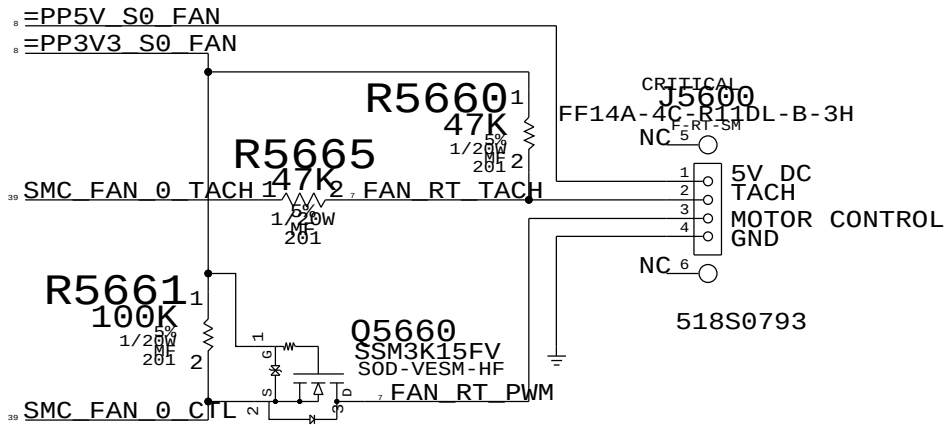
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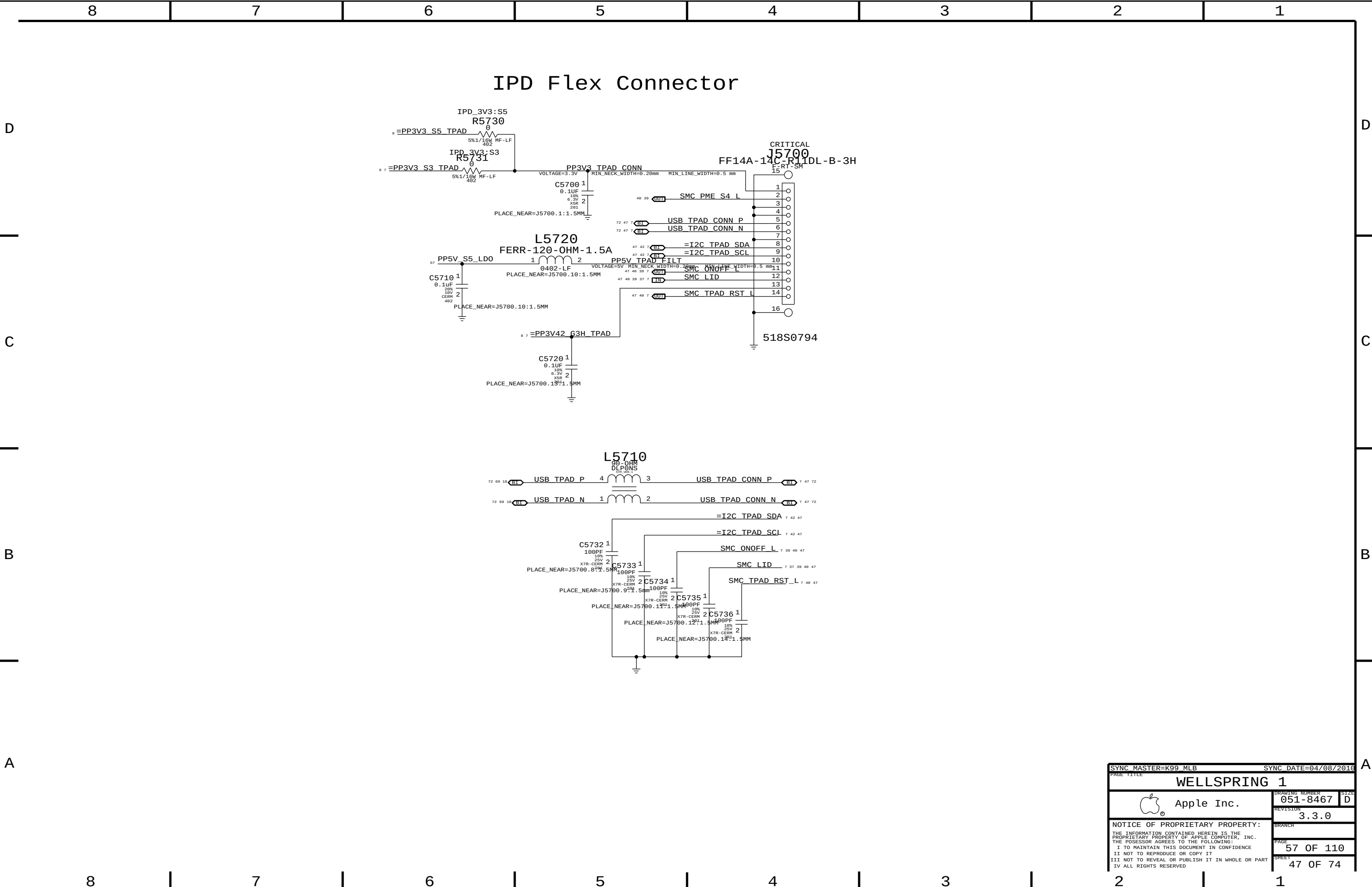
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PAGE TITLE		Voltage & Current Sensing	
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FAN CONNECTOR



SYNC_MASTER=K99_MLB		SYNC_DATE=04/08/2016	
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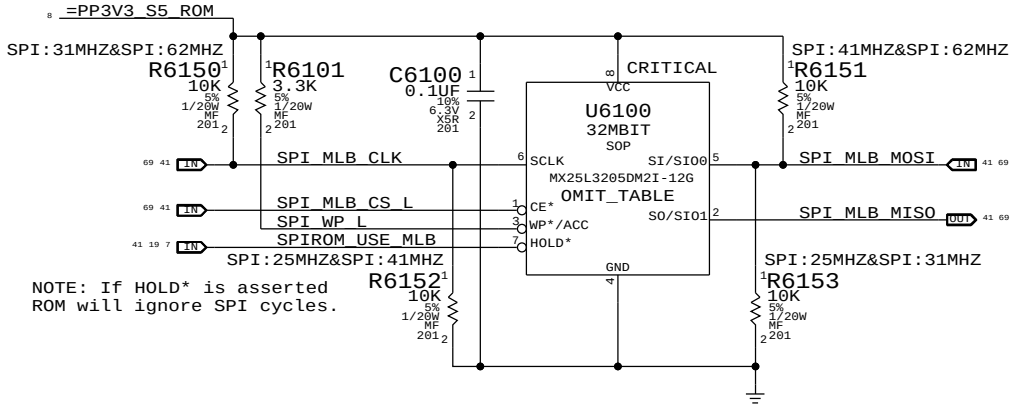
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MCP89 SPI Frequency Select		
Frequency	SPI_MOSI	SPI_CLK
25.0 MHz	0	0
31.2 MHz	0	1
41.7 MHz	1	0
62.5 MHz	1	1
NOTE: 42 & 62 MHz use FAST_READ command.		

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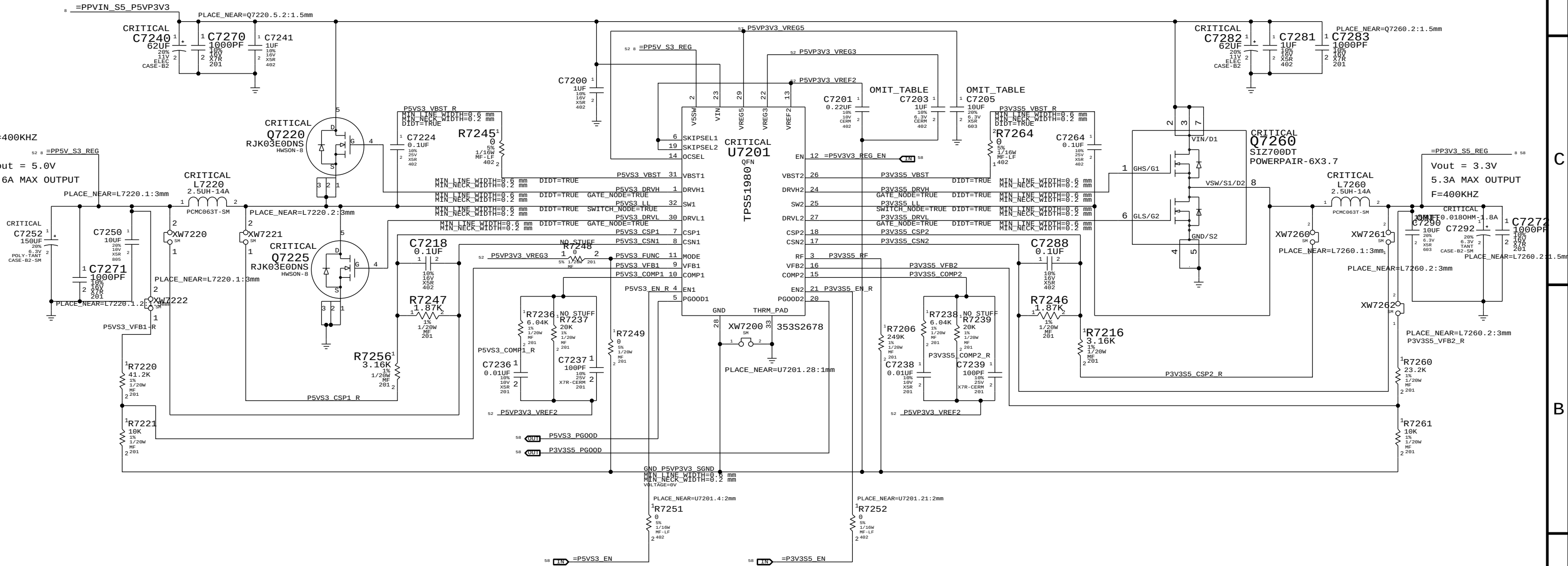
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SYNC_MASTER=K99_MLB		SYNC_DATE=04/08/2016	
PAGE TITLE			
5V / 3.3V Power Supply			
 Apple Inc.		DRAWING NUMBER	SIZE
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		3.3.0	
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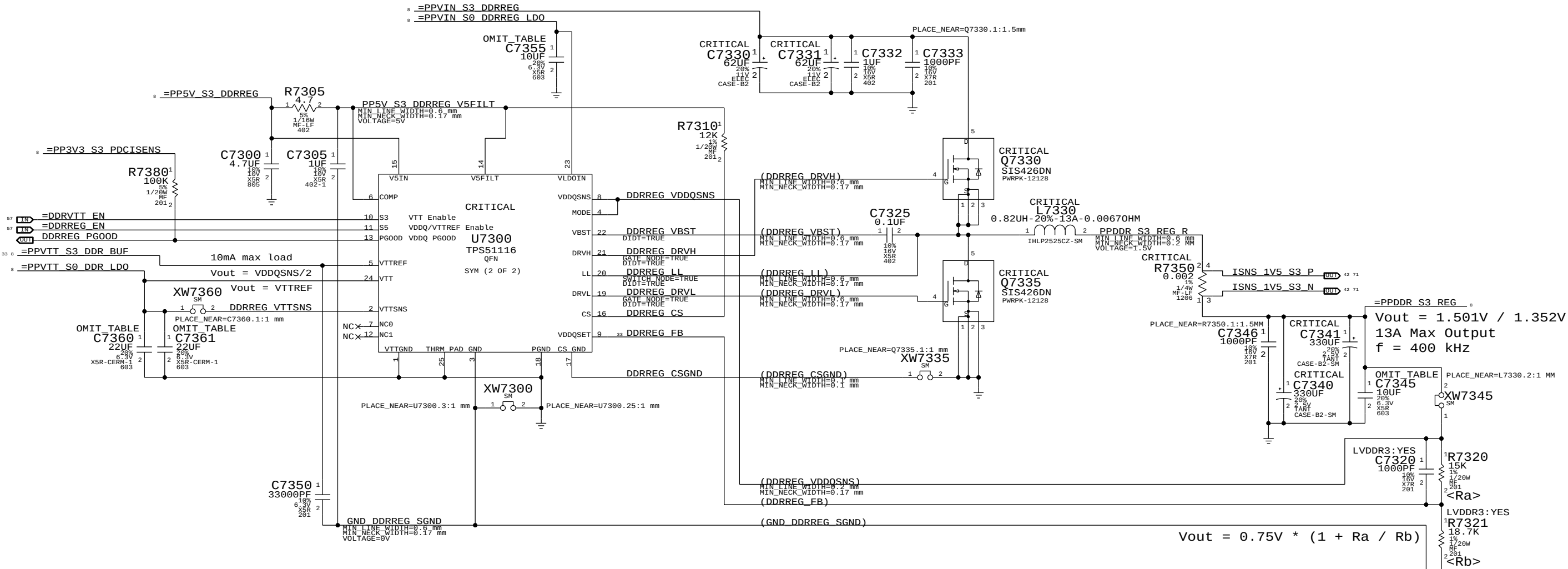
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8 7 6 5 4 3 2 1



Use LVDDR3:YES for fixed 1.35V operation or LVDDR3:NO for fixed 1.5V operation.

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0331	1	RES, 15K, 1%, 1/16W, MF-LF, 0402	R7321		LVDDR3:NO

SYNC MASTER=K16 MLB

SYNC DATE=06/01/2016

1.5V/1.35V LVDDR3 Supply

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051-8467

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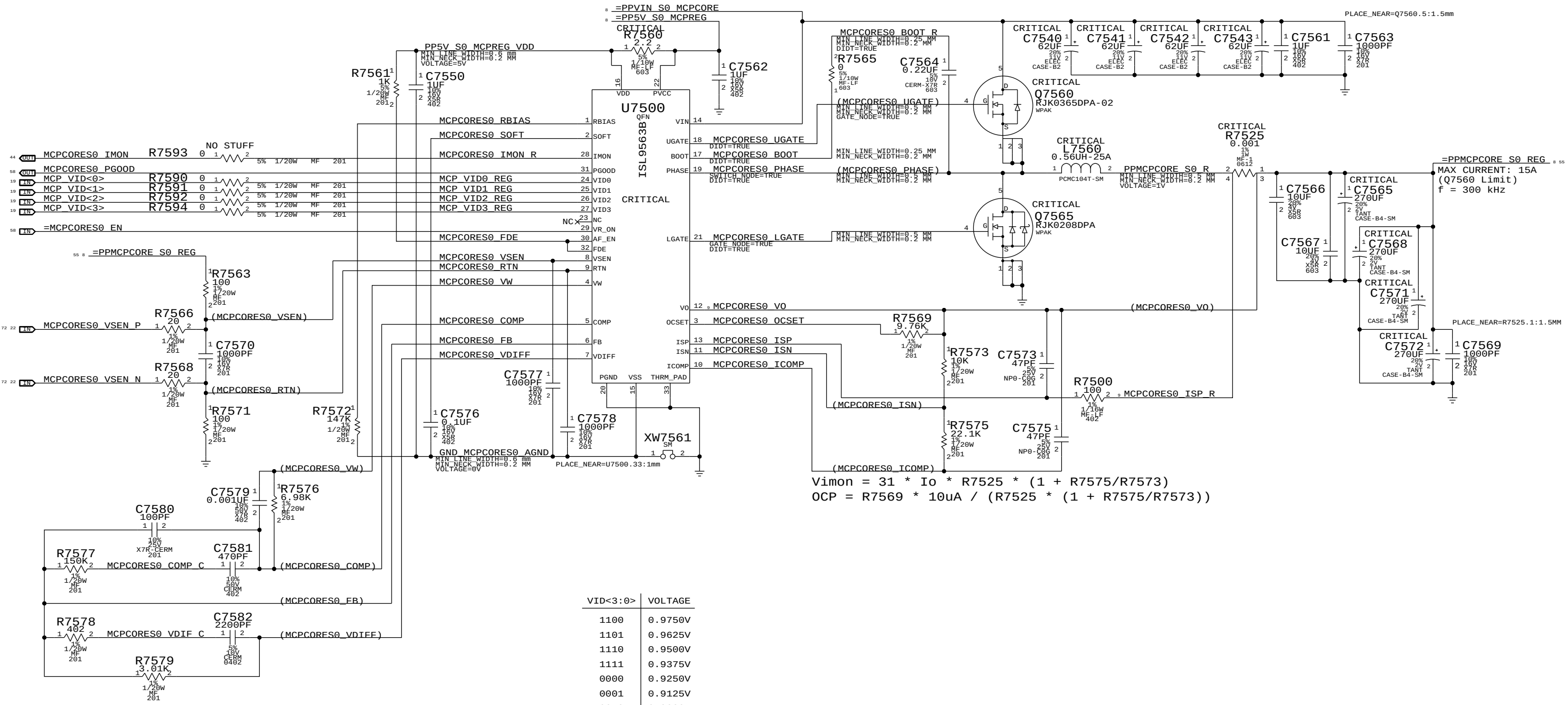
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A



$$V_{im} = 31 \cdot I_o \cdot R_{7525} \cdot (1 + R_{7575}/R_{7573})$$
$$OCP = R_{7569} \cdot 10\mu A / (R_{7525} \cdot (1 + R_{7575}/R_{7573}))$$

K6 NOTES : XOR AND INVERTER IS REMOVED, CANNOT SYNC THIS PAGE FROM T27

SYNC_MASTER=(K99_MLB) SYNC_DATE=(02/11/2010)

MCP VCore Regulator

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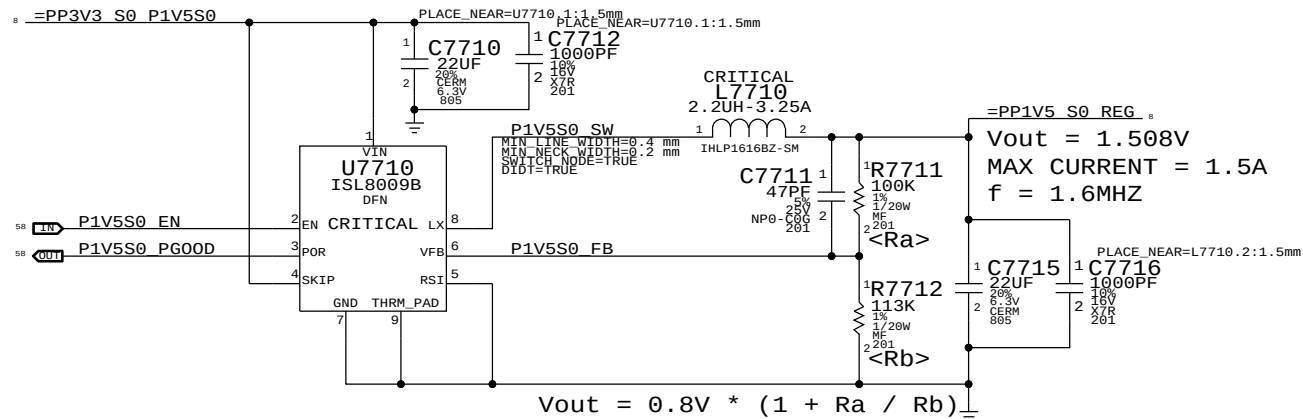
REVISION
3.3.0

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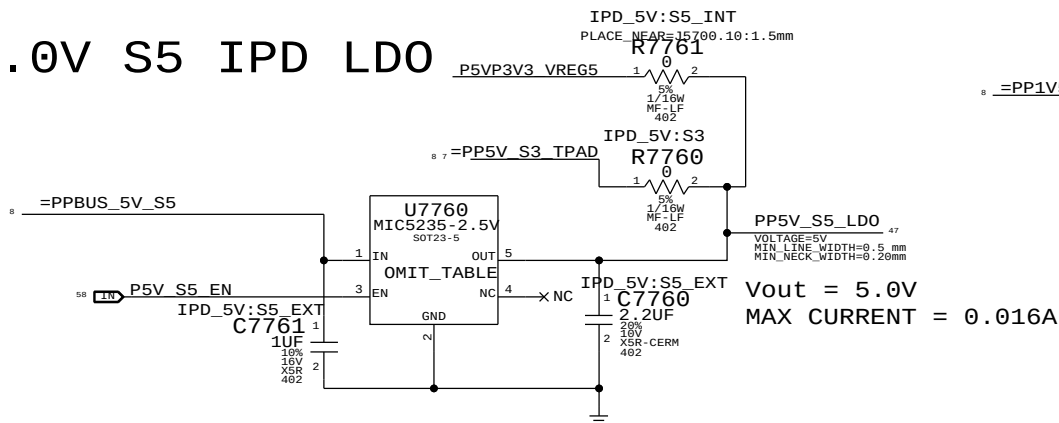
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1.5V S0 Regulator

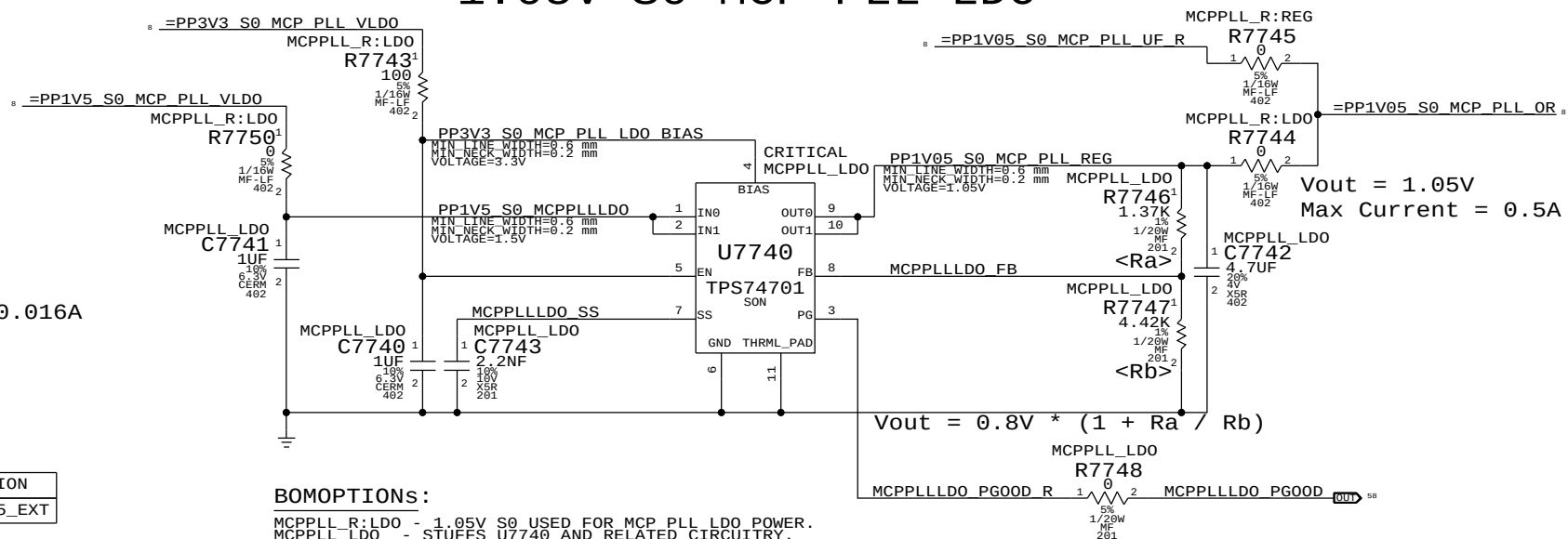


5.0V S5 IPD LDO



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S3034	1	IC, LDO, MIC5235, 5V, 1%, 150MA, SOT23-5	U7760		IPD_5V:S5_EXT

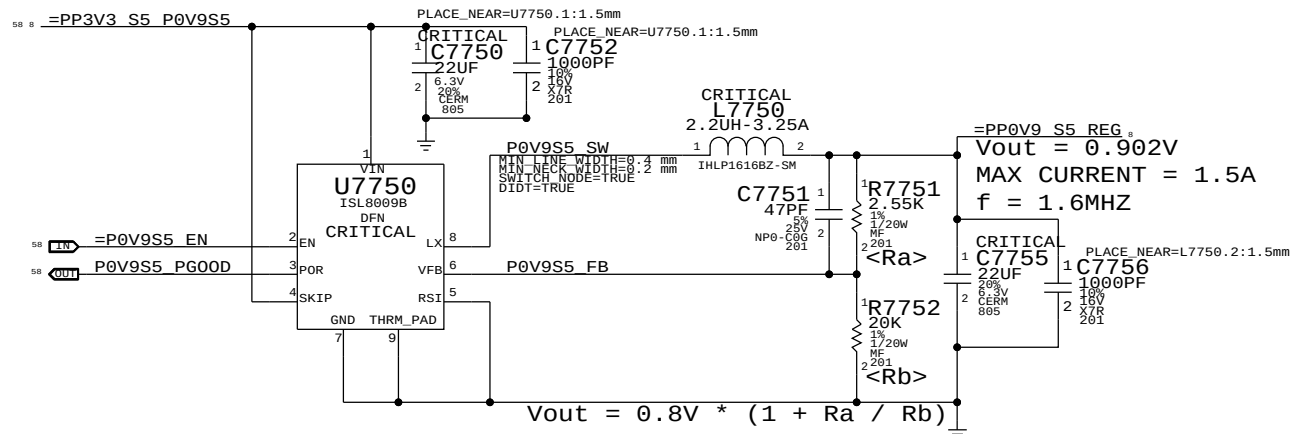
1.05V S0 MCP PLL LDO



BOMOPTIONS:

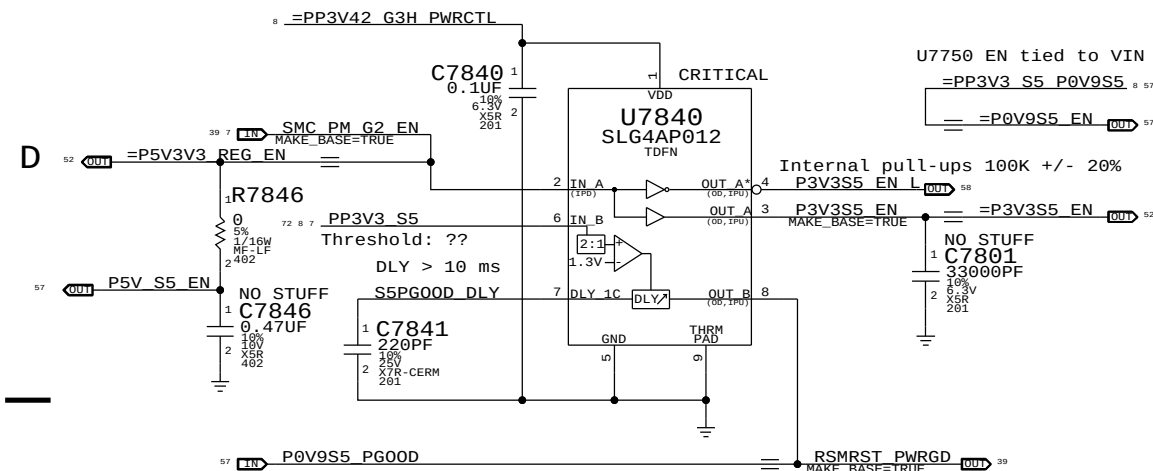
MCPPLL_R:LDO - 1.05V S0 USED FOR MCP PLL LDO POWER.
MCPPLL_LDO - STUFFS U7740 AND RELATED CIRCUITRY.
TO USE U7740, MCPPLL_R:LDO AND MCPPLL_LDO MUST BE ACTIVE.
TO USE 1.05V S0, MCPPLL_R:REG MUST BE ACTIVE, MCPPLL_LDO CAN BE ACTIVE, MCPPLL_R:LDO MUST BE INACTIVE.

MCP 0.9V S5 (AUXC) Switcher

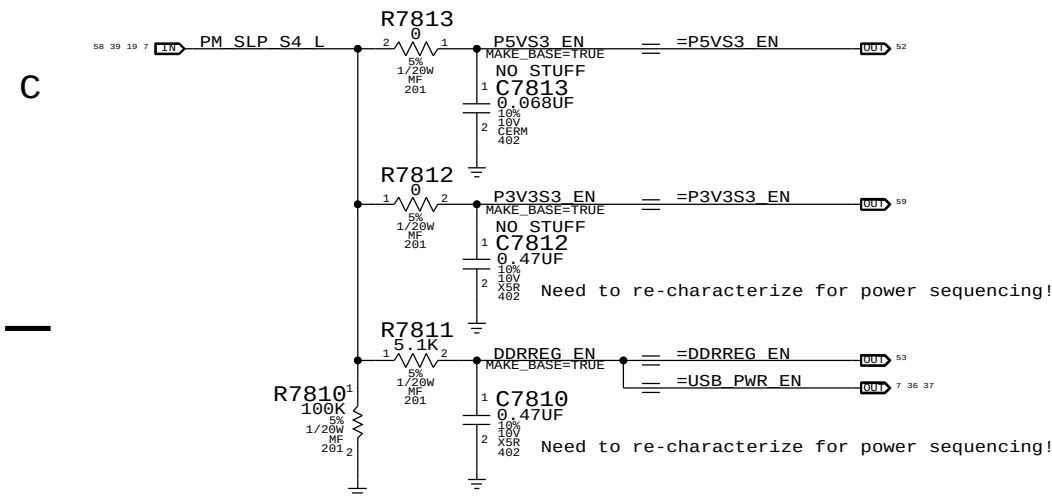


SYNC MASTER=K99 MLB		SYNC DATE=04/08/2016	
PAGE TITLE		Misc Power Supplies	
Apple Inc.		DRAWING NUMBER	051-8467
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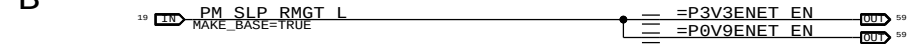
S5 Rail Enables & PG00D



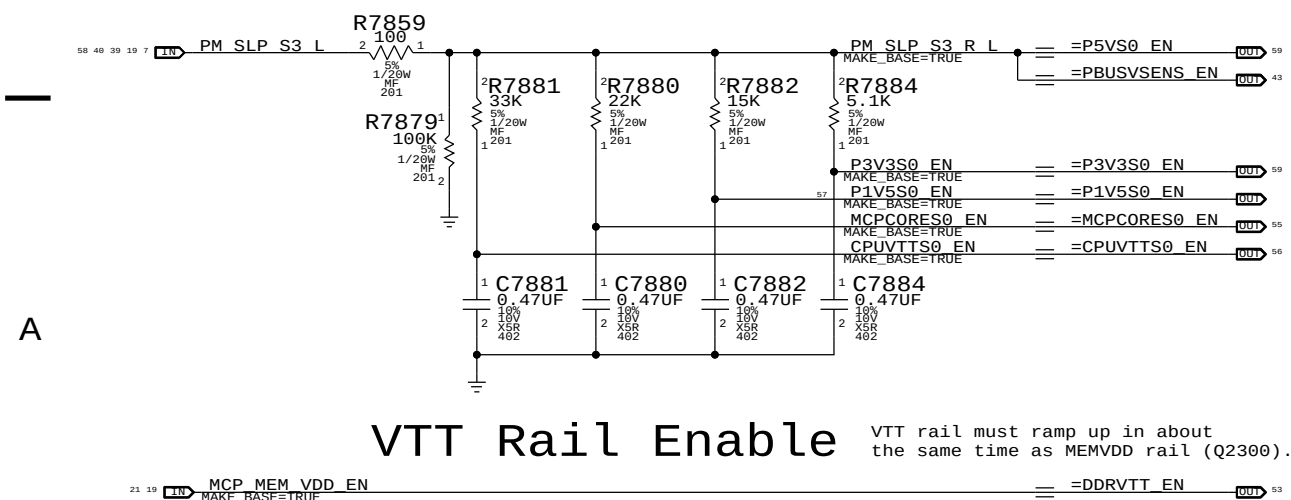
S3 Rail Enables



ENET Rail Enables



S0 Rail Enables

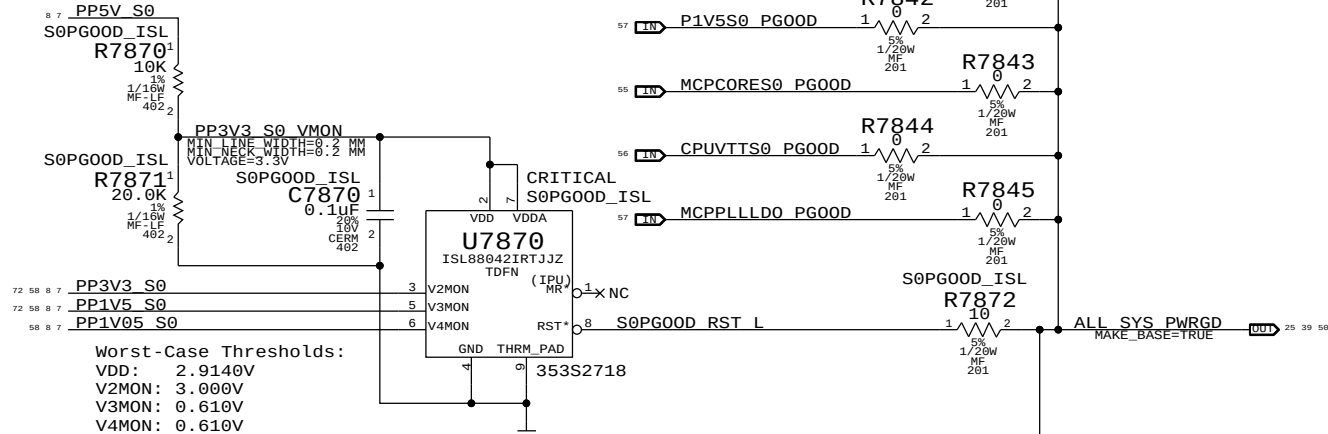


VTT Rail Enable

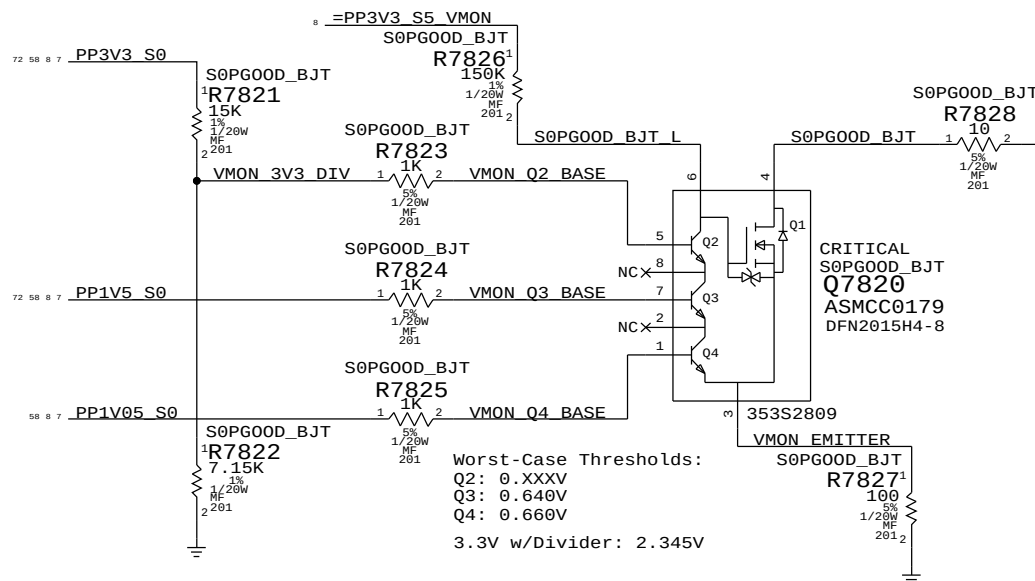
VTT rail must ramp up in about the same time as MEMVDD rail (Q2300).

S0 Rail PG00D Circuitry

S0 Rail PG00D (ISL Version)



S0 Rail PG00D (BJT Version)

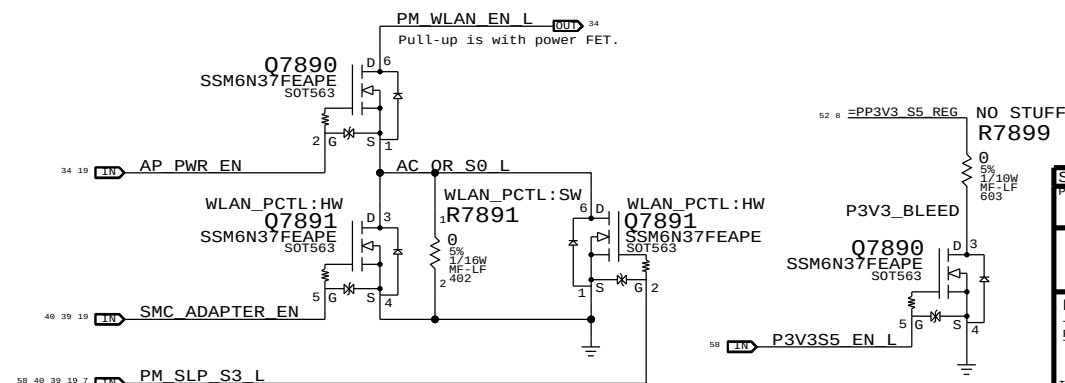


WLAN Enable Generation

```
"WLAN" = ("S3" && "AP_PWR_EN" && ("AC" || "S0"))
```

NOTE: S3 term is guaranteed by S3 pull-up on open-drain AP_PWR_EN signal.

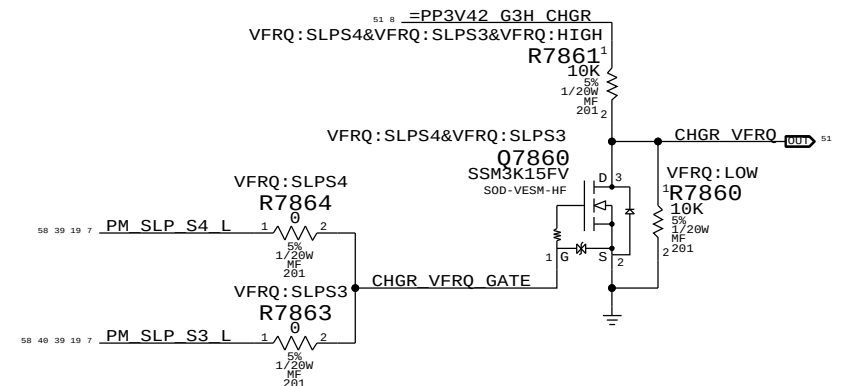
NOTE: "AC" term valid only when Q7891 is stuffed



Power Control Signals

State	SMC_PM_G2_ENABLE	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1
Sleep (S3)	1	1	0
Soft-Off (S5)	1	0	0
Battery Off (G3Hot)	0	0	0

ISL6259 Frequency Select



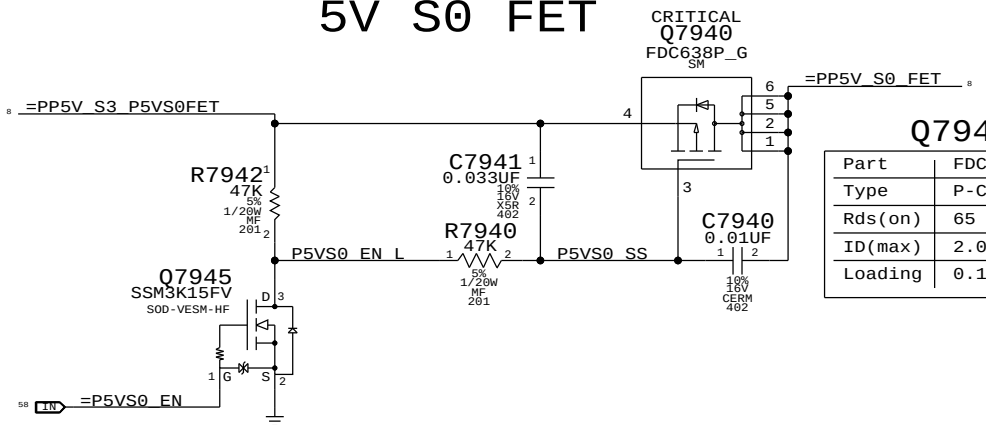
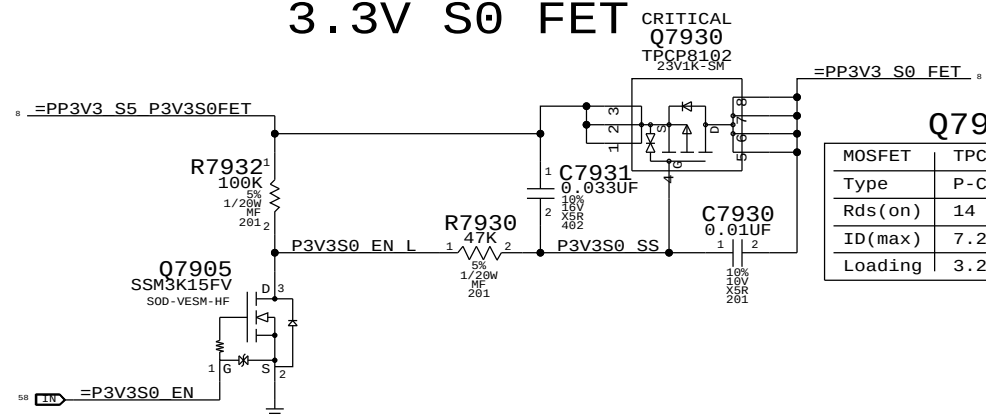
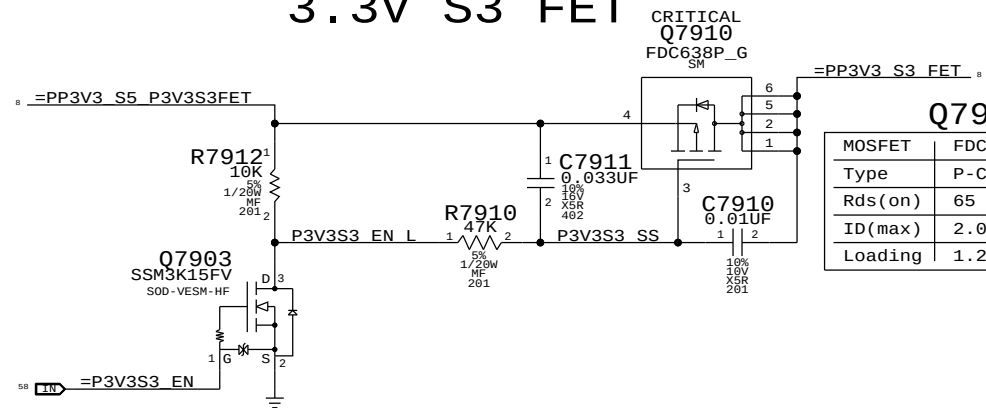
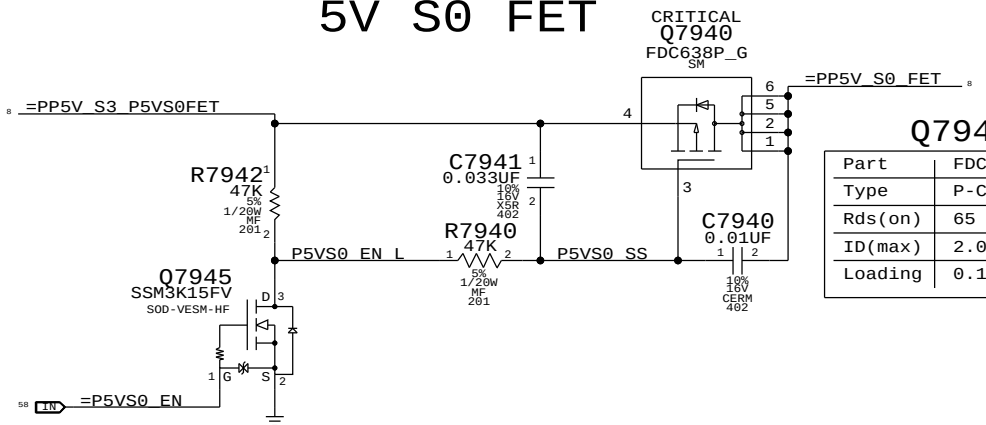
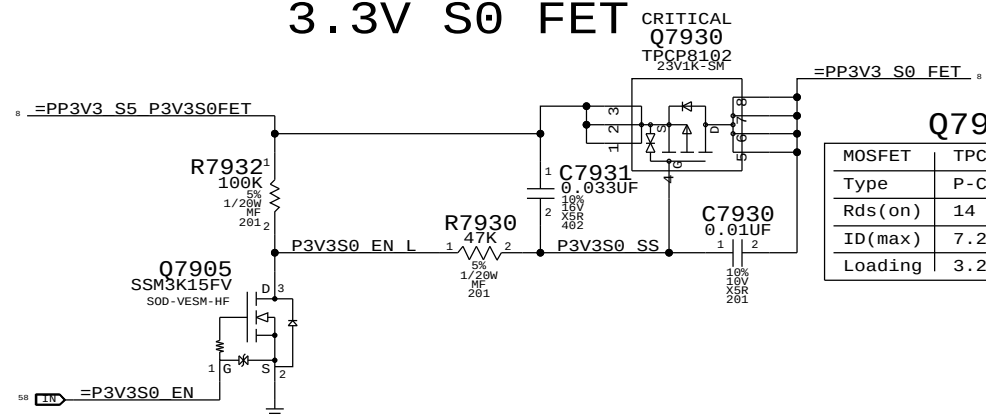
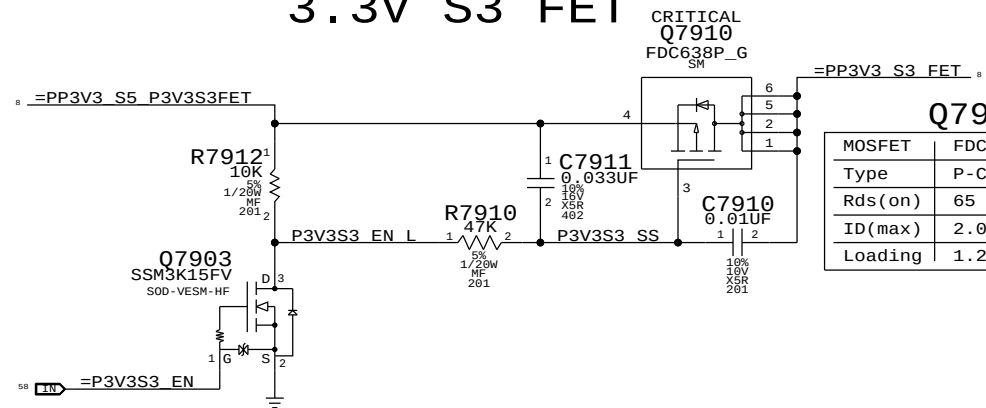
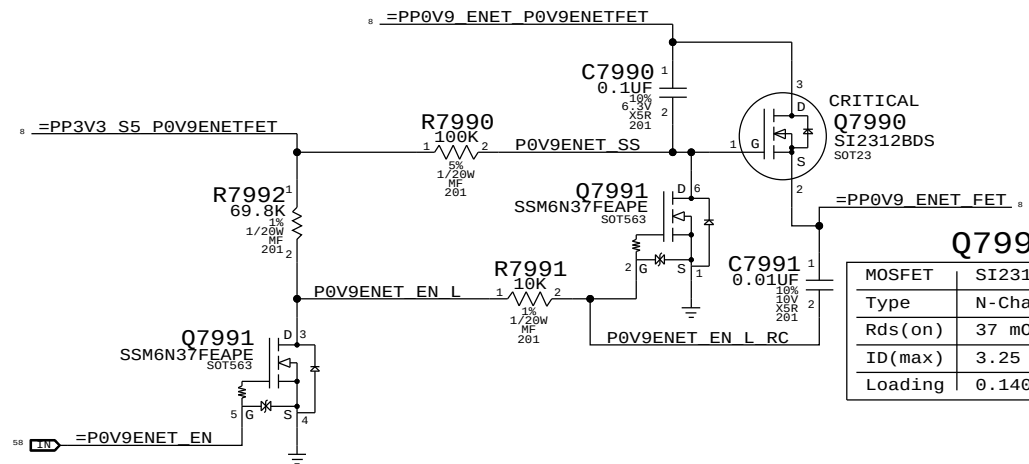
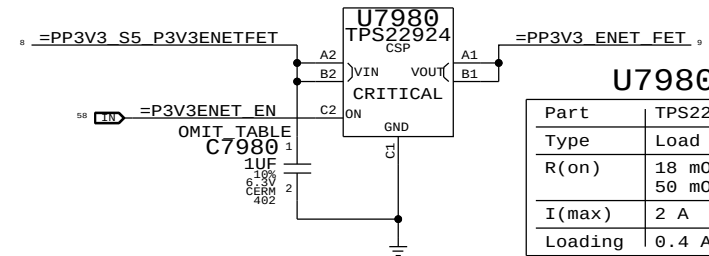
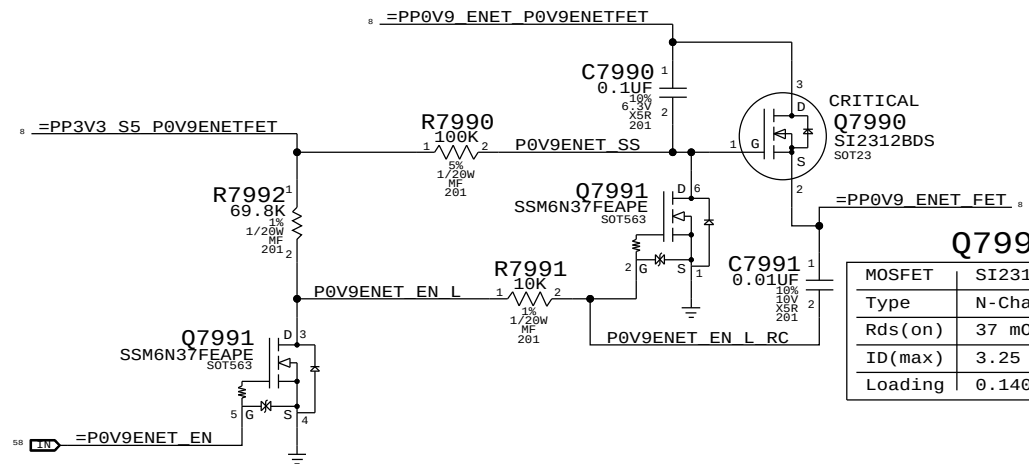
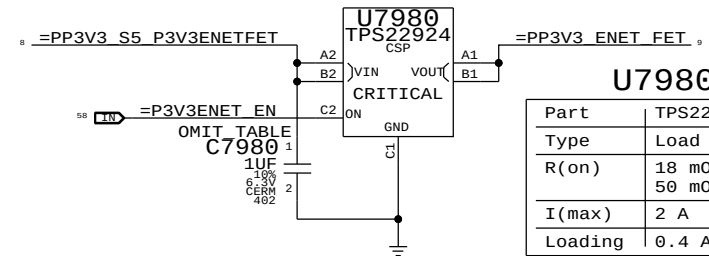
PAGE TITLE Power Sequencing



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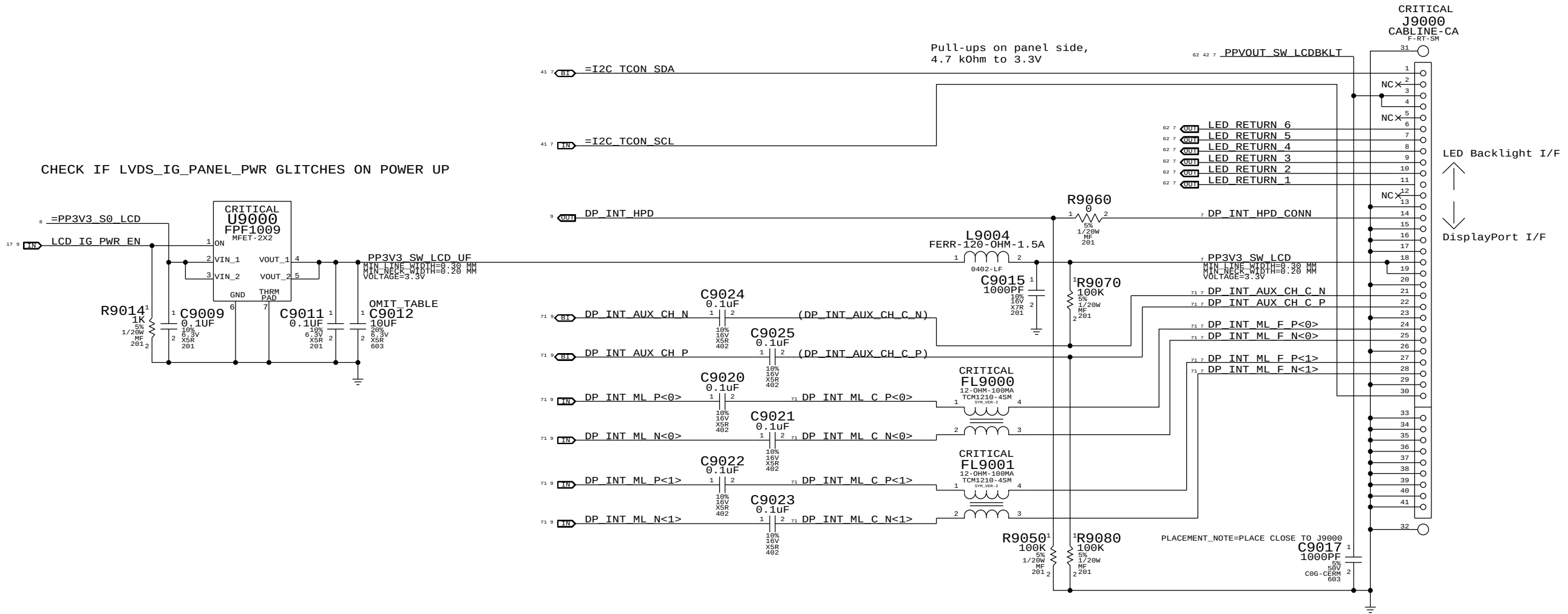
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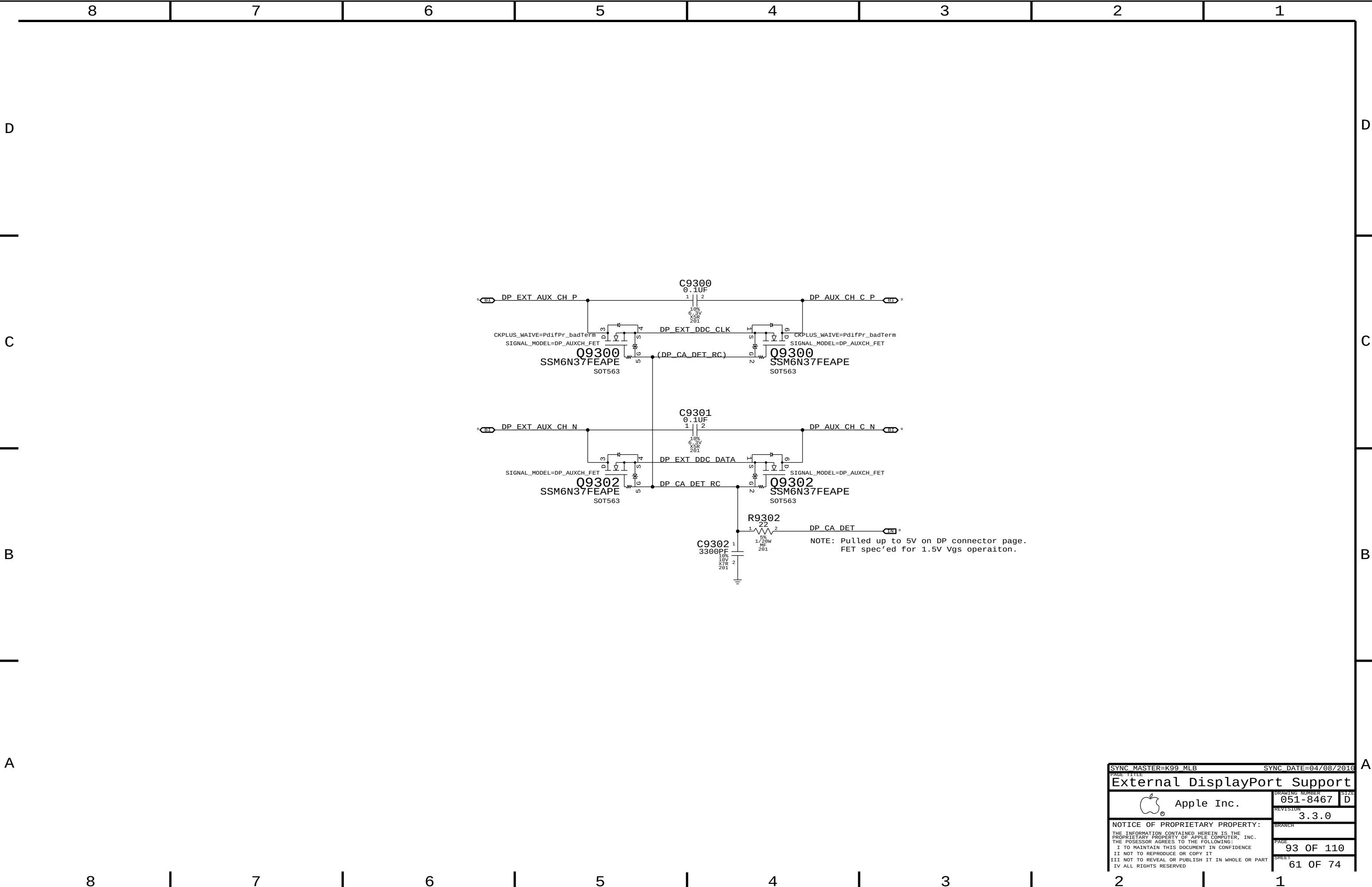
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LCD Connector
Internal DP Connector: 518S0787



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Internal DisplayPort Connector			
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External DisplayPort Support

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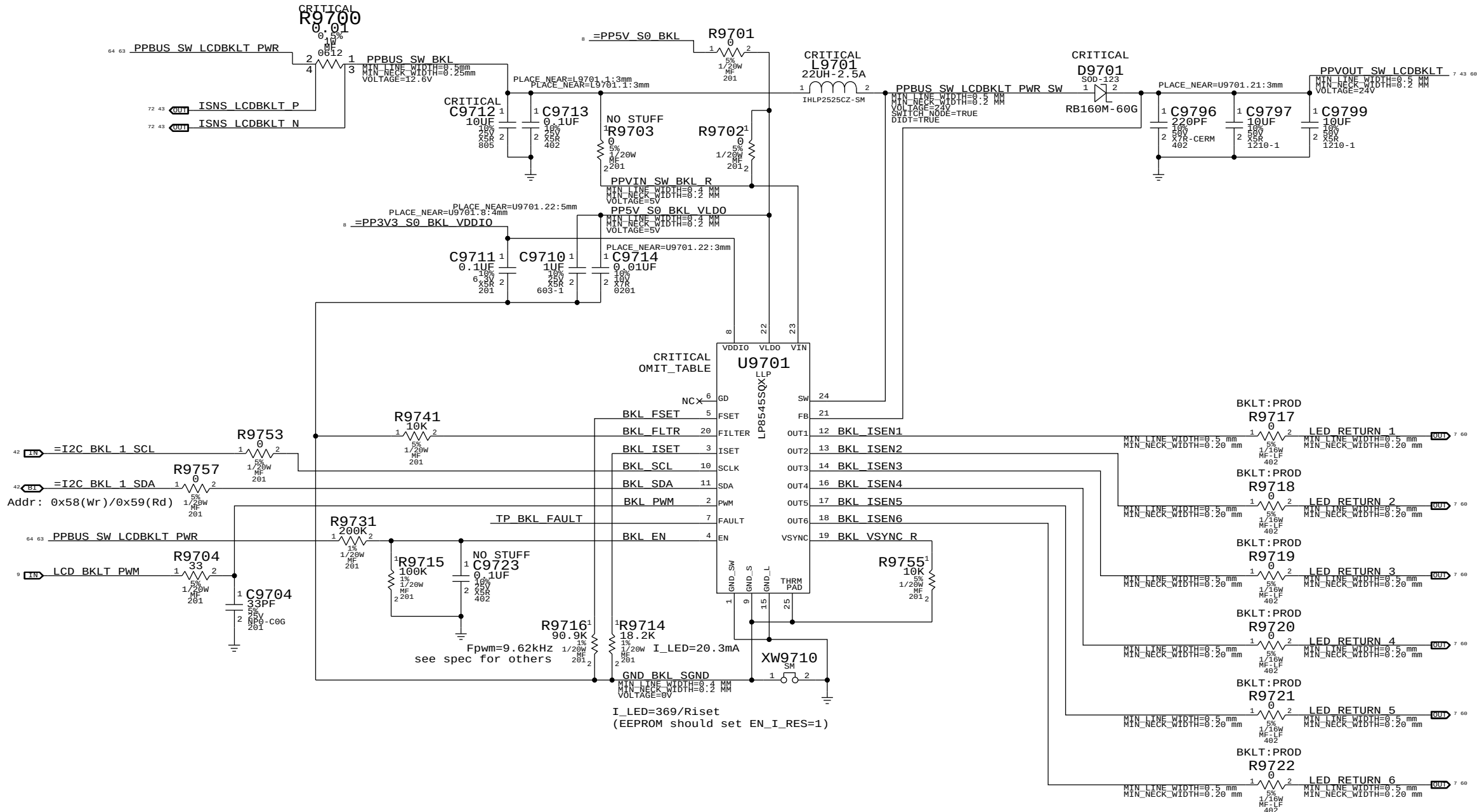


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FOR LP8543:
STUFF R9741
NO STUFF R9740, C9740, C9741, R9754

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
103S0198	3	RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402, SM	R9717, R9718, R9719		BKLT:ENG
103S0198	3	RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402, SM	R9720, R9721, R9722		BKLT:ENG
353S2896	1	IC, LP8545, LED BKLT CTRLR, PRODUCTIO, LLP24	U9701	CRITICAL	PROJ:K16
353S2967	1	IC, LP8545, LED BKLT CTRLR, LLP24, K99 VER	U9701	CRITICAL	PROJ:K99

10.2 ohm resistors for current measurement on LED strings.

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LCD Backlight Driver			
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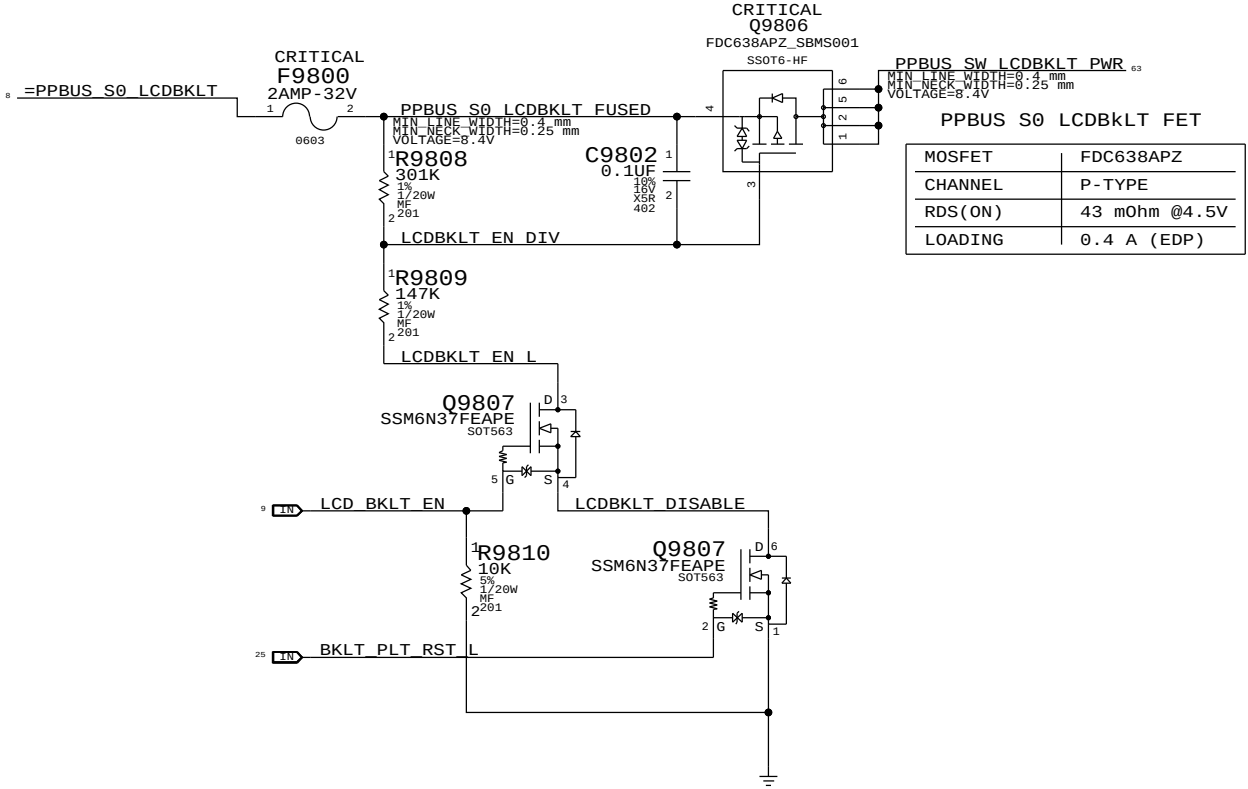
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SYNC_DATE=04/08/2016

LCD Backlight Support

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ADDITIONAL CPU VCORE HF DECOUPLING
40x 1uF 0402



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Additional CPU/GPU Decoupling				
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FSB (Front-Side Bus) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FSB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
FSB_DSTB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FSB_DATA	*	=2X_DIELECTRIC	?
FSB_DSTB	*	=3X_DIELECTRIC	?
FSB_ADDR	*	=STANDARD	?
FSB_ADSTB	*	=2X_DIELECTRIC	?
FSB_1X	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FSB_DATA	TOP,BOTTOM	=4X_DIELECTRIC	?
FSB_DSTB	TOP,BOTTOM	=5X_DIELECTRIC	?
FSB_ADDR	TOP,BOTTOM	=3X_DIELECTRIC	?
FSB_ADSTB	TOP,BOTTOM	=4X_DIELECTRIC	?
FSB_1X	TOP,BOTTOM	=3X_DIELECTRIC	?

All 4x/2x/1x FSB signals with impedance requirements are 50-ohm single-ended.

FSB 4X signals / groups shown in signal table on right.
Signals within each 4x group should be matched within 5 ps of strobe.
DSTB# complementary pairs should be matched within 1 ps of each other, all DSTB#s matched to +/- 135 ps.
Spacing is 2x dielectric between DATA#, DINV# signals, with 3x dielectric spacing to the DSTB#s.

FSB 2X signals / groups shown in signal table on right.
Signals within each 2x group should be matched within 20 ps. ADTSB#s should be matched +/- 270 ps.
Spacing is 1x dielectric between ADDR#, REQ# signals, with 2x dielectric spacing to ADSTB#.

FSB 1X signals shown in signal table on right.

Intel Design Guide recommends FSB signals be routed only on internal layers.

NOTE: Intel Design Guide allows closer spacing if signal lengths can be shortened.

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.1
SOURCE: Santa Rosa Platform DG, Rev 1.5 (#22294), Sections 4.2 & 4.3

CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?
CPU_8MIL	*	8 MIL	?
CPU_COMP	*	25 MIL	?
CPU_GTLREF	*	25 MIL	?
CPU_ITP	*	=2:1_SPACING	?
CPU_VCCSENSE	*	25 MIL	?

SR DG recommends at least 25 mils, >50 mils preferred

Most CPU signals with impedance requirements are 55-ohm single-ended.
Some signals require 27.4-ohm single-ended impedance.

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.1
SOURCE: Santa Rosa Platform DG, Rev 0.9 (#20517), Sections 4.4 & 5.8.2.4

MCP FSB COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_FSB_COMP	*	8 MIL	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.1.4

FSB Clock Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_FSB_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_FSB	*	=3X_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2.5

CPU / FSB Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		PROPERTY	VALUE
	PHYSICAL	SPACING		
FSB_DATA_GROUP0	FSB_55S	FSB_DATA	FSB D L<15..0>	7 10 14
FSB_DATA_GROUP0	FSB_55S	FSB_DATA	FSB DINV L<0>	7 10 14
FSB_DSTB0	FSB_DSTB_55S	FSB_DSTB	FSB DSTB L P<0>	7 10 14
FSB_DSTB0	FSB_DSTB_55S	FSB_DSTB	FSB DSTB L N<0>	7 10 14
FSB_DATA_GROUP1	FSB_55S	FSB_DATA	FSB D L<31..16>	7 10 14
FSB_DATA_GROUP1	FSB_55S	FSB_DATA	FSB DINV L<1>	7 10 14
FSB_DSTB1	FSB_DSTB_55S	FSB_DSTB	FSB DSTB L P<1>	7 10 14
FSB_DSTB1	FSB_DSTB_55S	FSB_DSTB	FSB DSTB L N<1>	7 10 14
FSB_DATA_GROUP2	FSB_55S	FSB_DATA	FSB D L<47..32>	7 10 14
FSB_DATA_GROUP2	FSB_55S	FSB_DATA	FSB DINV L<2>	7 10 14
FSB_DSTB2	FSB_DSTB_55S	FSB_DSTB	FSB DSTB L P<2>	7 10 14
FSB_DSTB2	FSB_DSTB_55S	FSB_DSTB	FSB DSTB L N<2>	7 10 14
FSB_DATA_GROUP3	FSB_55S	FSB_DATA	FSB D L<63..48>	7 10 14
FSB_DATA_GROUP3	FSB_55S	FSB_DATA	FSB DINV L<3>	7 10 14
FSB_DSTB3	FSB_DSTB_55S	FSB_DSTB	FSB DSTB L P<3>	7 10 14
FSB_DSTB3	FSB_DSTB_55S	FSB_DSTB	FSB DSTB L N<3>	7 10 14
FSB_ADDR_GROUP0	FSB_55S	FSB_ADDR	FSB A L<16..3>	7 10 14
FSB_ADDR_GROUP0	FSB_55S	FSB_ADDR	FSB REQ L<4..0>	7 10 14
FSB_ADSTB0	FSB_55S	FSB_ADSTB	FSB ADSTB L<0>	7 10 14
FSB_ADDR_GROUP1	FSB_55S	FSB_ADDR	FSB A L<35..17>	7 10 14
FSB_ADSTB1	FSB_55S	FSB_ADSTB	FSB ADSTB L<1>	7 10 14
FSB_1X	FSB_55S	FSB_1X	FSB ADS L	7 10 14
FSB_BREQ0_L	FSB_55S	FSB_1X	FSB_BREQ0_L	10 14
FSB_1X	FSB_55S	FSB_1X	FSB BNR L	10 14
FSB_1X	FSB_55S	FSB_1X	FSB BPRT L	10 14
FSB_1X	FSB_55S	FSB_1X	FSB DBSY L	10 14
FSB_1X	FSB_55S	FSB_1X	FSB DEFER L	10 14
FSB_1X	FSB_55S	FSB_1X	FSB DRDY L	10 14
FSB_1X	FSB_55S	FSB_1X	FSB HIT L	7 10 14
FSB_1X	FSB_55S	FSB_1X	FSB HITM L	7 10 14
FSB_1X	FSB_55S	FSB_1X	FSB LOCK L	7 10 14
FSB_CPURST_L	FSB_55S	FSB_1X	FSB_CPURST_L	10 13 14
FSB_1X	FSB_55S	FSB_1X	FSB RS L<2..0>	10 14
FSB_1X	FSB_55S	FSB_1X	FSB TRDY L	10 14
CPU_ASYNC	CPU_55S	CPU_AGTL	CPU A20M L	10 14
CPU_BSEL	CPU_55S	CPU_AGTL	CPU BSEL<2..0>	9 10
CPU_FERR_L	CPU_55S	CPU_8MTL	CPU FERR L	10 14
CPU_ASYNC	CPU_55S	CPU_AGTL	CPU IGNNE L	10 14
CPU_TWIT_L	CPU_55S	CPU_AGTL	CPU INIT L	10 14
CPU_ASYNC_R	CPU_55S	CPU_AGTL	CPU INTR	10 14
CPU_ASYNC_R	CPU_55S	CPU_AGTL	CPU NMI	10 14
CPU_PROCHOT_L	CPU_55S	CPU_AGTL	CPU PROCHOT L	10 14 40
CPU_PWRGD	CPU_55S	CPU_AGTL	CPU PWRGD	10 13 14
CPU_ASYNC	CPU_55S	CPU_AGTL	CPU SMI L	10 14
CPU_ASYNC	CPU_55S	CPU_AGTL	CPU STPCLK L	10 14
PM_THRMTRIP_L	CPU_55S	CPU_8MTL	PM THRMTRIP L	10 14 40
FSB_CPUSLP_L	CPU_55S	CPU_AGTL	FSB_CPUSLP L	10 14
CPU_FROM_SB	CPU_55S	CPU_AGTL	CPU DPSLP L	10 14
CPU_DPRSTP_L	CPU_55S	CPU_AGTL	CPU DPRSTP L	10 14 54
CPU_ASYNC	CPU_55S	CPU_AGTL	FSB DPWR L	10 14
FSB_CLK_CPU	CLK_FSB_100D	CLK_FSB	FSB CLK CPU P	10 14
FSB_CLK_CPU	CLK_FSB_100D	CLK_FSB	FSB CLK CPU N	10 14
FSB_CLK_ITP	CLK_FSB_100D	CLK_FSB	FSB CLK ITP P	13 14
FSB_CLK_ITP	CLK_FSB_100D	CLK_FSB	FSB CLK ITP N	13 14
FSB_CLK_MCP	CLK_FSB_100D	CLK_FSB	FSB CLK MCP P	14
FSB_CLK_MCP	CLK_FSB_100D	CLK_FSB	FSB CLK MCP N	14
CPU_IERR_L	CPU_55S		CPU IERR L	10
PM DPRSLPVR	CPU_55S	CPU_AGTL	PM DPRSLPVR	14 54
(See above)	CPU_55S	CPU_AGTL	IMVP DPRSLPVR	
MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP BCLK VML COMP VDD	14
MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP BCLK VML COMP GND	14
MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP CPU COMP VCC	14
MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP CPU COMP GND	14
CPU_GTLREF	CPU_50S	CPU_GTLREF	CPU GTLREF	10 33
CPU_COMP	CPU_50S	CPU_COMP	CPU COMP<3>	10
CPU_COMP	CPU_27P4S	CPU_COMP	CPU COMP<2>	10
CPU_COMP	CPU_50S	CPU_COMP	CPU COMP<1>	10
CPU_COMP	CPU_27P4S	CPU_COMP	CPU COMP<0>	10
XDP_TDI	CPU_55S	CPU_ITP	XDP TDI	10 13
XDP_TDO	CPU_55S	CPU_ITP	XDP TDO	10 13
XDP_TMS	CPU_55S	CPU_ITP	XDP TMS	10 13
XDP_TCK	CPU_55S	CPU_ITP	XDP TCK	10 13
XDP_TRST_L	CPU_55S	CPU_ITP	XDP TRST L	10 13
XDP_BPM_L	CPU_55S	CPU_ITP	XDP BPM L<4..0>	10 13
XDP_BPM_L5	CPU_55S	CPU_ITP	XDP BPM L<5>	10 13
(FSB_CPURST_L)	CPU_55S	CPU_ITP	XDP CPURST L	13
	CPU_55S	CPU_8MTL	CPU VID<6..0>	11 12 54
	CPU_55S	CPU_8MTL	IMVP6 VID<6..0>	12
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU VCCSENSE P	11 54
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU VCCSENSE N	11 54
(CPU_VCCSENSE)	CPU_27P4S	CPU_VCCSENSE	IMVP6 VSEN P	54
(CPU_VCCSENSE)	CPU_27P4S	CPU_VCCSENSE	IMVP6 VSEN N	54

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF

D

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=2:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DATA2DATA	*	=1.5:1_SPACING	?
MEM_DATA2MEM	*	=3:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_20THER	*	25 MIL	?

NV DG says 3x inner, 4x outer
NV DG says 2x inner, 4x outer
NV DG says 2x inner, 4x outer
NV DG says 2x inner, 4x outer
NV DG says 2x inner, 4x outer
NV DG says 2x inner, 4x outer
NV DG says 2x inner, 4x outer
NV DG says 2x inner, 4x outer
NV DG says 4x inner, 5x outer

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM
MEM_CLK	MEM_DATA	*	MEM_CLK2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_DATA	*	MEM_CMD2MEM
MEM_CMD	MEM_DQS	*	MEM_CMD2MEM

C

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DATA	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DATA	MEM_CLK	*	MEM_DATA2MEM
MEM_DATA	MEM_CTRL	*	MEM_DATA2MEM
MEM_DATA	MEM_CMD	*	MEM_DATA2MEM
MEM_DATA	MEM_DATA	*	MEM_DATA2DATA
MEM_DATA	MEM_DQS	*	MEM_DATA2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DATA	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	*	*	MEM_20THER
MEM_CTRL	*	*	MEM_20THER
MEM_CMD	*	*	MEM_20THER
MEM_DATA	*	*	MEM_20THER
MEM_DQS	*	*	MEM_20THER

DDR3:
DQ signals should be matched within 5 ps of associated DQS pair.
DQS intra-pair matching should be within 1 ps, inter-pair matching should be within 360 ps
No DQS to clock matching requirement.
CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 2 ps.
CMD/CTRL signals should be matched within 150 ps.
All memory signals maximum length is 1.030 ps.

B

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.2.3
SOURCE: Santa Rosa Platform DG, Rev 1.0 (#21112), Section 6.2

MCP MEM COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_MEM_COMP	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_MEM_COMP	*	=2x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.2.2

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Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
MEM_A_CLK	MEM_70D	MEM_CLK	MEM A CLK P<5..0>	9 15 26 27 32
MEM_A_CLK	MEM_70D	MEM_CLK	MEM A CLK N<5..0>	9 15 26 27 32
MEM_A_CKE	MEM_50S	MEM_CTRL	MEM A CKE<3..0>	15 21 26 27 32
MEM_A_CNTL	MEM_50S	MEM_CTRL	MEM A CS L<3..0>	15 26 27 32
MEM_A_CNTL	MEM_50S	MEM_CTRL	MEM A ODT<3..0>	15 26 27 32
MEM_A_CMD	MEM_50S	MEM_CMD	MEM A A<15..0>	9 15 26 27 32
MEM_A_CMD	MEM_50S	MEM_CMD	MEM A BA<2..0>	15 26 27 32
MEM_A_CMD	MEM_50S	MEM_CMD	MEM A RAS L	15 26 27 32
MEM_A_CMD	MEM_50S	MEM_CMD	MEM A CAS L	15 26 27 32
MEM_A_CMD	MEM_50S	MEM_CMD	MEM A WE L	15 26 27 32
MEM_A_DQ_BYTE0	MEM_55S	MEM_DATA	MEM A DQ<7..0>	15 26
MEM_A_DQ_BYTE1	MEM_55S	MEM_DATA	MEM A DQ<15..8>	15 26
MEM_A_DQ_BYTE2	MEM_55S	MEM_DATA	MEM A DQ<23..16>	15 26
MEM_A_DQ_BYTE3	MEM_55S	MEM_DATA	MEM A DQ<31..24>	15 26
MEM_A_DQ_BYTE4	MEM_55S	MEM_DATA	MEM A DQ<39..32>	15 27
MEM_A_DQ_BYTE5	MEM_55S	MEM_DATA	MEM A DQ<47..40>	15 27
MEM_A_DQ_BYTE6	MEM_55S	MEM_DATA	MEM A DQ<55..48>	15 27
MEM_A_DQ_BYTE7	MEM_55S	MEM_DATA	MEM A DQ<63..56>	15 27
MEM_A_DQ_BYTE0	MEM_55S	MEM_DATA	MEM A DM<0>	15 26
MEM_A_DQ_BYTE1	MEM_55S	MEM_DATA	MEM A DM<1>	15 26
MEM_A_DQ_BYTE2	MEM_55S	MEM_DATA	MEM A DM<2>	15 26
MEM_A_DQ_BYTE3	MEM_55S	MEM_DATA	MEM A DM<3>	15 26
MEM_A_DQ_BYTE4	MEM_55S	MEM_DATA	MEM A DM<4>	15 27
MEM_A_DQ_BYTE5	MEM_55S	MEM_DATA	MEM A DM<5>	15 27
MEM_A_DQ_BYTE6	MEM_55S	MEM_DATA	MEM A DM<6>	15 27
MEM_A_DQ_BYTE7	MEM_55S	MEM_DATA	MEM A DM<7>	15 27
MEM_A_DQS0	MEM_70D	MEM_DQS	MEM A DQS P<0>	15 26
MEM_A_DQS0	MEM_70D	MEM_DQS	MEM A DQS N<0>	15 26
MEM_A_DQS1	MEM_70D	MEM_DQS	MEM A DQS P<1>	15 26
MEM_A_DQS1	MEM_70D	MEM_DQS	MEM A DQS N<1>	15 26
MEM_A_DQS2	MEM_70D	MEM_DQS	MEM A DQS P<2>	15 26
MEM_A_DQS2	MEM_70D	MEM_DQS	MEM A DQS N<2>	15 26
MEM_A_DQS3	MEM_70D	MEM_DQS	MEM A DQS P<3>	15 26
MEM_A_DQS3	MEM_70D	MEM_DQS	MEM A DQS N<3>	15 26
MEM_A_DQS4	MEM_70D	MEM_DQS	MEM A DQS P<4>	15 27
MEM_A_DQS4	MEM_70D	MEM_DQS	MEM A DQS N<4>	15 27
MEM_A_DQS5	MEM_70D	MEM_DQS	MEM A DQS P<5>	15 27
MEM_A_DQS5	MEM_70D	MEM_DQS	MEM A DQS N<5>	15 27
MEM_A_DQS6	MEM_70D	MEM_DQS	MEM A DQS P<6>	15 27
MEM_A_DQS6	MEM_70D	MEM_DQS	MEM A DQS N<6>	15 27
MEM_A_DQS7	MEM_70D	MEM_DQS	MEM A DQS P<7>	15 27
MEM_A_DQS7	MEM_70D	MEM_DQS	MEM A DQS N<7>	15 27
MEM_B_CLK	MEM_70D	MEM_CLK	MEM B CLK P<5..0>	9 15 28 29 32
MEM_B_CLK	MEM_70D	MEM_CLK	MEM B CLK N<5..0>	9 15 28 29 32
MEM_B_CKE	MEM_50S	MEM_CTRL	MEM B CKE<3..0>	15 21 28 29 32
MEM_B_CNTL	MEM_50S	MEM_CTRL	MEM B CS L<3..0>	15 28 29 32
MEM_B_CNTL	MEM_50S	MEM_CTRL	MEM B ODT<3..0>	15 28 29 32
MEM_B_CMD	MEM_50S	MEM_CMD	MEM B A<15..0>	9 15 28 29 32
MEM_B_CMD	MEM_50S	MEM_CMD	MEM B BA<2..0>	15 28 29 32
MEM_B_CMD	MEM_50S	MEM_CMD	MEM B RAS L	15 28 29 32
MEM_B_CMD	MEM_50S	MEM_CMD	MEM B CAS L	15 28 29 32
MEM_B_CMD	MEM_50S	MEM_CMD	MEM B WE L	15 28 29 32
MEM_B_DQ_BYTE0	MEM_55S	MEM_DATA	MEM B DQ<7..0>	15 28
MEM_B_DQ_BYTE1	MEM_55S	MEM_DATA	MEM B DQ<15..8>	15 28
MEM_B_DQ_BYTE2	MEM_55S	MEM_DATA	MEM B DQ<23..16>	15 28
MEM_B_DQ_BYTE3	MEM_55S	MEM_DATA	MEM B DQ<31..24>	15 28
MEM_B_DQ_BYTE4	MEM_55S	MEM_DATA	MEM B DQ<39..32>	15 29
MEM_B_DQ_BYTE5	MEM_55S	MEM_DATA	MEM B DQ<47..40>	15 29
MEM_B_DQ_BYTE6	MEM_55S	MEM_DATA	MEM B DQ<55..48>	15 29
MEM_B_DQ_BYTE7	MEM_55S	MEM_DATA	MEM B DQ<63..56>	15 29
MEM_B_DQ_BYTE0	MEM_55S	MEM_DATA	MEM B DM<0>	15 28
MEM_B_DQ_BYTE1	MEM_55S	MEM_DATA	MEM B DM<1>	15 28
MEM_B_DQ_BYTE2	MEM_55S	MEM_DATA	MEM B DM<2>	15 28
MEM_B_DQ_BYTE3	MEM_55S	MEM_DATA	MEM B DM<3>	15 28
MEM_B_DQ_BYTE4	MEM_55S	MEM_DATA	MEM B DM<4>	15 29
MEM_B_DQ_BYTE5	MEM_55S	MEM_DATA	MEM B DM<5>	15 29
MEM_B_DQ_BYTE6	MEM_55S	MEM_DATA	MEM B DM<6>	15 29
MEM_B_DQ_BYTE7	MEM_55S	MEM_DATA	MEM B DM<7>	15 29
MEM_B_DQS0	MEM_70D	MEM_DQS	MEM B DQS P<0>	15 28
MEM_B_DQS0	MEM_70D	MEM_DQS	MEM B DQS N<0>	15 28
MEM_B_DQS1	MEM_70D	MEM_DQS	MEM B DQS P<1>	15 28
MEM_B_DQS1	MEM_70D	MEM_DQS	MEM B DQS N<1>	15 28
MEM_B_DQS2	MEM_70D	MEM_DQS	MEM B DQS P<2>	15 28
MEM_B_DQS2	MEM_70D	MEM_DQS	MEM B DQS N<2>	15 28
MEM_B_DQS3	MEM_70D	MEM_DQS	MEM B DQS P<3>	15 28
MEM_B_DQS3	MEM_70D	MEM_DQS	MEM B DQS N<3>	15 28
MEM_B_DQS4	MEM_70D	MEM_DQS	MEM B DQS P<4>	15 29
MEM_B_DQS4	MEM_70D	MEM_DQS	MEM B DQS N<4>	15 29
MEM_B_DQS5	MEM_70D	MEM_DQS	MEM B DQS P<5>	15 29
MEM_B_DQS5	MEM_70D	MEM_DQS	MEM B DQS N<5>	15 29
MEM_B_DQS6	MEM_70D	MEM_DQS	MEM B DQS P<6>	15 29
MEM_B_DQS6	MEM_70D	MEM_DQS	MEM B DQS N<6>	15 29
MEM_B_DQS7	MEM_70D	MEM_DQS	MEM B DQS P<7>	15 29
MEM_B_DQS7	MEM_70D	MEM_DQS	MEM B DQS N<7>	15 29
MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM COMP VDD	15
MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM COMP GND	15

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A

SYNC MASTER=K99 MLB

SYNC DATE=04/08/2016

PAGE TITLE

Memory Constraints

 Apple Inc.

DRAWING NUMBER

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SIZE

D

REVISION

3.3.0

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SHEET

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PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
CLK_PCIE_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?	PCIE	TOP,BOTTOM	=4X_DIELECTRIC	?
CLK_PCIE	*	20_MIL	?				
MCP_PEX_COMP	*	8_MIL	?				

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.3

NEED PCIe Gen1/Gen2 notes!

Analog Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CRT_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CRT	*	20_MIL	?	CRT	CRT	*	CRT_2CRT
CRT_2CRT	*	15_MIL	?				
CRT_2CLK	*	50_MIL	?				
CRT_2SWITCHER	*	250_MIL	?				
CRT_SYNC	*	=4X_DIELECTRIC	?				
MCP_DAC_COMP	*	=2X_DIELECTRIC	?				

CRT signal single-ended impedance varies by location:
- 37.5-ohm from MCP to first termination resistor.
- 50-ohm from first to second termination resistor.
- 75-ohm from output of three-pole filter to connector (if possible).
R/G/B signals should be matched as close as possible and < 10 inches.
SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.4.1.

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
DP_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
LVDS_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
MCP_DV_COMP	*	Y	20_MIL	20_MIL	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
DISPLAYPORT	*	=3X_DIELECTRIC	?	DISPLAYPORT	TOP,BOTTOM	=4X_DIELECTRIC	?
LVDS	*	=3X_DIELECTRIC	?	LVDS	TOP,BOTTOM	=4X_DIELECTRIC	?

LVDS intra-pair matching should be 5 mils. Pairs should be matched within 100 mils.
NOTE: NV DG recommends 90 ohm differential for LVDS, but cable/display assume 100 ohm.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 100 ps.
DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max trace length: LVDS 10 inches, DP 8.5 inches.
SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.4.2

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SATA	*	=3X_DIELECTRIC	?	SATA	TOP,BOTTOM	=4X_DIELECTRIC	?
SATA_TERM	*	8_MIL	?				

SATA intra-pair matching should be 1 ps.
Max trace length: 12 inches for SATA Gen1/Gen2, TBD for SATA Gen3.
SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.6

MCP89 Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
	PCIE_90D	PCIE	PEG R2D P<15..0>
	PCIE_90D	PCIE	PEG R2D N<15..0>
PEG_R2D	PCIE_90D	PCIE	PEG R2D C P<15..0>
	PCIE_90D	PCIE	PEG R2D C N<15..0>
PEG_D2R	PCIE_90D	PCIE	PEG D2R P<15..0>
	PCIE_90D	PCIE	PEG D2R N<15..0>
	PCIE_90D	PCIE	PEG D2R C P<15..0>
	PCIE_90D	PCIE	PEG D2R C N<15..0>
	PCIE_90D	PCIE	PCIE AP R2D P
	PCIE_90D	PCIE	PCIE AP R2D N
PCIE_AP_R2D	PCIE_90D	PCIE	PCIE AP R2D C P
	PCIE_90D	PCIE	PCIE AP R2D C N
PCIE_AP_D2R	PCIE_90D	PCIE	PCIE AP D2R P
	PCIE_90D	PCIE	PCIE AP D2R N
	PCIE_90D	PCIE	PCIE ENET R2D P
	PCIE_90D	PCIE	PCIE ENET R2D N
PCIE_ENET_R2D	PCIE_90D	PCIE	PCIE ENET R2D C P
	PCIE_90D	PCIE	PCIE ENET R2D C N
PCIE_ENET_D2R	PCIE_90D	PCIE	PCIE ENET D2R P
	PCIE_90D	PCIE	PCIE ENET D2R N
	PCIE_90D	PCIE	PCIE ENET D2R C P
	PCIE_90D	PCIE	PCIE ENET D2R C N
	PCIE_90D	PCIE	PCIE FW R2D P
	PCIE_90D	PCIE	PCIE FW R2D N
PCIE_FW_R2D	PCIE_90D	PCIE	PCIE FW R2D C P
	PCIE_90D	PCIE	PCIE FW R2D C N
PCIE_FW_D2R	PCIE_90D	PCIE	PCIE FW D2R P
	PCIE_90D	PCIE	PCIE FW D2R N
	PCIE_90D	PCIE	PCIE FW D2R C P
	PCIE_90D	PCIE	PCIE FW D2R C N
MCP_PEG_REECLK	CLK_PCIE_100D	CLK_PCIE	PEG CLK100M P
	CLK_PCIE_100D	CLK_PCIE	PEG CLK100M N
MCP_PE1_REECLK	CLK_PCIE_100D	CLK_PCIE	PCIE CLK100M AP P
	CLK_PCIE_100D	CLK_PCIE	PCIE CLK100M AP N
MCP_PE2_REECLK	CLK_PCIE_100D	CLK_PCIE	PCIE CLK100M ENET P
	CLK_PCIE_100D	CLK_PCIE	PCIE CLK100M ENET N
MCP_PE3_REECLK	CLK_PCIE_100D	CLK_PCIE	PCIE CLK100M FW P
	CLK_PCIE_100D	CLK_PCIE	PCIE CLK100M FW N
MCP_PEX_CLK_COMP		MCP_PEX_COMP	MCP PEX0_TERM
CRT_RED	CRT_50S	CRT	CRT IG_R_C_PR
CRT_GREEN	CRT_50S	CRT	CRT IG_G_Y_Y
CRT_BLUE	CRT_50S	CRT	CRT IG_B_COMP_PB
CRT_SYNC	CRT_50S	CRT_SYNC	CRT IG_HSYNC
CRT_SYNC	CRT_50S	CRT_SYNC	CRT IG_VSYNC
MCP_DAC_RSET		MCP_DAC_COMP	MCP_TV_DAC_RSET
MCP_DAC_VREF		MCP_DAC_COMP	MCP_TV_DAC_VREF
DP_INT_ML	DP_90D	DISPLAYPORT	DP IG_ML1_P<1..0>
DP_INT_ML	DP_90D	DISPLAYPORT	DP IG_ML1_N<1..0>
DP_INT_AUX_CH	DP_90D	DISPLAYPORT	DP IG_AUX_CH1_P
DP_INT_AUX_CH	DP_90D	DISPLAYPORT	DP IG_AUX_CH1_N
DP_EXT_ML	DP_90D	DISPLAYPORT	DP IG_ML0_P<3..0>
DP_EXT_ML	DP_90D	DISPLAYPORT	DP IG_ML0_N<3..0>
DP_EXT_AUX_CH	DP_90D	DISPLAYPORT	DP IG_AUX_CH0_P
DP_EXT_AUX_CH	DP_90D	DISPLAYPORT	DP IG_AUX_CH0_N
MCP_TMDS0_RSET	MCP_DV_COMP		MCP_TMDS0_RSET
MCP_TMDS0_VPROBE			MCP_TMDS0_VPROBE
LVDS_IG_A_CLK	LVDS_100D	LVDS	LVDS IG_A_CLK_P
LVDS_IG_A_CLK	LVDS_100D	LVDS	LVDS IG_A_CLK_N
LVDS_IG_A_DATA	LVDS_100D	LVDS	LVDS IG_A_DATA_P<2..0>
LVDS_IG_A_DATA	LVDS_100D	LVDS	LVDS IG_A_DATA_N<2..0>
LVDS_IG_A_DATA3	LVDS_100D	LVDS	LVDS IG_A_DATA_P<3>
LVDS_IG_A_DATA3	LVDS_100D	LVDS	LVDS IG_A_DATA_N<3>
LVDS_IG_B_CLK	LVDS_100D	LVDS	LVDS IG_B_CLK_P
LVDS_IG_B_CLK	LVDS_100D	LVDS	LVDS IG_B_CLK_N
LVDS_IG_B_DATA	LVDS_100D	LVDS	LVDS IG_B_DATA_P<2..0>
LVDS_IG_B_DATA	LVDS_100D	LVDS	LVDS IG_B_DATA_N<2..0>
LVDS_IG_B_DATA3	LVDS_100D	LVDS	LVDS IG_B_DATA_P<3>
LVDS_IG_B_DATA3	LVDS_100D	LVDS	LVDS IG_B_DATA_N<3>
MCP_IEPAB_RSET	MCP_DV_COMP		MCP_IFPAB_RSET
MCP_IEPAB_VPROBE			MCP_IFPAB_VPROBE
SATA_HDD_R2D	SATA_90D	SATA	SATA HDD_R2D_C_P
	SATA_90D	SATA	SATA HDD_R2D_C_N
	SATA_90D	SATA	SATA HDD_R2D_P
	SATA_90D	SATA	SATA HDD_R2D_N
SATA_HDD_D2R	SATA_90D	SATA	SATA HDD_D2R_P
	SATA_90D	SATA	SATA HDD_D2R_N
	SATA_90D	SATA	SATA HDD_D2R_C_P
	SATA_90D	SATA	SATA HDD_D2R_C_N
SATA_ODD_R2D	SATA_90D	SATA	SATA ODD_R2D_C_P
	SATA_90D	SATA	SATA ODD_R2D_C_N
	SATA_90D	SATA	SATA ODD_R2D_P
	SATA_90D	SATA	SATA ODD_R2D_N
SATA_ODD_D2R	SATA_90D	SATA	SATA ODD_D2R_P
	SATA_90D	SATA	SATA ODD_D2R_N
	SATA_90D	SATA	SATA ODD_D2R_C_P
	SATA_90D	SATA	SATA ODD_D2R_C_N
MCP_SATA_TERM		SATA_TERM	MCP_SATA_TERM

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MCP Constraints 1

Apple Inc.

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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=1.5x_DIELECTRIC	?
CLK_LPC	*	=2x_DIELECTRIC	?

D SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.7

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_USB_RBIAS	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?	USB	TOP, BOTTOM	=4x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.8

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.9

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?
MCP_HDA_COMP	*	8 MIL	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.10

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=1.5x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.11

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=1.5x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.12

MCP89 Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
IPC_AD	IPC 55S	IPC	LPC AD<3..0>	7 19 39 41
IPC_FRAME_L	IPC 55S	IPC	LPC FRAME_L	7 19 39 41
IPC_RESET_L	IPC 55S	IPC	LPC RESET_L	19 25
MCP_IPC_CLK0	CLK_IPC 55S	CLK_IPC	LPC CLK33M SMC_R	19 25
	CLK_IPC 55S	CLK_IPC	LPC CLK33M SMC	25 39
	CLK_IPC 55S	CLK_IPC	LPC CLK33M LPCPLUS	7 25 41
USB_EXT_A	USB_90D	USB	USB EXT_A_P	18 36
	USB_90D	USB	USB EXT_A_N	18 36
	USB_90D	USB	USB EXT_A_MUXED_P	36 72
	USB_90D	USB	USB EXT_A_MUXED_N	36 72
USB_MINI	USB_90D	USB	USB MINI_P	9 18
	USB_90D	USB	USB MINI_N	9 18
USB_EXTD	USB_90D	USB	USB EXT_D_P	7 18 37
	USB_90D	USB	USB EXT_D_N	7 18 37
USB_CAMERA	USB_90D	USB	USB CAMERA_P	7 18 37
	USB_90D	USB	USB CAMERA_N	7 18 37
USB_RT	USB_90D	USB	USB BT_P	7 18 34
	USB_90D	USB	USB BT_N	7 18 34
USB_TPAD	USB_90D	USB	USB TPAD_P	18 47 72
	USB_90D	USB	USB TPAD_N	18 47 72
USB_IR	USB_90D	USB	USB IR_P	
	USB_90D	USB	USB IR_N	
USB_EXTR	USB_90D	USB	USB EXTB_P	
	USB_90D	USB	USB EXTB_N	
USB_T57	USB_90D	USB	USB T57_P	
	USB_90D	USB	USB T57_N	
USB_EXTC	USB_90D	USB	USB EXTC_P	9 18
	USB_90D	USB	USB EXTC_N	9 18
USB_SDCARD	USB_90D	USB	USB SDCARD_P	18 38
	USB_90D	USB	USB SDCARD_N	18 38
USB_WM	USB_90D	USB	USB WM_P	
	USB_90D	USB	USB WM_N	
MCP_USB_RBIA_S	MCP_USB_RBIA_S		MCP_USB_RBIA_S_GND	18
SMBUS_MCP_0_CLK	SMB 55S	SMB	SMBUS MCP_0_CLK	19 42
SMBUS_MCP_0_DATA	SMB 55S	SMB	SMBUS MCP_0_DATA	19 42
(SMBUS_SMC_MGMT_SCL)	SMB 55S	SMB	SMBUS MCP_1_CLK	19 42
(SMBUS_SMC_MGMT_SDA)	SMB 55S	SMB	SMBUS MCP_1_DATA	19 42
HDA_BIT_CLK	HDA 55S	HDA	HDA BIT_CLK	7 19 37
	HDA 55S	HDA	HDA BIT_CLK_R	19 37
HDA_SYNC	HDA 55S	HDA	HDA SYNC	7 19 37
	HDA 55S	HDA	HDA SYNC_R	19 37
HDA_RST_L	HDA 55S	HDA	HDA RST_R_L	19
	HDA 55S	HDA	HDA RST_L	7 19 37
HDA_SDIN0	HDA 55S	HDA	HDA SDIN0	7 19 37
	HDA 55S	HDA	HDA SDIN_CODEC	
HDA_SDOUT	HDA 55S	HDA	HDA SDOUT	7 19 37
	HDA 55S	HDA	HDA SDOUT_R	19
MCP_HDA_PULLDN_COMP		MCP_HDA_COMP	MCP_HDA_PULLDN_COMP	19
MCP_SUS_CLK	CLK_SLOW 55S	CLK_SLOW	PM_CLK32K_SUSCLK_R	19 25
	CLK_SLOW 55S	CLK_SLOW	PM_CLK32K_SUSCLK	25 39
SPT_CLK	SPT 55S	SPT	SPI_CLK_R	19 41
	SPT 55S	SPT	SPI_CLK	41
SPT_MOSI	SPT 55S	SPT	SPI_MOSI_R	19 41
	SPT 55S	SPT	SPI_MOSI	41
SPT_MISO	SPT 55S	SPT	SPI_MISO	19 41
SPT_CS0	SPT 55S	SPT	SPI_CS0_R_L	19 41
	SPT 55S	SPT	SPI_CS0_L	41
	SPT 55S	SPT	SPI_MLB_CLK	41 48
	SPT 55S	SPT	SPI_MLB_MOSI	41 48
	SPT 55S	SPT	SPI_MLB_MISO	41 48
	SPT 55S	SPT	SPI_MLB_CS_L	41 48
	SPT 55S	SPT	SPI_ALT_CLK	7 41
	SPT 55S	SPT	SPI_ALT_MOSI	7 41
	SPT 55S	SPT	SPI_ALT_MISO	7 41
	SPT 55S	SPT	SPI_ALT_CS_L	7 41

MCP RGMII (Ethernet) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_MII_COMP	*	=STANDARD	7.5 MIL	7.5 MIL	=STANDARD	=STANDARD	=STANDARD
ENET_MII_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_BUF0_CLK	*	=3:1_SPACING	?
ENET_MII	*	12 MIL	?

D SOURCE: MCP73 Interface DG (DG-02974-001_v01), Sections 2.7.2 & 2.7.4

88E1116R (Ethernet PHY) Constraints

[illegible]

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_MDI	*	25 MIL	?

SOURCE: MCP73 Interface DG (DG-02974-001_v01), Section 2.7.4

SD Card Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SD_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_INTERFACE	*	=3X_DIELECTRIC	?

RGMII Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	MCP_MII_COMP	MCP_MII_COMP		MCP MII COMP VDD 18
	MCP_MII_COMP	MCP_MII_COMP		MCP MII COMP GND 18
	MCP_CLK25M_BUF0	ENET_MII_55S	MCP_BUF0_CLK	MCP CLK25M BUF0 R
		ENET_MII_55S	MCP_BUF0_CLK	RTL8211 CLK25M CKXTAL1
	ENET_INTR_I	ENET_MII_55S	ENET_MII	ENET INTR L
	ENET_MDIO	ENET_MII_55S	ENET_MII	ENET MDIO 9 18
	ENET_MDC	ENET_MII_55S	ENET_MII	ENET MDC
	ENET_PWDOWN_I	ENET_MII_55S	ENET_MII	ENET PWDOWN L
	ENET_RXCLK	ENET_MII_55S	ENET_MII	ENET CLK125M RXCLK R
		ENET_MII_55S	ENET_MII	ENET CLK125M RXCLK 9 18
		ENET_MII_55S	ENET_MII	ENET RXD_R<3..0>
	ENET_RXD_STRAP	ENET_MII_55S	ENET_MII	ENET RXD<0> 9 18
	ENET_RXD_STRAP	ENET_MII_55S	ENET_MII	ENET RXD<3..1> 9 18
	ENET_RXD	ENET_MII_55S	ENET_MII	ENET RX_CTRL 9 18
	ENET_TXCLK	ENET_MII_55S	ENET_MII	ENET CLK125M TXCLK
	ENET_TXD	ENET_MII_55S	ENET_MII	ENET TXD<0>
	ENET_TXD	ENET_MII_55S	ENET_MII	ENET TXD<3..1>
	ENET_TXD	ENET_MII_55S	ENET_MII	ENET TX_CTRL
		ENET_MII_55S	ENET_MII	ENET_RESET L

Ethernet Net Properties

		NET_TYPE		
ELECTRICAL_CONSTRAINT_SET		PHYSICAL	SPACING	
	ENET_MDI	ENET MDI 1000	ENET MDI	ENET MDI P<3..0>
		ENET MDI 1000	ENET MDI	ENET MDI N<3..0>

SD Card Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
  	SD_DATA	SD_55S	SD_INTERFACE	SD D<4..0> SDCONN DATA<4..0> BCM57765_CR DATA<4>
		SD_55S	SD_INTERFACE	
		SD_55S	SD_INTERFACE	
  	SD_DATA_R	SD_55S	SD_INTERFACE	SD D<7..5> SDCONN DATA<7..5> BCM57765_CR DATA<7..5>
		SD_55S	SD_INTERFACE	
		SD_55S	SD_INTERFACE	
  	SD_CLK	SD_55S	SD_INTERFACE	SD CLK SD CLK_R SDCONN CLK
		SD_55S	SD_INTERFACE	
		SD_55S	SD_INTERFACE	
  	SD_CMD	SD_55S	SD_INTERFACE	SD CMD SDCONN CMD BCM57765_CR CMD
		SD_55S	SD_INTERFACE	
		SD_55S	SD_INTERFACE	

NOTE: SD_D<7..5> are different to support BCM5764M/BCM57765 co-layout.



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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_1T01_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_1T01_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=1:1_SPACING	?
THERM	*	=1:1_SPACING	?
AUDIO	*	=1:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENETCONN	*	25 MILS	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
MEM_POWER	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

C

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	*	GND_P2MM
MEM_CMD	GND	*	GND_P2MM
MEM_CTRL	GND	*	GND_P2MM
MEM_DATA	GND	*	GND_P2MM
MEM_DQS	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_POWER	*	PWR_P2MM
MEM_CMD	MEM_POWER	*	PWR_P2MM
MEM_CTRL	MEM_POWER	*	PWR_P2MM
MEM_DATA	MEM_POWER	*	PWR_P2MM
MEM_DQS	MEM_POWER	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
PCIE	GND	*	GND_P2MM
SATA	GND	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SATA	SB_POWER	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_FSB	GND	*	GND_P2MM
CPU_COMP	GND	*	GND_P2MM
CPU_GTLREF	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LVDS	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_MDI	GND	*	GND_P2MM

SD CARD READER LAYOUT RELAXATIONS

B

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SD_55S_OVERRIDE	*	OVERRIDE	=STANDARD_OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE

MCP Fanout Constraint Relaxations

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S_OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM_OVERRIDE	5.8 MM_OVERRIDE	OVERRIDE	OVERRIDE
MCP_DV_COMP_OVERRIDE	TOP_OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM_OVERRIDE	500 MIL_OVERRIDE	OVERRIDE	OVERRIDE
MCP_MEM_COMP_OVERRIDE	TOP_OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM_OVERRIDE	500 MIL_OVERRIDE	OVERRIDE	OVERRIDE
MCP_MII_COMP_OVERRIDE	TOP_OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM_OVERRIDE	500 MIL_OVERRIDE	OVERRIDE	OVERRIDE
MCP_USB_RBIAIS_OVERRIDE	TOP_OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM_OVERRIDE	500 MIL_OVERRIDE	OVERRIDE	OVERRIDE
MCP_DV_COMP_OVERRIDE	*	OVERRIDE	OVERRIDE	0.25 MM_OVERRIDE	250 MIL_OVERRIDE	OVERRIDE	OVERRIDE

A

Misc Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
(USB_EXT_A)	USB_90D	USB	USB_EXT_A_MUXED_P 36 69
(USB_EXT_A)	USB_90D	USB	USB_EXT_A_MUXED_N 36 69
(USB_EXT_A)	USB_90D	USB	USB_LT1_P 36
(USB_EXT_A)	USB_90D	USB	USB_LT1_N 36
(USB_TPAD)	USB_90D	USB	USB_TPAD_P 18 47 69
(USB_TPAD)	USB_90D	USB	USB_TPAD_N 18 47 69
(USB_TPAD)	USB_90D	USB	USB_TPAD_CONN_P 7 47
(USB_TPAD)	USB_90D	USB	USB_TPAD_CONN_N 7 47
SMBUS_SMC_MGMT_SDA	SMB_55S	SMB	I2C_SMC_SMS_SDA_R
SMBUS_SMC_MGMT_SCL	SMB_55S	SMB	I2C_SMC_SMS_SCL_R
	SMB_55S	SMB	I2C_TCON_SCL 42
	SMB_55S	SMB	I2C_TCON_SDA 42
	SMB_55S	SMB	I2C_TCON_SCL_CONN 60
	SMB_55S	SMB	I2C_TCON_SDA_CONN 60

Graphics Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
	DP_90D	DISPLAYPORT	DP_INT_ML_P<1..0> 9 60
	DP_90D	DISPLAYPORT	DP_INT_ML_N<1..0> 9 60
	DP_90D	DISPLAYPORT	DP_INT_ML_C_P<1..0> 60
	DP_90D	DISPLAYPORT	DP_INT_ML_C_N<1..0> 60
	DP_90D	DISPLAYPORT	DP_INT_ML_F_P<1..0> 7 60
	DP_90D	DISPLAYPORT	DP_INT_ML_F_N<1..0> 7 60
	DP_90D	DISPLAYPORT	DP_INT_AUX_CH_C_P 7 60
	DP_90D	DISPLAYPORT	DP_INT_AUX_CH_C_N 7 60
	DP_90D	DISPLAYPORT	DP_INT_AUX_CH_P 9 60
	DP_90D	DISPLAYPORT	DP_INT_AUX_CH_N 9 60
(DP_EXT_ML)	DP_90D	DISPLAYPORT	DP_EXT_ML_P<3..0> 9 62
	DP_90D	DISPLAYPORT	DP_EXT_ML_N<3..0> 9 62
	DP_90D	DISPLAYPORT	DP_EXT_ML_C_P<3..0> 62
	DP_90D	DISPLAYPORT	DP_EXT_ML_C_N<3..0> 62
	DP_90D	DISPLAYPORT	DP_EXT_ML_F_P<3..0> 62
	DP_90D	DISPLAYPORT	DP_EXT_ML_F_N<3..0> 62
(DP_EXT_AUX_CH)	DP_90D	DISPLAYPORT	DP_EXT_AUX_CH_C_P 9 62
	DP_90D	DISPLAYPORT	DP_EXT_AUX_CH_C_N 9 62

Power Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
CPUTHMSNS_D2	THERM_1T01_55S	THERM	DRAMTHMSNS_D2_P 45
	THERM_1T01_55S	THERM	DRAMTHMSNS_D2_N 45
CPU_THERMD	THERM_1T01_55S	THERM	CPU_THERMD_P 10 45
	THERM_1T01_55S	THERM	CPU_THERMD_N 10 45
MCPTHMSNS_D2	THERM_1T01_55S	THERM	MLBR_THMDIODE_P 45
	THERM_1T01_55S	THERM	MLBR_THMDIODE_N 45
MCP_THMDIODE	THERM_1T01_55S	THERM	MCP_THMDIODE_P 19 45
	THERM_1T01_55S	THERM	MCP_THMDIODE_N 19 45
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_1V5_S3_P 43 53
	SENSE_1T01_55S	SENSE	ISNS_1V5_S3_N 43 53
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_AIRPORT_P 34 43
	SENSE_1T01_55S	SENSE	ISNS_AIRPORT_N 34 43
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_CSREG_P 44
	SENSE_1T01_55S	SENSE	ISNS_CSREG_N 44
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_HDD_P 35 43
	SENSE_1T01_55S	SENSE	ISNS_HDD_N 35 43
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	ISNS_LCDBKLT_P 43 63
	SENSE_1T01_55S	SENSE	ISNS_LCDBKLT_N 43 63
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	CPUVTTIS0_CS_P 56
	SENSE_1T01_55S	SENSE	CPUVTTIS0_CS_N 56
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	IMVP6_CS_P 54
	SENSE_1T01_55S	SENSE	IMVP6_CS_N 54
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	IMVP6_CS_R_P 54
	SENSE_1T01_55S	SENSE	IMVP6_CS_R_N 54
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	CPU_VTTSENSE_P 56
	SENSE_1T01_55S	SENSE	CPU_VTTSENSE_N 56
SENSE_DIEFPATR	SENSE_1T01_55S	SENSE	MCPCORES0_VSEN_P 22 55
	SENSE_1T01_55S	SENSE	MCPCORES0_VSEN_N 22 55
		MEM_POWER	PP1V5R1V35_S3 7 8
		SB_POWER	PP3V3_S5 7 8 58
		SB_POWER	PP3V3_S0 7 8 58
		SB_POWER	PP1V5_S0 7 8 58
		GND	GND

Audio Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
SPKRAMP_INR	DIEFPATR	AUDIO	SPKRAMP_INR_P 7 37 49
	DIEFPATR	AUDIO	SPKRAMP_INR_N 7 37 49
MAX98300_R	DIEFPATR	AUDIO	MAX98300_R_P 49
	DIEFPATR	AUDIO	MAX98300_R_N 49

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K16/K99 Specific Constraints			
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K99 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM	NO_TYPE, BGA	MM	15.5.1

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	0.100 MM	0.076 MM	30 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	12.7 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
27P4_OHM_SE	ISL3, ISL10	Y	0.250 MM	0.250 MM			
27P4_OHM_SE	ISL4, ISL9	Y	0.250 MM	0.250 MM			

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
40_OHM_SE	TOP,BOTTOM	Y	0.170 MM	0.170 MM			
40_OHM_SE ISL3, ISL4, ISL9, ISL10		Y	0.140 MM	0.140 MM			

PHYSICAL_RULE_SET	LAYER	ALL W ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	TOP,BOTTOM	Y	0.110 MM	0.110 MM			
50_OHM_SE ISL3, ISL4, ISL9, ISL10		Y	0.090 MM	0.090 MM			

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	TOP,BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE ISL3, ISL4, ISL9, ISL10		Y	0.076 MM	0.076 MM			

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
70_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
70_OHM_DIFF	TOP,BOTTOM	Y	0.175 MM	0.175 MM		0.130 MM	0.130 MM
70_OHM_DIFF	ISL3,ISL10	Y	0.135 MM	0.135 MM		0.130 MM	0.130 MM
70_OHM_DIFF	ISL4,ISL9	Y	0.155 MM	0.155 MM		0.130 MM	0.130 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
80_OHM_DIFF	TOP,BOTTOM	Y	0.140 MM	0.140 MM		0.160 MM	0.160 MM
80_OHM_DIFF	ISL3,ISL10	Y	0.109 MM	0.109 MM		0.160 MM	0.160 MM
80_OHM_DIFF	ISL4,ISL9	Y	0.125 MM	0.125 MM		0.160 MM	0.160 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
75_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
75_OHM_DIFF	TOP,BOTTOM	Y	0.160 MM	0.160 MM		0.160 MM	0.160 MM
75_OHM_DIFF	ISL3, ISL10	Y	0.120 MM	0.120 MM		0.140 MM	0.140 MM
75_OHM_DIFF	ISL4, ISL9	Y	0.140 MM	0.140 MM		0.140 MM	0.140 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	TOP, BOTTOM	Y	0.115 MM	0.115 MM		0.210 MM	0.210 MM
90_OHM_DIFF	ISL3, ISL10	Y	0.089 MM	0.089 MM		0.210 MM	0.210 MM
90_OHM_DIFF	ISL4, ISL9	Y	0.105 MM	0.105 MM		0.210 MM	0.210 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
95_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
95_OHM_DIFF	TOP,BOTTOM	Y	0.115 MM	0.115 MM		0.210 MM	0.210 MM
95_OHM_DIFF	ISL3,ISL10	Y	0.089 MM	0.089 MM		0.210 MM	0.210 MM
95_OHM_DIFF	ISL4,ISL9	Y	0.105 MM	0.105 MM		0.210 MM	0.210 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	TOP, BOTTOM	Y	0.091 MM	0.091 MM		0.200 MM	0.200 MM
100_OHM_DIFF	ISL3, ISL10	Y	0.075 MM	0.075 MM		0.300 MM	0.300 MM
100_OHM_DIFF	ISL4, ISL9	Y	0.085 MM	0.085 MM		0.200 MM	0.200 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	0.1 MM	?
BGA_P2MM	*	0.2 MM	?
BGA_P3MM	*	0.3 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?
1.5:1_SPACING	*	0.15 MM	?
1.8:1_SPACING	*	0.18 MM	?
2:1_SPACING	*	0.2 MM	?
2.28:1_SPACING	*	0.228 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
PP1V5_MEM	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.2 MM	1000
PwR_P2MM	*	0.2 MM	1000

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
NB_STATIC	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.140 MM	?
3X_DIELECTRIC	*	0.210 MM	?
4X_DIELECTRIC	*	0.280 MM	?
1.5X_DIELECTRIC	*	0.105 MM	?
5X_DIELECTRIC	*	0.350 MM	?

8		7		6		5		4		3		2		1			
1UF 0402 CAPACITOR VENDOR TABLES FOR ACOUSTICS																	
SAMSUNG						MURATA						TAIYO YUDEN					
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0629	2	CAP, 1UF, 6.3V, 10%, 0402	C7203,C7000	CRITICAL	SS_CAP_1UF	138S0628	2	CAP, 1UF, 6.3V, 10%, 0402	C7203,C7000	CRITICAL	MU_CAP_1UF	138S0630	2	CAP, 1UF, 6.3V, 10%, 0402	C7203,C7000	CRITICAL	TY_CAP_1UF
2.2UF 0402 CAPACITOR VENDOR TABLES FOR ACOUSTICS																	
SAMSUNG						MURATA						TAIYO YUDEN					
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C1241,C1242,C1243,C1244,C1246,C1247,C1248	CRITICAL	SS_CAP_2_2UF	138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C1241,C1242,C1243,C1244,C1246,C1247,C1248	CRITICAL	MU_CAP_2_2UF	138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C1241,C1242,C1243,C1244,C1246,C1247,C1248	CRITICAL	TY_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C1251,C1252,C1253,C1254,C1256,C1257,C1258	CRITICAL	SS_CAP_2_2UF	138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C1251,C1252,C1253,C1254,C1256,C1257,C1258	CRITICAL	MU_CAP_2_2UF	138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C1251,C1252,C1253,C1254,C1256,C1257,C1258	CRITICAL	TY_CAP_2_2UF
138S0632	8	CAP, 2.2UF, 6.3V, 20%, 0402	C1269,C1261,C1262,C1263,C1264,C1265,C1266	CRITICAL	SS_CAP_2_2UF	138S0633	8	CAP, 2.2UF, 6.3V, 20%, 0402	C1269,C1261,C1262,C1263,C1264,C1265,C1266	CRITICAL	MU_CAP_2_2UF	138S0634	8	CAP, 2.2UF, 6.3V, 20%, 0402	C1269,C1261,C1262,C1263,C1264,C1265,C1266	CRITICAL	TY_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9001,C9002,C9003,C9004,C9005,C9006,C9007,C9008	CRITICAL	SS_CAP_2_2UF	138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9001,C9002,C9003,C9004,C9005,C9006,C9007,C9008	CRITICAL	MU_CAP_2_2UF	138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9001,C9002,C9003,C9004,C9005,C9006,C9007,C9008	CRITICAL	TY_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9011,C9012,C9013,C9014,C9015,C9016,C9017,C9018	CRITICAL	SS_CAP_2_2UF	138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9011,C9012,C9013,C9014,C9015,C9016,C9017,C9018	CRITICAL	MU_CAP_2_2UF	138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9011,C9012,C9013,C9014,C9015,C9016,C9017,C9018	CRITICAL	TY_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9021,C9022,C9023,C9024,C9025,C9026,C9027,C9028	CRITICAL	SS_CAP_2_2UF	138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9021,C9022,C9023,C9024,C9025,C9026,C9027,C9028	CRITICAL	MU_CAP_2_2UF	138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9021,C9022,C9023,C9024,C9025,C9026,C9027,C9028	CRITICAL	TY_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9031,C9032,C9033,C9034,C9035,C9036,C9037,C9038	CRITICAL	SS_CAP_2_2UF	138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9031,C9032,C9033,C9034,C9035,C9036,C9037,C9038	CRITICAL	MU_CAP_2_2UF	138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9031,C9032,C9033,C9034,C9035,C9036,C9037,C9038	CRITICAL	TY_CAP_2_2UF
138S0632	12	CAP, 2.2UF, 6.3V, 20%, 0402	C1289,C1286,C1287,C1288,C1291,C1292,C1293,C1294	CRITICAL	SS_CAP_2_2UF	138S0633	12	CAP, 2.2UF, 6.3V, 20%, 0402	C1289,C1286,C1287,C1288,C1291,C1292,C1293,C1294	CRITICAL	MU_CAP_2_2UF	138S0634	12	CAP, 2.2UF, 6.3V, 20%, 0402	C1289,C1286,C1287,C1288,C1291,C1292,C1293,C1294	CRITICAL	TY_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3051,C3054,C3055,C3056,C3058,C3059,C3061,C3064	CRITICAL	SS_CAP_2_2UF	138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3051,C3054,C3055,C3056,C3058,C3059,C3061,C3064	CRITICAL	MU_CAP_2_2UF	138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3051,C3054,C3055,C3056,C3058,C3059,C3061,C3064	CRITICAL	TY_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3021,C3024,C3025,C3026,C3031,C3034,C3035,C3036	CRITICAL	SS_CAP_2_2UF	138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3021,C3024,C3025,C3026,C3031,C3034,C3035,C3036	CRITICAL	MU_CAP_2_2UF	138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3021,C3024,C3025,C3026,C3031,C3034,C3035,C3036	CRITICAL	TY_CAP_2_2UF
138S0632	8	CAP, 2.2UF, 6.3V, 20%, 0402	C3042,C3044,C3045,C3046,C3050,C3051,C3054,C3055	CRITICAL	SS_CAP_2_2UF	138S0633	8	CAP, 2.2UF, 6.3V, 20%, 0402	C3042,C3044,C3045,C3046,C3050,C3051,C3054,C3055	CRITICAL	MU_CAP_2_2UF	138S0634	8	CAP, 2.2UF, 6.3V, 20%, 0402	C3042,C3044,C3045,C3046,C3050,C3051,C3054,C3055	CRITICAL	TY_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3001,C3004,C3005,C3010,C3011,C3012,C3014,C3015	CRITICAL	SS_CAP_2_2UF	138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3001,C3004,C3005,C3010,C3011,C3012,C3014,C3015	CRITICAL	MU_CAP_2_2UF	138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3001,C3004,C3005,C3010,C3011,C3012,C3014,C3015	CRITICAL	TY_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3021,C3024,C3025,C3030,C3031,C3034,C3035,C3036	CRITICAL	SS_CAP_2_2UF	138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3021,C3024,C3025,C3030,C3031,C3034,C3035,C3036	CRITICAL	MU_CAP_2_2UF	138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3021,C3024,C3025,C3030,C3031,C3034,C3035,C3036	CRITICAL	TY_CAP_2_2UF
138S0632	9	CAP, 2.2UF, 6.3V, 20%, 0402	C3012,C3044,C3045,C3046,C3050,C3051,C3054,C3055	CRITICAL	SS_CAP_2_2UF	138S0633	9	CAP, 2.2UF, 6.3V, 20%, 0402	C3012,C3044,C3045,C3046,C3050,C3051,C3054,C3055	CRITICAL	MU_CAP_2_2UF	138S0634	9	CAP, 2.2UF, 6.3V, 20%, 0402	C3012,C3044,C3045,C3046,C3050,C3051,C3054,C3055	CRITICAL	TY_CAP_2_2UF
10UF 0603 CAPACITOR VENDOR TABLES FOR ACOUSTICS																	
SAMSUNG						MURATA						TAIYO YUDEN					
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0626	1	CAP, 10UF, 6.3V, 20%, 0603	C1280	CRITICAL	SS_CAP_10UF	138S0625	1	CAP, 10UF, 6.3V, 20%, 0603	C1280	CRITICAL	MU_CAP_10UF	138S0627	1	CAP, 10UF, 6.3V, 20%, 0603	C1280	CRITICAL	TY_CAP_10UF
138S0626	8	CAP, 10UF, 6.3V, 20%, 0603	C4006,C4005,C5025,C7205,C7206,C7345,C7355,C7356	CRITICAL	SS_CAP_10UF	138S0625	8	CAP, 10UF, 6.3V, 20%, 0603	C4006,C4005,C5025,C7205,C7206,C7345,C7355,C7356	CRITICAL	MU_CAP_10UF	138S0627	8	CAP, 10UF, 6.3V, 20%, 0603	C4006,C4005,C5025,C7205,C7206,C7345,C7355,C7356	CRITICAL	TY_CAP_10UF
138S0626	8	CAP, 10UF, 6.3V, 20%, 0603	C9012,C9406,C2500,C2520,C2560,C2567,C2600,C2601	CRITICAL	SS_CAP_10UF	138S0625	8	CAP, 10UF, 6.3V, 20%, 0603	C9012,C9406,C2500,C2520,C2560,C2567,C2600,C2601	CRITICAL	MU_CAP_10UF	138S0627	8	CAP, 10UF, 6.3V, 20%, 0603	C9012,C9406,C2500,C2520,C2560,C2567,C2600,C2601	CRITICAL	TY_CAP_10UF
22UF 0603 CAPACITOR VENDOR TABLES FOR ACOUSTICS																	
SAMSUNG						MURATA						TAIYO YUDEN					
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0635	4	CAP, 22UF, 6.3V, 20%, 0603	C1210,C1214,C1217,C1218	CRITICAL	SS_CAP_22UF	138S0676	4	CAP, 22UF, 6.3V, 20%, 0603	C1210,C1214,C1217,C1218	CRITICAL	MU_CAP_22UF	138S0688	4	CAP, 22UF, 6.3V, 20%, 0603	C1210,C1214,C1217,C1218	CRITICAL	TY_CAP_22UF
138S0635	3	CAP, 22UF, 6.3V, 20%, 0603	C1223,C1226,C1227	CRITICAL	SS_CAP_22UF	138S0676	3	CAP, 22UF, 6.3V, 20%, 0603	C1223,C1226,C1227	CRITICAL	MU_CAP_22UF	138S0688	3	CAP, 22UF, 6.3V, 20%, 0603	C1223,C1226,C1227	CRITICAL	TY_CAP_22UF
138S0635	5	CAP, 22UF, 6.3V, 20%, 0603	C1230,C4002,C7300,C7361,C4000	CRITICAL	SS_CAP_22UF	138S0676	5	CAP, 22UF, 6.3V, 20%, 0603	C1230,C4002,C7300,C7361,C4000	CRITICAL	MU_CAP_22UF	138S0688	5	CAP, 22UF, 6.3V, 20%, 0603	C1230,C4002,C7300,C7361,C4000	CRITICAL	TY_CAP_22UF
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